Statistics
An Alternative Approach

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1 The Great Schism

A long-standing debate ran through the last century concerning the reality of mathematical objects. Following Cantor's work on various notions of infinity, a rift gradually opened between, on the one hand, idealist mathematicians who admitted and manipulated infinite objects (cardinals, continuity, etc.), and, on the other hand, proponents of a constructive approach in which all objects are explicitly shaped by finite, discrete, and empirically verifiable algorithms (e.g. Rowe, 2022). Brouwer, in particular, pointed out around 1907 the inability of idealist mathematical objects to carry any numerical meaning. The ensuing discussions were intense, but gradually most (though not all) mathematical theorems found a constructive proof, framed in the logical connectors and quantifiers specific to this philosophy.

Surprisingly, this debate did not extend to one of the most recent branches of mathematics: statistics. The mathematical formalization of statistics was forged during the same period, from the 1920s until about 1945. As we show in the section below, it is purely idealist. The curriculum of the modern statistician is imbued with idealities concerning several hypothetical infinite objects (called *population*, *probability on the Borel σ-algebra*, *continuous variable*, *sampling distribution*, among others). The very conception of a *statistical datum* arises from an idealized hypothetical mechanism.

Yet, some mathematicians of the time did point to a potential pitfall for statistics. Thus, the probabilist most often cited by statisticians, Kolmogorov, noted: "in describing any observable random process we can obtain only finite fields of probability"[1] (Kolmogorov, 1933). It is equally well established that if one were to use probabilities of idealized events (such as the Borel σ-algebra, for instance), these cannot be interpreted as "actual and observable events"[2] (Kolmogorov, 1933). Similarly, the mathematician of measure theory most frequently cited by statisticians, Borel, was highly critical of the statisticians' ideal of reproducibility of data: "It is, however, easy to see [that these mathematical assertions] are contrary to common sense and absolutely unacceptable for the simple reason that, if n is very large, it is inconceivable that an experiment could be repeated n times" (É. Borel, 1949).

The question, then, is whether statistics—or "data science," as it is now called—can be built from finite objects. Is there a path to conceive statistics without resorting to the construction of an ideal hypothetical world, and to produce instead a system with empirical meaning? This is the object of this course.

We will first identify the three fundamental idealities at work in modern statistics, as well as a fourth ideality employed when the social sciences make use of statistics, as in the case of econometrics.

[1] Bei einer Beschreibung irgendwelcher wirklich beobachtbarer zufälliger Prozesse kann man nur endliche Wahrscheinlichkeitsfelder erhalten (Kolmogorov, 1933, Chapter 2, §1, trans. N. Morrison)

[2] reelle und [...] beobachtbare Ereignisse (*ibid., §2*)

2 Ronald Fisher (1922) : What is a Datum?

2.1 Data reduction

It was in 1922, in a foundational article, that Ronald Fisher proposed to give a logical meaning to statistical activity. Acknowledging its frequent use, Fisher lamented that its basic principles were "in the state of obscurity" and that "fundamental problems have been ignored and fundamental paradoxes left unresolved" (Fisher, 1922, p. 310). What were these dark corners Fisher sought to illuminate? His diagnosis can be summarized as follows: while everyone is aware of the natural imperfection of data — *erroribus obnoxiae manent*, a notion spreading since the work of Gauss (1821) — how is it possible to define precise statistical concepts? Fisher pointed, for instance, to terminological confusions in applied statistics: "it is customary to apply the same name, *mean, standard deviation, correlation coefficient,* etc., both to the true value which we should like to know, but can only estimate, and to the particular value at which we happen to arrive by our methods of estimation" (p. 311). Thus, there would be an object, the "true value," which is unfortunately accessible to humans only through a mediator, the "estimate," and for Fisher there reigns a confusion between these two notions. But how can this "true value" be defined? In other words, what exactly do statistical calculations and procedures such as the mean or correlation aim at? Fisher offered a new answer to this question, one whose impact on our contemporary conception of *data* remains spectacular.

In the second section of his article, he specified that the statistician's task is the *reduction of data*: "A quantity of data, which usually by its mere bulk is incapable of entering the mind, is to be replaced by relatively few quantities which shall adequately represent the whole, or which, in other words, shall contain as much as possible, ideally the whole, of the relevant information contained in the original data." (p. 311) We readily recognize this concern for compression, now a core element in the training of every student in statistics or data science. But what does data reduction aim at? To answer this question, Fisher envisioned (we emphasize) "a *hypothetical infinite population*, of which the actual data are regarded as constituting a random sample. The law of distribution of this hypothetical population is specified by relatively few parameters, which are sufficient to describe it exhaustively in respect of all qualities under discussion" (p. 311). Let us first explicate this conception, which constitutes the cornerstone of modern statistical analysis.

The "infinite population" describes a set in which reside *all* possible realizations of a random experiment. If I am interested, for example, in the result of rolling three distinct six-sided dice, the population would then contain all possible infinite sequences of triplets (x,y,z) whose components are the results of each of the three dice. The datum we receive after the random experiment "rolling 3 six-sided dice 10 times" is the extraction of 10 successive elements from this infinite set. Fisher's conception consists in making this hypothetical "infinite population" equivalent to a "law of distribution," otherwise called a "probability distribution." In our example, this distribution governs the frequency of occurrence of each face of each die in the "infinite population." Fisher is explicit on this point: "when we say that the probability of throwing a five with a die is one-sixth, [...] of a hypothetical population of an infinite number of throws, with the die in its original condition, exactly one-sixth will be fives" (p. 312). Without concerning itself with the manipulation of infinite objects or their division, this vision of probability lies at the very foundation of the mathematical theory of statistics, as attested by Cramér (1946, §13.5, p. 149) in a classic reference. The *parameters* of this distribution are the probabilities, i.e., limiting frequencies of occurrence of each symbol on each die. In our example, 15 parameters suffice to describe the population. This is the meaning of the expression

$$\{P_X^\theta; \theta \in \Theta\}$$

which inaugurates every statistics textbook, indicating that at the beginning of the study of a variable X stood an infinite population, identified with a probability distribution (limiting frequencies) denoted P_X^θ, itself governed by one or several parameters denoted θ. The set Θ is the domain of possible variation of the parameters (between 0 and 1 in the case of the die), but it is understood that a *statistical experiment* arises from a population P_X^θ where θ is a fixed parameter (for instance 1/6 for each symbol in the case of a fair die), but unfortunately hidden from the human world.

Fisher also noted that the distribution function could be continuous, if the possible values taken by the experiment are numerous. If a die has six faces, an experiment such as your salary in five years or the price of gas in a month has many possible values. So many, in fact, that the statistician considers a second infinity: "To reach a true curve, not only would an infinite number of individuals have to be placed in each class, but the number of classes (arrays) into which the population is divided must be made infinite" (Fisher, 1922, p. 312).

The meaning of operations such as the mean or correlation is thus that of a reduction of data made possible by the mechanism of data production and its link with the population: "the process of the reduction of data is, even in the simplest cases, performed by interpreting the available observations as a sample from a hypothetical infinite population." This mechanism bears the name of "sampling." It is "random" in the sense that if the mechanism is activated a second time, the sample obtained may take different values. Each sample coming from the same population, it contains the trace of the law of that population.

Fisher further specified the program to implement data reduction. It is divided into three problems: the *specification*, the *estimation*, and the *distribution*. The problem of *specification* is "the choice of the mathematical form of the population" (p. 313), which includes the formula of the probability distribution of the infinite population. Here one specifies the mechanism of data production, for instance "independent and identically distributed." The problem of *estimation* is a calculation aiming to approximate the parameters of the infinite population. Here the statistician concentrates efforts on qualifying the quality of the calculation, through notions such as bias, sufficiency, or efficiency. The problem of *distribution* includes the "discussions of the distribution of statistics derived from samples" (p. 313). Note the plural in "samples," indicating that the distribution of the estimator is computed from several samplings.

This program, now widely taught, culminates in the theory of hypothesis testing (Neyman, 1937), which seeks to assess whether the particular sample before our eyes conforms, or not, to a hypothesis posited about the value of the parameter of the infinite population. Let us now examine how this conception of the world aligns with sensible reality.

2.2 The Infinite Population and Endless Reproduction

In its ideal vision of the world, statistics first imagined a hypothetical population. The population embodies the infinity of possible situations that may at any moment be encountered through an experiment. Note that this ideality neglects individuals, since it is only the law of the population that is sought by the statistician. We propose to call this phenomenon the "*erasure of the data support*." Let us explain this point.

In his book originally published in 1925 and cited here in its 1970 edition, Fisher gives the following example:

> "[...] since no observational record can completely specify a human being, the populations studied are always to some extent abstractions. If we have records of the stature of 10,000 recruits, it is rather the population of statures than the population of recruits that is open to study. Nevertheless, in a real sense, statistics is the study of populations, or aggregates of individuals, rather than of individuals. Scientific theories which involve the properties of large aggregates of individuals, and not necessarily the properties of the individuals themselves, such as the Kinetic Theory of Gases, the Theory of Natural Selection, or the chemical Theory of Mass Action, are essentially statistical arguments, and are liable to misinterpretation as soon as the statistical nature of the argument is lost sight of. In Wave Mechanics this is now clearly recognised. Statistical methods are essential to social studies, and it is principally by the aid of such methods that these studies may be raised to the rank of sciences" (Fisher, 1970, 14th edition)

One might think that the term "population" refers to the concept of "soldier of the British army." This is not the case, as the statistician's attention shifts from the soldier to the measurement of his height, with no reference to the individuals themselves. These heights form a sample of 10,000 numbers, extracted from the population.

Let us examine the consequences of the erasure of the data support from a logical standpoint. If I record the ages of a group of people, they would typically be written in the form of Table A in Table 1.1. A statistician interested in the distribution of age in this type of training center would rather present them in the form of Table B;

Table A

Who	**Age**
John	45
Simone	56
Ahmed	48
Joe	39
Carla	50

Table B

	X
X_1	45
X_2	56
X_3	48
X_4	39
X_5	50

TABLE 1.1: Two arrangements of a dataset concerning the ages of five individuals.

In Table B, the particular individuals are erased, highlighting the indifference of our encounter with them. Naturally, the notation is the vehicle for a key idea. Conceptually, what must be retained here is that the statistician focuses on *the value of the variable* (here the value of age), with no further perception of the source of this value. From a logical standpoint, the focus is on the image space of the "variable" function, while obliterating the origin of this function. We believe this perspective constitutes a historical bifurcation from the logic summarized in Chapter 1 of Kolmogorov (1933), which defines experimental data starting from the individuals themselves. We will return in detail to this bifurcation in subsection 3.

A second ideality is at work in our reception of data. The five ages of Table B could have been otherwise had we extracted another sequence from the population set. This potential for reproduction *ad infinitum* of data lies at the very basis of "statistical inference," enabling judgments about the quality of an estimator, the significance of a parameter, or the effect of one variable on another.

In the previous excerpt, Fisher's reference to the natural sciences introduces a confusion about these two idealities—the population and reproduction—which is worth examining, since it sheds light on modern discourse. Fisher mentions the "properties" of "large aggregates of individuals," taking care to indicate that these are not the properties of the individuals themselves. We may suppose, given his examples, that these refer as much to molecules as to human beings. A large aggregate of individuals standing before me, or a large number of molecules contained in a balloon, is not the same thing as the idea of the infinite population containing, for example, all possible sequences of velocities and directions taken by a molecule. These aggregates do *not* constitute the statistical population. They consist of a large number of *reproductions*, potentially infinite, drawn from one and the same population. The properties of their aggregation are governed by the Central Limit Theorem, which asserts, in its simplest version, that each value—the velocity of a molecule, the height of an individual—constitutes an independent and identically distributed reproduction from the population. More general versions of the Central Limit Theorem exist, accommodating various assumptions in the reproduction of data, but never escaping this essential element: data are an extract, among an infinity of other potential extracts, from the population.

This idea of the statistical property of aggregates is already present in earlier reasoning, beginning of course with the work of Quetelet (1848) leading to the concept of the social state, although, to our knowledge, Quetelet made no mention of the mechanism of data production from an infinite population, nor of the Central Limit Theorem. In statistical physics, after the first works of Boltzmann and Gibbs, a formal clarification was put forth by Khinchin in favor of probability theory, whose principal task, according to the author, is "the study of group phenomena [and] the discovery of [...] general laws as are implied by the gross character of the phenomena and depend comparatively little on the nature of the individual objects" (Khinchin, 1949, pp. 1-2).

The focus on aggregates and the example of soldiers show that the statistician's concern shifted from tangible observations to the ideality of the hypothetical infinite population. The aim is not to say something about these 10,000 soldiers or the 10^{23} molecules of gas in a balloon, but to say something about the parameter governing the process of data production. The *erasure of the data support* and the reproduction *ad infinitum* have the effect of forgetting the very context of data production and measurement. Statistics thus reaches a level of generality that offers a basis of comparison for *all* the world's data, since the support is no longer at stake, only the values are. It therefore allows reasoning across disciplines, which, from Fisher's time until today, sustains the metaphor of certain scientific currents wishing to render the social sciences as rigorous, in terms of mathematical analysis, as the natural sciences. Beyond this excerpt from Fisher, this reasoning is also found, for example, in the foundations of the econometric society, whose principal aim is to promote the study of economic problems that are "penetrated by constructive and rigorous thinking similar to that which has come to dominate in the natural sciences" (Frisch, 1933).

3 Harald Cramér (1946) : A Probability Space Aligned with Statistical Idealities

The success of Fisher's proposal owes much to the work of Harald Cramér who, in his now classic book, set out to bring together the work of the "brilliant schools of British and American statisticians, among whom the name of Professor R.A. Fisher should be mentioned in the foremost place" and that of French and Russian mathematicians who had developed the mathematical theory of probability. Let us first examine the essential conceptual shifts that Cramér contributed to probability in order to align it with Fisher's conceptions.

First, concerning the foundations of probability theory, Cramér noted that several opinions were represented in the literature (Laplace, von Mises, Kolmogorov, Keynes). Strangely, he considered himself within the school of Kolmogorov who, he wrote, "chooses the [...] starting-point as the frequency school, but avoids postulating the existence of definite limits of frequency ratios, and introduces the probability of an event simply as a number associated with that event" (Cramér, 1946, p. 151), in other words a frequency ν/n. While Kolmogorov (1933) is perfectly clear on this point, it is curious to observe that Cramér did not manage to hold to this conception without appealing to a limiting process. Further on, he introduced *continuous* distribution functions while continuing to call them "frequency curves." He admitted (our emphasis): "In statistical applications, variables of the continuous type occur when we are concerned with the measurement of quantities which, *within certain limit*, may assume any value. [...] When, for theoretical purposes, variables of this kind are considered as continuous, *a certain mathematical idealization* of actually observed facts is thus already implied" (p. 170).

This uneasy justification for moving toward the continuous is well known to anyone teaching classical descriptive statistics. When a sequence of 10 prices or 10 ages is shown to students, we have the greatest difficulty persuading them to forget discrete diagrams in order to accept that the variable is continuous. This inaugural classification in every descriptive statistics curriculum between discrete and continuous variables is a third ideality constructed by modern statistics. Logically speaking, this shift implies taking the limit of the frequency ν/n, which does not set Cramér's and statisticians' concept of probability at a great distance from that of von Mises.

In this regard, a conceptual shift occurred between Kolmogorov (1933) and Cramér (1946). For the former, if we are under the conditions, denoted $\mathfrak{S}$, of repetition of an experiment under identical circumstances, one considers the set E of elementary events, i.e., all observable outcomes of an experiment. A "random experiment" is a partition of the set of elementary events E. When Cramér established his probabilistic structures, he used the same symbol $\mathfrak{S}$ to denote "random experiments" (in the plural), observing that "in a sequence of random experiments, it is not possible to predict individual results" (Cramér, 1946, p. 141). This semantic shift in the use of the symbol $\mathfrak{S}$ is also a mathematical shift, since random experiments become the mechanism of data production, and it will later be necessary for Cramér to specify this mechanism (p. 149).

The technical definition of randomness was thus connected to the mechanism of data production. It accords with the three idealities introduced by Fisher. This shift underlies the current definition of the probability space $(\Omega, \mathcal{A}, P)$, where $\mathcal{A}$ is a σ-algebra on the universe Ω and P is a σ-additive measure on $\mathcal{A}$. A random variable X is then defined as a measurable function on the probability space. It induces a probability measure P_X equal to the composition $P \circ X^{-1}$. The reality of the initial set Ω is generally not discussed, all the statistician's effort being devoted to the specification or estimation of

P_X, which represents the mechanism of data production. The exclusive focus on P_X is the mathematical translation of the phenomenon of *erasure of the data support*.

The orientation of attention toward the mechanism of data production rather than the data themselves is explicitly assumed by statistical theory: "Often we are not interested in the individuals as such, but only in the values of the variable characteristic X and their distribution among the members" (Cramér, 1946, p. 324). To conceptualize this situation, Cramér invoked the image of the urn, which still illustrates all our standard textbooks in statistics and probability (italics in the original):

> we may replace the parent population by an urn containing one ticket for each member of the population, with the corresponding value of ξ inscribed on it. The [random] experiment $\mathcal{E}$ will then consist in drawing at random a ticket, noting the value inscribed, and replacing the ticket in the urn [...] *As a matter of illustration,* we may thus interpret any type of random experiment $\mathcal{E}$ as the random selection of an individual from an *infinite parent population*[...]. We then imagine an urn containing an infinite number of tickets, on each of which a certain number is written, in such a way that the distribution of these numbers is identical with the distribution of the random variable X associated with $\mathcal{E}$ (Cramér, 1946, p. 324)

With this image, the mechanism of data production itself is idealized. But the most fundamental element in this story is the focus now placed on the value taken by the random variable (the argument of P_X), with no further consideration for the domain of the probability function P. At this point, the remainder of the statistics curriculum consists in defining a priori and without any particular context probability measures P_X which then become the starting point of statistical analysis. These probability measures bear names such as Normal, Bernoulli, Binomial, Geometric, Chi-square, etc. Depending on the statistician's preconceptions, one of these laws will be designated to govern a priori the distribution of values in the hypothetical infinite population, each law having a certain plasticity represented by a parameter to be discovered. Thus, if a statistical statement invites you with "Let data $X_1, \ldots, X_n$ be drawn from a Normal law $N(\mu,\sigma^2)$," you should expect to perform all sorts of calculations to calibrate (estimate) the values of the mean μ and variance σ^2 to the *data* that the population has kindly revealed.

But if the starting point becomes the probability measure P_X, might it still be possible to uncover the fundamental set Ω from which it originates? The answer to this question would give tangible reality to Cramér's construction. Alas, as we shall see in the following paragraph, the path leads to an impasse.

Ironically, the core of the argument is to be found in... Kolmogorov (1933, Chapter 2, Section 3). To our knowledge, the only subsequent work to revisit this question is Chow and Teicher (1997, Section 6.3). We therefore start from a probability measure P_X. This generates a function $G(x) = P_X(]-\infty,x])$, which is a distribution function, i.e., a real function, non-decreasing, left-continuous, and possessing a right-hand limit everywhere (a so-called "cadlag" function). We ask from which probability space $(\Omega, \mathcal{A}, P)$ and which random variable $X : \Omega \to \mathbb{R}$ this measure originates. We then set, for P, the relation $P(]-\infty,x]) = G(x)$, where we have used the notation $P(]-\infty,x]) = P(\{\omega \in \Omega : X(\omega) \leq x\}) = P(X^{-1}]-\infty,x])$. We find that the solution consists in setting $\Omega = \mathbb{R}$ and taking $\mathcal{A}$ as the set of Borel subsets of $\mathbb{R}$. The probability P is σ-additive and defined here on the algebra of intervals. It must be extended to the Borel sets, which is achieved by Carathéodory's extension theorem. In so doing, we have found that the random variable X is none other than... the identity: $X(\omega) = \omega$! This trivial solution shows, from a logical standpoint, that the very support of the data — the individuals, the molecules — has been eliminated from the field of analysis.

4 Andreï Kolmogorov (1933): Probability and Experiment

4.1 Contrast between Kolmogorov and the Paradigm of Statistics

Most, if not all, textbooks on Probability and Statistics introduce the axioms of probability, with possible slight variations. For example, Karr (1993) begins by defining a *random experiment* as consisting of three elements:

1. Sample space: the set Ω of all (conceptually) possible outcomes.
2. Outcomes: elements ω of the sample space, also referred to as *sample points* or *realizations*.
3. Events: subsets of Ω for which probability is defined.

Thereafter, Karr (1993) defines several set operations between events, and then a σ-algebra on Ω is defined because "the family of sets for which probability is defined must be a σ-algebra" (p. 21): a σ-algebra on Ω is a family of subsets of Ω containing Ω and closed under countable unions and complementation. The pair $(\Omega,\mathcal{F})$, where $\mathcal{F}$ is a σ-algebra on Ω, is called a *measurable space*.

A *probability* on the measurable space $(\Omega,\mathcal{F})$ is a function $P : \mathcal{F} \longrightarrow [0,1]$ such that

1. $P(\Omega) = 1$.
2. Whenever $A_1,A_2,\ldots$ are (pairwise) disjoint sets in $\mathcal{F}$,
$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$
This last property is called *σ-additivity*.

The triplet $(\Omega,\mathcal{F},P)$ is called a *probability space*.

Both the properties characterizing a measurable space $(\Omega,\mathcal{F})$ and those characterizing the probability are typically viewed as the *axioms of Kolmogorov*. Once the axioms have been introduced, the following two standard exercises are proposed:

- **Exercise 1:** P is finitely additive: if $A_1,\ldots,A_n$ are (pairwise) disjoint, then
$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i).$$
As an illustration, Chow and Teicher (1997, p. 19) states that "Clearly, countable additivity implies finite additivity."
- **Exercise 2:** If $A_1 \supset A_2 \supset \cdots$ such that $A = \bigcap_{n=1}^{\infty} A_n$, then $P(A_n)$ converges to $P(A)$ as $n \to \infty$. This condition is called *σ-continuity*. In the case of Chow and Teicher (1997), the equivalence is stated in Theorem 2(iv) of Section 1.5.

Whenever this presentation of a probability space is attributed to Kolmogorov, what is assumed is that Kolmogorov's text contains, more or less, this very presentation. In other words, reading Kolmogorov (1933) is taken to be equivalent to reading what we have taught and/or learned in a graduate-level statistics course. However, the presentation in Kolmogorov (1933) is entirely different:

1. In Chapter 1, Kolmogorov (1933) introduces the axioms in the context of a *finite sample space* Ω:
   (a) Instead of working with a σ-algebra, he accordingly considers an algebra of subsets of Ω.
   (b) The axioms of a probability function are $P(\Omega) = 1$ and the finite additivity of P.
2. In Chapter 2, Kolmogorov (1933) considers the case of an *infinite sample space*. In this context, he introduces σ-continuity as an additional axiom.

Thus, what the current presentation treats as mere exercises, the original Kolmogorovian presentation makes a clear distinction between the axioms in the context of a finite sample space and those in the context of an infinite sample space. It could be suggested that, since this was the first time an axiomatic presentation was introduced, it was pedagogically natural to begin with what we might now label the "simple case." However, there is an argument contained in Section 1 of Chapter 1 of Kolmogorov (1933) that, to the best of our knowledge, none of these presentations have mentioned or discussed, namely that the axioms of the probability function are *consistent*.

This remark suggests that Kolmogorov's text should be read more attentively. As a result of such a reading, one would conclude that Kolmogorov closely follows Hilbert's formalism.

4.2 Why Kolmogorov wrote his book

Traditionally, it is stated that Kolmogorov's main motivation for writing his book was to establish the axioms of probability. Moreover, it is often stated that Kolmogorov used the measure-theoretic framework to definitively formulate these axioms. In a sense, this is correct, as Kolmogorov himself mentioned in the preface to his book:

> The purpose of this monograph is to give an axiomatic foundation for the theory of probability. The author set himself the task of putting in their natural place, among the general notions of modern mathematics, the basic concepts of probability theory — concepts which until recently were considered to be quite peculiar (Kolmogorov, 1956, p. v) (Most of the time, we quote the English translation of Kolmogorov (1933), but when we highlight a specific German word or expression, we refer to the original Kolmogorov (1933).).

Kolmogorov acknowledges that this task would have been hopeless without Lebesgue's theories of measure and integration. This was due to the analogies between the measure of a set and the probability of an event, as well as between the integral of a function and the mathematical expectation of a random variable. In spite of this, Kolmogorov's diagnosis was that "there was lacking a complete exposition of the whole system, free of extraneous complications."

However, this was not his main motivation, precisely because, for him, it was necessary to focus probability theory on problems different from those of measure theory. This is why he explicitly highlights problems that fall outside the scope of ideas familiar to measure theory:

> They are the following: Probability distributions in infinite-dimensional spaces (Chapter III, § 4); differentiation and integration of mathematical expectation

> with respect to a parameter (Chapter IV, § 5); and *especially the theory of conditional probabilities and conditional expectations (Chapter V).* It should be emphasized that these new problems arose, of necessity, from some perfectly concrete physical problems (p. v; the italics are ours).

It is clear that the main objective of Kolmogorov's book is related to conditional probability and conditional expectation. This theory is developed, as Kolmogorov himself points out, in Chapter 5. As we saw in the previous sections, Cramér and Neyman follow Kolmogorov, and, in particular, they define the conditional probability of an event given a second one, provided that the latter has positive probability. Why, then, emphasize a theory of this probabilistic object?

The answer can be found in a first reading of Chapter 5. In Section 1, Kolmogorov defines the conditional expectation of a measurable function using the Radon–Nikodym theorem, which allows him to ensure its existence. More precisely, let $(\Omega, \mathfrak{F}, P)$ be a probability space (with Ω of infinite cardinality). Let

$$L^1(\Omega, \mathfrak{F}, P) = \left\{ f : \Omega \longrightarrow \mathbb{R} : f \text{ is a measurable function and } \int_{\Omega} f \, dP < \infty \right\}.$$

Let $\mathfrak{B} \subset \mathfrak{F}$ be a sub-σ-field. By the Radon–Nikodym theorem, there exists a P-unique $\mathfrak{B}$-measurable integrable function $\widetilde{f}$ such that

$$\int_A f \, dP = \int_A \widetilde{f} \, dP_{\mathfrak{B}} \qquad A \in \mathfrak{B},$$

where $P_{\mathfrak{B}}$ is the trace of P on $\mathfrak{B}$. Kolmogorov defines the conditional expectation of f given $\mathfrak{B}$ as

$$E(f \mid \mathfrak{B}) \doteq \widetilde{f},$$

so that the conditional expectation is an operator

$$E(\cdot \mid \mathfrak{B}) : L^1(\Omega, \mathfrak{F}, P) \longrightarrow L^1(\Omega, \mathfrak{B}, P_{\mathfrak{B}})$$

such that, to each $f \in L^1(\Omega, \mathfrak{F}, P)$, there is associated $E(f \mid \mathfrak{B})$, defined through the representation

$$\int_A f \, dP = \int_A E(f \mid \mathfrak{B}) \, dP_{\mathfrak{B}} \qquad A \in \mathfrak{B}.$$

Once Kolmogorov had defined conditional expectation using the Radon–Nikodym theorem, he explained Borel's paradox. In what follows, we first explain the paradox; thereafter, we summarize Kolmogorov's explanation of it, which will allow us to return to Chapter 1 of his book in order to uncover a contribution of Kolmogorov that has not been recognized by the standard literature. By doing so, we will answer the question of how Kolmogorov conceived the idea of defining conditional expectation and conditional probability in this way.

4.3 Borel's Paradox

Let us begin by presenting the standard theory for computing conditional probability when both random variables are absolutely continuous. More precisely, let X and Y be a pair of random variables on a probability space $(\Omega,\mathfrak{F},P)$ having an absolutely continuous distribution function $F_{X,Y}$ with density $f_{X,Y}$, so that

$$P(X < x, Y < y) = F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,v) \, du \, dv.$$

If F_X and F_Y are the marginal distributions of X and Y, and f_X, f_Y are their respective densities, then

$$\begin{aligned}
F_X(x) &= F_{X,Y}(x,+\infty) = \int_{-\infty}^{x} f_X(u)\,du = \int_{-\infty}^{x} \left(\int_{-\infty}^{+\infty} f_{X,Y}(u,v)\,dv \right) du, \\
F_Y(y) &= F_{X,Y}(+\infty,y) = \int_{-\infty}^{y} f_Y(v)\,dv = \int_{-\infty}^{y} \left(\int_{-\infty}^{+\infty} f_{X,Y}(u,v)\,du \right) dv.
\end{aligned}$$

Applying the standard definition of conditional probability, we obtain:

$$\begin{aligned}
P(X \in A \mid Y \in B) &= \frac{P(X \in A \cap Y \in B)}{P(Y \in B)} \\
&= \frac{\displaystyle\int_A \int_B f_{X,Y}(u,v)\,du\,dv}{\displaystyle\int_B f_Y(v)\,dv}.
\end{aligned}$$

For $A = [a_1,a_2)$ and $B = [b,b+h)$ with $h > 0$, we get

$$P(a_1 \le X < a_2 \mid b \le Y < b+h) = \int_{a_1}^{a_2} \left(\frac{\displaystyle\int_b^{b+h} f_{X,Y}(u,v)\,dv}{\displaystyle\int_b^{b+h} f_Y(v)\,dv} \right) du. \tag{1.1}$$

Since both $P(a_1 \le X < a_2 \mid b \le Y < b+h)$ and $P(b \le Y < b+h)$ tend to zero as $h \to 0$, the left-hand side of (1.1) results in an indeterminate form. These two probabilities become $P(Y = b)$ and $P(a_1 \le X < a_2, Y = b)$, which are positive only if Y has a discrete part in its distribution at b. Since these probabilities are zero in the present case, we apply a form of L'Hôpital's rule to evaluate the right-hand side of (1.1). Thus,

$$\begin{aligned}
P(a_1 \le X < a_2 \mid Y = b) &= \lim_{h \to 0} P(a_1 \le X < a_2 \mid b \le Y < b+h) \\
&= \lim_{h \to 0} \int_{a_1}^{a_2} \left(\frac{\frac{1}{h}\int_b^{b+h} f_{X,Y}(u,v)\,dv}{\frac{1}{h}\int_b^{b+h} f_Y(v)\,dv} \right) du \\
&= \int_{a_1}^{a_2} \frac{f_{X,Y}(u,b)}{f_Y(b)}\,du.
\end{aligned}$$

Here, we have taken the limit inside the integral and used a form of the Lebesgue version of the fundamental theorem of integral calculus. Now, set

$$f_{X|Y}(u \mid b) = \begin{cases} f_{X,Y}(u,b)/f_Y(b), & \text{if } f_Y(b) > 0, \\ \alpha, & \text{if } f_Y(b) = 0, \end{cases}$$

where $\alpha \ge 0$ is an arbitrary number. Thus,

$$P(a_1 \le X < a_2 \mid Y = b) = \int_{a_1}^{a_2} f_{X|Y}(u \mid b)\,du.$$

Application to Borel's Paradox

Let us apply this standard definition to the sphere example developed by E. Borel (1909) and analyzed by Kolmogorov (1933). Suppose we randomly pick a point ω on the surface of the Earth, considered as a perfect sphere. Suppose that ω falls on an arc C comprising half of a great circle. Consider a subarc D occupying $\frac{1}{4}$ of C. What is the probability that ω falls on D, given that it falls on C?

We fix a system of spherical coordinates for the unit sphere:

$$x = \cos\theta\cos\phi; \quad y = \cos\theta\sin\phi, \quad z = \sin\theta,$$

where $\theta \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\phi \in [0,2\pi)$. Here, θ represents latitude and ϕ represents longitude. Thus, the equator corresponds to $\theta = 0$, the North Pole to $\theta = \frac{\pi}{2}$, and each meridian to some particular ϕ. Within this coordinate system, the surface area element has magnitude

$$\cos\theta \, d\theta \, d\phi.$$

If A contains all points such that $\theta_1 \leq \theta \leq \theta_2$ and $\phi_1 \leq \phi \leq \phi_2$, then the probability that a randomly chosen point falls inside A is given by

$$P(A) = \int_{\phi_1}^{\phi_2} \int_{\theta_1}^{\theta_2} \frac{\cos\theta}{4\pi} \, d\theta \, d\phi.$$

We now introduce the random variables Θ and Φ, where Θ measures latitude and Φ measures longitude. For example, $\Phi = \phi$ determines the meridian corresponding to longitude ϕ. The joint probability density function for (Θ,Φ) is

$$p(\theta,\phi) = \frac{\cos\theta}{4\pi}.$$

We compute the conditional probability density functions $p(\phi \mid \Theta = \theta)$ and $p(\theta \mid \Phi = \phi)$:

$$\begin{aligned}
p(\phi \mid \Theta = \theta) &= \frac{p(\theta,\phi)}{p_\Theta(\theta)} \\
&= \frac{\frac{\cos\theta}{4\pi}}{\int_0^{2\pi} \frac{\cos\theta}{4\pi} \, d\phi} \\
&= \frac{\frac{\cos\theta}{4\pi}}{\frac{\cos\theta}{2}} \\
&= \frac{1}{2\pi}, \\
p(\theta \mid \Phi = \phi) &= \frac{p(\theta,\phi)}{p_\Phi(\phi)} \\
&= \frac{\frac{\cos\theta}{4\pi}}{\int_{-\pi}^{\pi} \frac{\cos\theta}{4\pi} \, d\theta} \\
&= \frac{\frac{\cos\theta}{4\pi}}{\frac{1}{2\pi}} \\
&= \frac{\cos\theta}{2}.
\end{aligned}$$

The first probability density function corresponds to the arc-length solution, meaning that we may choose coordinates so that C comprises half of the equator. Since we picked ω randomly, it is equally likely to fall anywhere on the equator; thus, the conditional probability that ω falls on some subarc is proportional to the subarc's length; see Figure 1.1.

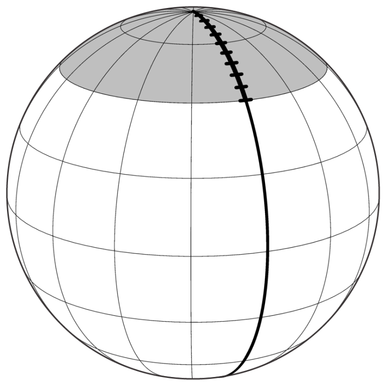


FIG. 1.1: Arc-length solution

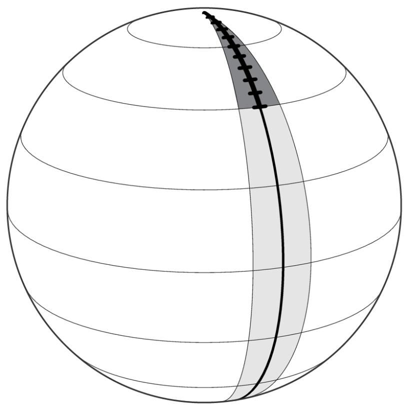


FIG. 1.2: Area solution

The second probability density function corresponds to the surface-area solution, namely that we may choose coordinates so that C is a meridian and the North Pole is the endpoint shared by D and C.

4.4 Kolmogorovian Concept of Conditional Probability

After briefly exhibiting Borel's paradox, Kolmogorov provides the following explanation:

> This shows that the concept of a conditional probability with regard to an isolated given hypothesis whose probability equals 0 is inadmissible. For we can obtain a probability distribution for Θ on the meridian circle only if we regard this circle as an element of the decomposition of the entire spherical surface into meridian circles with the given poles (Kolmogorov, 1956, p. 45).

The origin of the paradox lies not only in conditioning on a zero-probability event but also in the very conception of such a notion by conditioning on a single isolated event. In fact, Kolmogorov emphasizes that it is possible to find the probability distribution for Θ on the meridian circle "only if we regard this circle as an element of the decomposition of the entire spherical surface into meridian circles with the given poles."

To grasp the depth of this statement, let us reread Chapter 1 of Kolmogorov's text, in which he develops the axioms in the case where what is usually called the sample space is finite.

Let us begin with some considerations as to why Kolmogorov chose to present the axioms in the finite case. First, Kolmogorov is aware of the necessity of developing probability theory on an axiomatic basis. This means that "after we have defined the elements to be studied and their basic relations, and have stated the axioms by which these relations are to be governed, all further exposition must be based exclusively on these axioms, independent of the *usual concrete meaning of these elements and their relations* (*und keine Rücksicht auf die jeweilige konkrete Bedeutung dieser Gegenstände und Beziehungen nehmen darf*)" (Kolmogorov, 1956, p. 1; italics are ours). It should be mentioned that the English translation conveys the German meaning, but there are some inaccuracies:

1. *"independent of"* suggests mere separation, whereas the German phrase *"keine Rücksicht auf"* implies an active exclusion of meaning.
2. *"usual concrete meaning"* does not fully capture *"jeweilige konkrete Bedeutung"*, which refers to the specific meaning in each case rather than something usual.
3. *"elements"* is a looser translation of *"Gegenstände"*, which in mathematical contexts is better rendered as *"objects"*.

A precise English translation would be:

> *and must not take into account the respective concrete meaning of these objects and their relations.*

The above requirement fits with Hilbert's *formalism*, as Kolmogorov himself stated in 1929 (we refer to the translation published in Kolmogorov (2006)):

> There are two theories promising to solve all the problems that trouble mathematicians. However, both of them would do so at a high price.
>
> *Formalism*, headed by Hilbert, is supposed to do so by transforming mathematics into a mere symbol game in which everything would be possible under the sole condition of proving that there are no contradictions in the game. Brouwer's *intuitionism*, on the contrary, intends to expel from mathematics everything that has no basis in intuition common to everybody (Kolmogorov, 2006, p. 380).
>
> [. . .]
>
> The most well-known description of the novel view on the structure of mathematical theory is that given in Hilbert's *Foundations of Geometry* at the end of the last and the beginning of this century. Hilbert states that geometry is concerned with *a system of things, conventionally called "points", "straight lines", "planes", connected with each other by relations of some undefined nature, which could be described in such a way as "a line passes through a point", etc. The nature of these objects and their relations by no means defines the substance of geometry.* Only the fact that these relations satisfy certain axioms (such as "a unique straight line passes through a pair of given points") is relevant for the development of geometry. Every system of objects and their relations that satisfies these 22 axioms has the same right to be called "space" from Hilbert's point of view (Kolmogorov, 2006, p. 380; italics are ours).

As can be seen, Kolmogorov has a clear understanding of the philosophy underlying formalism, as presented in *Foundations of Geometry.* This is precisely the philosophy he advances in 1933:

> In accordance with the above [namely, the formalist point of view], in § 1 the concept of a *field of probabilities* is defined as a system of sets that satisfies certain conditions. What the elements of this set represent is of no importance in the purely mathematical development of the theory of probability (cf. the introduction of basic geometric concepts in *Foundations of Geometry* by Hilbert, or the definitions of groups, rings, and fields in abstract algebra) (Kolmogorov, 1956, p. 1).

We emphasize his reference to Hilbert's *Foundations of Geometry.* Moreover, the other examples (groups, rings, etc.) are discussed from a formalist perspective by Kolmogorov in 1929; see Kolmogorov (2006, Section II).

It is relevant to mention that the axiomatization of probability theory was one of the 23 open problems Hilbert proposed at the International Congress of Mathematicians in 1900 in Paris:

> **6. Mathematical treatment of the axioms of physics.** The investigations on the foundations of geometry suggest the problem: *To treat in the same manner, by means of axioms, those physical sciences in which mathematics plays an important part; in the first rank are the theory of probabilities and mechanics.*
>
> As to the axioms of the theory of probabilities, it seems to me desirable that their logical investigation should be accompanied by a rigorous and satisfactory development of the method of mean values in mathematical physics, and in particular in the kinetic theory of gases (Hilbert, 1902, p. 449; we refer to the English translation; in the original German text); Hilbert (1900), the 6th problem is stated on p. 272).

We know that Kolmogorov read Hilbert (1900) not only from these references but also because he mentions it in a paper published in 1954 (although extemporaneous):

> In his famous lecture delivered at the Congress of 1900, Hilbert said that the unity of mathematics, the impossibility of breaking it up into independent branches, arises from the very essence of our science. The most convincing corroboration of the correctness of this idea is the appearance of new junction points at each new stage of mathematical development, where ideas and methods from the most diverse mathematical disciplines prove to be indispensable and enter into new linkages to solve totally new problems (?, ?, pp. 1–2).

It should also be mentioned that, after introducing the axioms in the finite case, Kolmogorov shows that they are consistent, as prescribed by the formalist philosophy advanced by Hilbert; below, we will discuss this point.

At first approach, it might appear that Kolmogorov (1933) is merely a formalist solution to the sixth problem stated by Hilbert, together with its "pure mathematical development." However, there are two details that show that Kolmogorov is not a fully committed supporter of formalism. The first detail is related to the expression, corrected above, according to which the further exposition of probability theory "must not take into account the respective concrete meaning of these objects (*Gegenstände*) and their relations." The emphasis on *objects* is not a naive issue but a fundamental one, related to the possibility of using mathematics in empirical contexts:

> With some approximation, we may formulate the recent opinions as follows. Hilbert proposed to keep only the formal part of mathematics, by his consistency hypothesis setting us free from the necessity to construct. On the contrary, Brouwer values mainly the constructive aspect, though it cannot give us the ultimate existence of infinite collections, which is needed for a free use of ways of reasoning that have become common in mathematics, and therefore he demands thorough reconsideration of the ways of mathematical proof.
>
> The emergence of these extreme points of view could be explained by the fact that the combination of the two aspects of set-theoretic mathematics caused great difficulties and even contradictions. The common source of these difficulties is the following. Mathematicians have been used to treating numbers, functions, and sets as if they were objects of the real world, similar to material objects.
>
> The very preference for the word *thing* (Ding) over the word *object* (Gegenstand) is rather typical in this sense. Hilbert, in his *Foundations of Geometry*, speaks about the system of "things," as do the majority of mathematicians. Meanwhile, this point of view in set theory led to a number of contradictions (Kolmogorov, 2006, pp. 382–383).

Kolmogorov illustrates this statement with an explanation of Russell's paradox: "Let us assume that all logical classes exist like telegraph posts, to which wires from all the things included in a class are attached. If a class is an element of itself, a post should play a double role, that of a class element and that of a post representing a class. The wire that is attached to it as an element of a class should return to it as a post representing the entire class of elements."

Using this "real-life example," Kolmogorov explains Russell's paradox, concluding that "all [of the explanations of this paradox] are reduced to the restriction that we should not consider the collection of all classes as a finished collection. In other words, *they are reduced to the denial of the legality of our analogy with real things*" (Kolmogorov, 2006, p. 383; italics are ours).

Therefore, we may conclude that the formalist point of view is applicable only when mathematical objects are not interpreted with respect to the real world.

The second detail, which becomes clearer in light of the previous comment, appears in the introduction of Chapter 1 of Kolmogorov (1933):

> Every axiomatic (abstract) theory admits, as is well known, *an unlimited number of concrete interpretations besides those from which it was derived.* Thus, we find applications in fields of science that have no relation to the concepts of random event and probability in the precise meaning of these words (Kolmogorov, 1956, p. 1; italics are ours).

Scheme of the argument:

1. Kolmogorov focuses on applications. In particular, when he introduces the σ-continuity axiom in Chapter 2, he explicitly states that real stochastic processes are finite. Infinite probability fields are mere idealizations.

2. He emphasizes that the axioms originate in concrete experience. Formalization initially allows freedom from such concrete meaning; in a second step, unlimited hypotheses enable the use of (finite) probability theory in various scientific fields without necessarily providing meanings to random events and probabilities. This undermines Cramér's and Koopmans' attempts.

We now turn to a first reading of Chapter 1 of Kolmogorov (1933).

Definition of Conditional Probability: Let $(M,\mathcal{M},P)$ be a finite probability space. Consider $\mathcal{C} \doteq \{C_1,C_2,C_3\}$ as a measurable partition, meaning that $C_1,C_2,C_3 \in \mathcal{M}$. For $B \in \mathcal{M}$, we define the function

$$P(B \mid \mathcal{C}) \quad = \quad e_1 \mathbb{1}_{\{C_1\}} + e_2 \mathbb{1}_{\{C_2\}} + e_3 \mathbb{1}_{\{C_3\}},$$

where e_1,e_2,e_3 are distinct real numbers. Thus,

$$P(B \mid \mathcal{C})(\omega) = e_1 \mathbb{1}_{\{C_1\}}(\omega) + e_2 \mathbb{1}_{\{C_2\}}(\omega) + e_3 \mathbb{1}_{\{C_3\}}(\omega).$$

For example, if $\omega \in C_1$, then

$$P(B \mid \mathcal{C})(\omega) = e_1.$$

We recall that

$$\mathbb{1}_{\emptyset}(\omega) = 0.$$

If an event C is such that $\{\omega \in M : \omega \in C\} = \emptyset$, then $P(C) = 0$. Consequently, the number denoted by $P(B \mid C)$ is defined arbitrarily (as long as it is finite), since any number multiplied by 0 equals 0.

We can observe that $P(B \mid \mathcal{C})$ is a function, called a simple function, meaning that it is a random variable defined on $(M,\sigma(\mathcal{C}))$, where

$$\sigma(\mathcal{C}) = \{\emptyset,C_1,C_2,C_3,C_1 \cup C_2,C_1 \cup C_3,C_2 \cup C_3,M\}.$$

It follows that

$$\begin{aligned} E[\,P(B \mid \mathcal{C})\,] &= E\left[e_1 \mathbb{1}_{\{C_1\}} + e_2 \mathbb{1}_{\{C_2\}} + e_3 \mathbb{1}_{\{C_3\}}\right] \\ &= e_1\, E[\mathbb{1}_{\{C_1\}}] + e_2\, E[\mathbb{1}_{\{C_2\}}] + e_3\, E[\mathbb{1}_{\{C_3\}}] \\ &= e_1 P(C_1) + e_2 P(C_2) + e_3 P(C_3). \end{aligned}$$

Kolmogorov (1933) defines conditional probability as a random variable. The first step is to define a partition of M:

$$M = C_1 + \cdots + C_n.$$

The second step is to define a real function x such that, for each event C_q, it takes a constant value a_q. The sum

$$E(x) = \sum_q a_q P(C_q)$$

is called the mathematical expectation of x.

A random variable for which the event C_q assumes the value

$$P(B \mid C_q)$$

is called the *conditional probability of the event B given the experiment $\mathfrak{U}$, denoted $P(B \mid \mathfrak{U})$*.

Returning to

$$P(B \mid \mathcal{C}) = e_1 \mathbb{1}_{\{C_1\}} + e_2 \mathbb{1}_{\{C_2\}} + e_3 \mathbb{1}_{\{C_3\}},$$

where

$$e_i \doteq P(B \mid C_i), \qquad i = 1,2,3,$$

we then have

$$
\begin{aligned}
E[\,P(B \mid \mathcal{C})\,] &= P(B \mid C_1)\,E(\mathbb{1}_{C_1}) + P(B \mid C_2)\,E(\mathbb{1}_{C_2}) + P(B \mid C_3)\,E(\mathbb{1}_{C_3}) \\
&= P(B \mid C_1)P(C_1) + P(B \mid C_2)P(C_2) + P(B \mid C_3)P(C_3) \\
&= P(B \cap C_1) + P(B \cap C_2) + P(B \cap C_3) \\
&= P(B) \\
&= E(\mathbb{1}_B).
\end{aligned}
$$

That is,

$$
\begin{aligned}
E[\,P(B \mid \mathcal{C})\,] &= E\,[E(\mathbb{1}_B \mid \mathcal{C})] \\
&= P(B \mid C_1)P(C_1) + P(B \mid C_2)P(C_2) + P(B \mid C_3)P(C_3) \\
&= E(\mathbb{1}_B).
\end{aligned}
$$

The theorem of total probability is equivalent to the Kolmogorovian definition of conditional probability, which in turn is the same as marginalizing a joint probability, and is also equivalent to the property of iterated expectation.

Thus, we have

$$E\,[E(\mathbb{1}_B \mid \mathcal{C})] = E(\mathbb{1}_B),$$

where

$$E(\mathbb{1}_B \mid \mathcal{C}) = P(B \mid C_1)\mathbb{1}_{\{C_1\}} + P(B \mid C_2)\mathbb{1}_{\{C_2\}} + P(B \mid C_3)\mathbb{1}_{\{C_3\}}.$$

The conditional expectation is a function of the partition $\mathcal{C}$, which is a partition of M. It is said that $E(\mathbb{1}_B \mid \mathcal{C})$ is measurable with respect to $\mathcal{C}$, meaning that it can be measured since it belongs to $\mathcal{M}$.

4.5 Why didn't Cramér follow Kolmogorov?

Because they define conditional probability with respect to an isolated event of positive probability. However, conditional probability can be defined with respect to a partition, and even if one member of the partition has probability equal to 0, it is still possible to define it correctly.

Moreover, it can be observed that the theorem of total probability corresponds to the Radon–Nikodym theorem in the finite case. This was Kolmogorov's inspiration for using such a theorem to ensure the *existence* of conditional expectation.

5 Tjalling Koopmans (1937) and Trygve Haavelmo (1944): Replicability in economics

To the work of Harald Cramér, which provided a rigorous mathematical structure to Fisher's ideas, was added the influential contribution of Jerzy Neyman (1937), who developed a theory of hypothesis testing aimed at posing a series of questions about the parameters of the population P_X^θ. The notions of infinite population, reproducibility of the sample, and tests found their natural examples in agriculture and biology. However, when it came to studying social phenomena, particularly economic ones, the reception of this theory was not immediate, since it was difficult to conceive that macroeconomic observations could be random realizations capable of taking other values under a different draw.

The appropriation of statistical ideas in economics occupies an important place in the early work of Tjalling Koopmans and, of course, of Trygve Haavelmo, who received the Nobel Prize in Economics in 1989 "for his clarification of the probability theory foundations of econometrics and his analyses of simultaneous economic structures." The challenge was to establish a link between economic theory and statistical data, which was at the time a delicate problem, since statistically observed correlations may be spurious and do not necessarily reflect a structural link between economic variables. Furthermore, as Haavelmo noted, the notions of probability and random variable seemed ill-suited to the needs of economists:

> Probability schemes, it is held, apply only to such phenomena as lottery drawings or, at best, to those series of observations where each observation may be considered as an independent drawing from one and the same "population." From this point of view it has been argued, e.g., that most economic time series do not conform well to any probability model, "because the successive observations are not independent." (Haavelmo, 1944, Preface)

It was therefore necessary to find a way to align statistical idealities with the conception of an economic datum that contains a certain persistence, as is the case, for example, with the observation of the price of coal or the individual consumption of a good over time.

The idea then arose that an economic datum has two components, one statistical in nature and the other economic. Simple statistical observations or calculations on the data have no meaning for the economist. But "the model attains economic meaning only after a corresponding system of quantities or objects in real economic life has been chosen or described, in order to be *identified with those* in the model" (Haavelmo, 1944, p. 8). To describe the economic component of the datum, a distinction was drawn between "true variables" and "observed variables." Alongside the statistician's hypothetical world of the population lies another hypothetical world belonging to the economist. It is in this second world that the "true variables" reside—such as, for example, the ideal price of coal or the ideal demand for a good. Ideal variables are denoted with an asterisk, y^*, and in the economist's ideal world the "true variable" is governed by constant laws, which Haavelmo writes as $y^* = f(x_1, \ldots, x_n)$. In this notation, the economist believes that the quantities x_i influence the ideal variable y^* through an economic relation, f, taking the form of a function suggested by economic theory. Of course, this clarification on the foundations of the economic datum does not come without a certain complicity from the universe: "Nature has a way of selecting jointly value-systems of the "true" variables such that these systems are as if the selection had been made by the rule defining our theoretical model." (Haavelmo, 1944, p. 9).

In the ideal world of economic theory, the relations are functional, meaning that to a set of quantities $x_1, \ldots, x_n$ is associated one and only one value of y^*. But in the tangible world of statistical observations, for that same set of variables $x_1, \ldots, x_n$, one may encounter *different* values of y (without an asterisk, since this pertains to the world of observed and no longer idealized data). This situation evokes measurement error: "the discrepencies $y - y^*$ for each individual may depend upon a great variety of factors, these factors may be different from one individual to another, and they may vary with time for each individual" (Haavelmo, 1944, p. 50). The idea had already germinated in the dissertation of Koopmans (1937), which made lengthy references to Fisher (1922). Analyzing macroeconomic data, Koopmans viewed data as the sum of a perfect economic model and an accidental error. The analogy with Gauss's propositions on natural phenomena and measurement errors had been raised in the founding issue of the leading journal of the

Econometric Society (Frisch, 1934). For Koopmans, "the point is that, mathematically, we can advance even if we assume a probability distribution only for the accidental errors in the variables, which prevent the regression equation from being exactly satisfied by the observations [...] each of the variables may be conceived as the sum of two components, a "*systematic component*" or "*true value*" and an "*erratic component*" or "*disturbance*" or "*accidental error*"" (Koopmans, 1937, p. 5, italics in the original). Two hypothetical worlds thus couple in the datum: a systematic component saves economic theory, while an erratic component stems from Fisher's hypothetical infinite population (Koopmans, 1937, p. 5). The parallel with observations and measurement errors in physics naturally reassured economists in their aim of reaching reasoning akin to that at work in the natural sciences. Our reception of the phenomenon is imperfect, tainted by an accidental error whose economic interpretation would later take many forms (economic shock, omitted variables, idiosyncratic component, ...).

The most striking element in this approach is the unquestioned confidence in the existence of the systematic component, although no logical foundation was advanced to grasp its empirical nature. In a historical context where models drawn from economic theory had not shown great predictive power, one may hypothesize that the additive structure was a compromise to account for the inadequacy of facts with models, while preserving the theoretical models, henceforth located in the hypothetical world of "true values."

The repeatability of observations, essential to Fisher's and Neyman's statistical program, nevertheless remained a difficulty in articulating the world of the statistical population with that of economic theory. How can the repeatability of longitudinal data such as the price of coal or the measurement of inflation over time be conceived? To carry out the third stage of the statistician's task according to Fisher, and to implement Neyman's hypothesis testing, it is necessary to compare data with *all possible reproductions* of the sample. This step allows one to determine the sampling distribution, which is necessary for the statistical treatment of the phenomenon under study. But how can one conceive of all possible reproductions of inflation? Koopmans noted (italics in the original):

> This conceptual scheme of the structure of the variables being accepted, it is clear that the expression "*repeated sampling*" requires an interpretation somewhat different from that prevailing in applications of sampling theory in the agricultural and biological field. In the latter domains in many cases repeated samples may in principle be obtained in any number. The distribution of a statistic "in repeated samples" is therefore something which, if only sufficiently extensive efforts are applied, may be found or controlled by experience. But in the conditions under which sampling theory is used here such a distribution is much more hypothetical in nature. The observations [...] constituting one sample, a repeated sample consists of a set of values which the variables would have assumed if in these years the systematic components had been the same and the erratic components had been other independent random drawings from the distribution they are supposed to have. (p. 7)

It was he who showed the way: "The observations [...] constituting one sample, a repeated sample consists of a set of values which the variables would have assumed if in these years the systematic components had been the same and the erratic components had been other independent random drawings from the distribution they are supposed to have." (p. 7) Replicability would therefore apply to the accidental error of the additive model. In Koopmans's work, this was a time series that could have followed a different trajectory if the sampling mechanism had been activated *on the same dates*.

6 From Observation to Data: Rényi, Lindley, and Psychometrics

In the late 1960s, psychometric theory underwent a profound shift in its probabilistic foundations. In their influential monograph, Lord and Novick abandoned the traditional foundations of psychological statistics and test theory based on von Mises' frequentist approach, and instead adopted the explication of measure-theoretic probability developed by Rényi and presented in a particularly clear and accessible form by Lindley (F. M. Lord & Novick, 1968; Novick, 1969; Rényi, 1962; Lindley, 1965).

A central contribution of Rényi's work lies in his interpretation of probability theory as a theory of experiments. Within this framework, a measurable space $(\Omega,\mathcal{A})$ is understood as representing an experiment (*Versuch*), where the nonempty set Ω denotes the set of all possible outcomes of the experiment, referred to as the *basic space*, and each element $\omega \in \Omega$ corresponds to a specific outcome (Rényi, 1970). Observations become data only insofar as they are situated within such a formally specified space of possible outcomes.

Lindley developed a closely related interpretation in his exposition of probability and statistics, describing a sample space as a collection of elementary events and defining events as sets of such elements (Lindley, 1965). The occurrence of an event is defined in terms of the occurrence of one of its constituent elementary events, thereby providing a precise account of how empirical observations are linked to abstract probabilistic structures.

One of the most significant aspects of Rényi's axiomatic system, as emphasized by Lindley, is the formulation of probability in terms of conditional probabilities rather than absolute probabilities. In contrast to Kolmogorov's approach—in which conditional probabilities are defined from unconditional ones—Rényi's system takes conditional probability as primitive (Lindley, 1965). This relational conception of probability proved especially relevant for psychometrics, where measurement models are inherently conditional on latent variables, experimental conditions, and background information.

By adopting this framework, Lord and Novick were able to reconceptualize psychological measurement in a way that aligned more closely with the inferential structure of test theory. The influence of Rényi and Lindley thus extends beyond technical formalism, contributing to a deeper understanding of how observations are structured and transformed into data within psychometric theory.

7 Lazarsfeld (1959): Replicability in humanities

Since these early intuitions, the idea has taken hold in all the social sciences. Thus, when defining the stochastic process $X_t(\omega)$ where $t \in \mathbb{N}$ and ω is an element of a set of possible reproductions Ω, Davidson (1994) made the following clarification (italics in the original):

> The entire infinite sequence corresponds to a *single* outcome ω of the underlying abstract probability space. In principle, a sampling exercise in this framework is the random drawing of a point in $\mathbb{R}^\infty$, called a realization or *sample path* of the random sequence; we may actually observe only a finite segment of this sequence, but the key idea is that a random experiment consists of drawing a complete realization. *Repeated sampling* means observing the same finite segment (relative to the origin of the index set) of different realizations, *not* different segments of the same realization. (p. 179)

This program is consistent with the indications of Fisher (1922): data are an extract taken from infinite sequences contained in the hypothetical population. But in the case of stochastic processes, the dependence between data imposes a set of more technical mathematical considerations in order to link observed data to the generating mechanism in the population (ergodic theorem and law of large numbers, in particular). Our second example comes from the statistical foundations of sociology and psychology. In a context of surveys or psychometric questions, to explain the principle of data reproduction *ad infinitum*, the following explanation by Lazarsfeld (1959) became famous:

> Suppose we ask an individual, Mr. Brown, repeatedly whether he is in favor of the United Nations; suppose further that after each question we “wash his brains” and ask him the same question again. Because Mr. Brown is not certain as to how he feels about the United Nations, he will sometimes give a favorable and sometimes an unfavorable answer. Having gone through this procedure many times, we then compute the proportion of times Mr. Brown was in favor of the United Nations. This we could also call the probability of Mr. Brown’s being in favor of the United Nations. (p. 493)

We have encountered many scientific circles perfectly familiar with this explanation, reproduced *in extenso* by F. Lord and Novick (1968, p. 29), a reference work in classical test theory, factor analysis, and IRT models. One cannot help but feel a certain fascination for the intellectual efforts deployed to adjust our world to the idealities of statistics. Despite the uniqueness of the encounter with any given datum, the reproducibility of the sample drawn from an imaginary population has thus become the reference framework by which every science may bring reality into its fold.

Part I

GEOMETRY OF DATA

Chapter 2

Database

1 Collecting Data

This course focuses on the processing of information that comes in the form of observed and collected data. Data today lie at the very heart of human activity. They are recorded and analyzed to address very diverse objectives of daily life. One might think of measurements of the presence of materials in air or water in the context of public health, the observation of solar electricity production in the framework of decarbonizing the economy, or the recording of sick leave in a company. In reality, we are all already familiar with the notion of data and aware of the issues of data management in the public sphere: data protection, fact checking, etc. Although data collection goes back to the earliest forms of human activity (crop counting, ...), its systematic use expanded in the wake of the Industrial Revolution and especially in the eighteenth century. Today, the collection and use of data are so massive that new names have been coined for their processing: data science, data mining, artificial intelligence, etc.

Seeking knowledge from experiments and sensible observations (through our senses) is an intellectual endeavor known as the *empirical method*. The word comes from the Greek ἐμπειρία, meaning "one who is guided by experience," and it is within this epistemology that this course is situated. By contrast, rationalism is another path to knowledge, privileging logic and deductive reasoning. This opposition runs throughout the history of Western thought since Antiquity.

When we say that we *collect* a datum, let us clarify that it is not necessarily a numerical datum. One might, for example, ask about the color of your coat, in which case the datum takes as values color categories: sky blue, vermilion red, lime green, etc. It is clear from this example that it will be important to define the values taken by the data. A pollster might limit themself to a few categories (blue, red, ...) while another might be more precise (sky blue, vermilion red, etc.). A physicist might also report the color by measuring frequency, in hertz. All these situations constitute observable data. Categories may also be more complex. If one asks "What did you think of the last James Bond film?", the answer is a relatively complex string of characters. It is the measurement of your feeling about the last James Bond movie—also an observed datum. The values taken by the data form an important set that we will describe in the following section.

The examples we have just cited are cases of direct measurement. These are carried out through an instrument (for example a frequency spectrometer), our senses (color), or a system that directly records what our senses perceive (hearing your opinion of a film and writing it down). But there are many everyday situations in which an indirect result is reported. Consider, for example, the unemployment rate, the carbon footprint of our university, or an intelligence quotient. In these latter examples, no direct measurement is available, and the result is produced after a series of calculations carried out from raw data. What these examples have in common is that the collected value is not directly observable by our senses (one does not observe the unemployment rate of society, nor the IQ of the human body, just as one does not place millions of sensors in the university to measure CO_2 emissions).

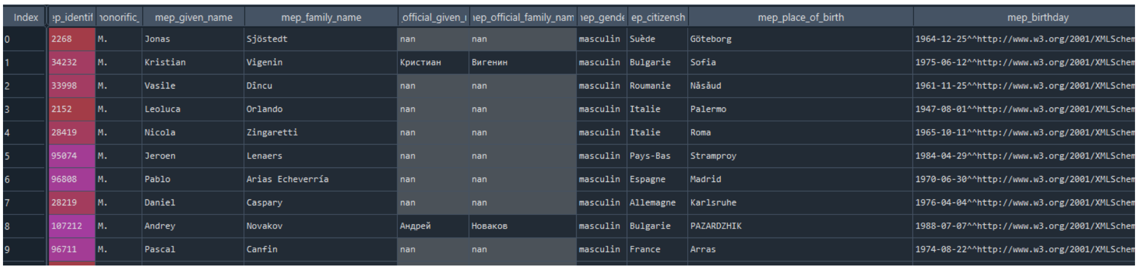
It is important to keep in mind this distinction between observable data, also called *raw data*, and the result of operations carried out from such data. Wherever possible, this course will deal with raw data, in order eventually to construct indirect results intended to quantify a concept that we will have previously defined (such as the concept of "unemployment" or of "intelligence quotient"). In principle, such a concept should be understood and perceived in the same way by everyone, though one may doubt whether this condition is met in most practical situations...

The recording of raw data is not without raising questions that influence the analysis one makes of them. Input errors, missing responses, measurement device errors, the existence of detection thresholds, lying, deliberate fraud... there are many reasons why a transcribed datum may fail to reflect the phenomenon it is meant to measure. In this course, we will see how it is possible, in certain situations, to identify or to take into account uncertainty in measurement.

2 The Constituents of a Database: Labels and Variables

Our work begins when a database is entrusted to us. As an example, we import from the website of the European Parliament[1] the database of Members of the European Parliament (MEPs) of the tenth legislature, which began in July 2024. We use Python in this course; an introduction to this language can be found in the appendices of this document. Figure 2.1 shows the first rows and columns of this database, and Table 2.1 summarizes all the variables contained in it. Presentation in tabular form is the simplest structure of a database. This structure is composed of two constituents:

1. **Labels**, usually arranged in rows. They identify the **statistical unit** of the database, i.e., the entity being measured. Labels are the identifiers of this unit. For example, in the case of Figure 2.1, the statistical unit is a specific Member of Parliament. It is convenient to identify this unit with a unique code; hence the database includes the column *identifier*, which is a unique identification number of the MEP.
2. **Variables**, usually arranged in columns. The name of the variable appears in the first row of the table, and the subsequent rows contain the values of the variable for the different statistical units of the database. For example, the MEP database includes the variable *family_name*. This column records, in turn, the family name of each statistical unit (that is, each MEP).



Index	ːp_identif	ɿonorific_	mep_given_name	mep_family_name	official_given_ɿ	ɿep_official_family_nam	ɿep_gende	ep_citizensh	mep_place_of_birth	mep_birthday
0	2268	M.	Jonas	Sjöstedt	nan	nan	masculin	Suède	Göteborg	1964-12-25^^http://www.w3.org/2001/XMLSchem
1	34232	M.	Kristian	Vigenin	Кристиан	Вигенин	masculin	Bulgarie	Sofia	1975-06-12^^http://www.w3.org/2001/XMLSchem
2	33998	M.	Vasile	Dîncu	nan	nan	masculin	Roumanie	Năsăud	1961-11-25^^http://www.w3.org/2001/XMLSchem
3	2152	M.	Leoluca	Orlando	nan	nan	masculin	Italie	Palermo	1947-08-01^^http://www.w3.org/2001/XMLSchem
4	28419	M.	Nicola	Zingaretti	nan	nan	masculin	Italie	Roma	1965-10-11^^http://www.w3.org/2001/XMLSchem
5	95074	M.	Jeroen	Lenaers	nan	nan	masculin	Pays-Bas	Stramproy	1984-04-29^^http://www.w3.org/2001/XMLSchem
6	96808	M.	Pablo	Arias Echeverría	nan	nan	masculin	Espagne	Madrid	1970-06-30^^http://www.w3.org/2001/XMLSchem
7	28219	M.	Daniel	Caspary	nan	nan	masculin	Allemagne	Karlsruhe	1976-04-04^^http://www.w3.org/2001/XMLSchem
8	107212	M.	Andrey	Novakov	Андрей	Новаков	masculin	Bulgarie	PAZARDZHIK	1988-07-07^^http://www.w3.org/2001/XMLSchem
9	96711	M.	Pascal	Canfin	nan	nan	masculin	France	Arras	1974-08-22^^http://www.w3.org/2001/XMLSchem

FIG. 2.1: First rows and columns of the MEP database (as displayed in Python in the *Variable Explorer* panel of *Spyder*).

A simple **database** is thus *the measurement of different variables on a set of labels.* In this course, we denote by Ω_n ("omega n") the set of labels. The index n indicates the

[1] https://data.europarl.europa.eu

Name of the variable (X)	**Meaning**	**Image ($\Im(X)$)**
mep_identifier	Identification number	
mep_honorific_prefix	Abbreviated title	
mep_given_name	First name	
mep_family_name	Last name	
mep_official_given_name	Official first name	
mep_official_family_name	Official last name	
mep_gender	Gender	{Male, Female, Unknown}
mep_citizenship	Nationality	
mep_place_of_birth	Place of birth	
mep_birthday	Date of birth	
mep_death_date	Date of death	
mep_image	Photograph	
mep_email	Email	
mep_homepage	Webpage	
mep_twitter	Twitter (X) page	
mep_facebook_page	Facebook page	
mep_facebook_profile	Facebook profile	
mep_instagram	Instagram page	
mep_URI	EU Parliament page	

TABLE 2.1: Variables of the MEP database of the tenth legislature

cardinality (the number of elements) of Ω_n. For example, in the European database there are $n = 719$ rows[2] (MEPs), and the label space is $\Omega_X = \{2268, 34232, 33998, \ldots\}$.

Variables are mathematical functions that assign a value to each label. For example, the variable *gender* assigns to each label, i.e., to each MEP, a measure of gender. The **image of a variable** X is the set $\Im(X)$ of possible values taken by this function. In the case of the gender of MEPs, the image of the variable is:

$$\Im(gender) = \{\text{Male}, \text{Female}, \text{Unknown}\}.$$

In general, a variable that we denote X is therefore a function

$$X : \Omega_n \longrightarrow \Im(X) : \omega \longrightarrow X(\omega) \ .$$

This function assigns to each element ω of Ω_n one and only one value taken from the image set $\Im(X)$.

To describe a variable, one must therefore begin by giving three pieces of information: its name, its meaning, and its image.

Exemple

In Table 2.1, the last column presents the Image set of the variables. You can complete it by exploring the MEP database (Exercise 2.1). In Python, if the database is named `EU`, use the code

```
EU['mep_gender'].unique()
```

[2] You are invited to find out why the number of MEPs is curiously 719 and not 720.

to explore, for example, the set of values taken by the variable *mep_gender*. Python's response to this code is

```
array(['male', 'female', 'unknown'], dtype=object)
```

The fact that variables are functions has an important consequence: they assign a *unique* value to each label of Ω_n. Here is a counterexample: if you ask a group of individuals "Which languages do you speak?", it is possible that the value of this question for a given individual is multiple: "French and Arabic." If the image of the variable consists of the languages spoken in the world, this answer is not unique since it contains two simultaneous values. The question posed does not constitute a variable, because a mathematical function cannot take two simultaneous values.

There are at least two ways to conceptualize this survey in order to construct valid variables. The first is to consider that "French and Arabic" constitutes a possible value of the variable, distinct from the value "French," from the value "Arabic," or from the value "Dutch and Arabic." This is a plausible vision for one who assigns a different weight to the use of multiple languages. Another approach is to ask individuals sequentially: "Do you speak French?" "Do you speak Arabic?" and thus construct two variables instead of one, each with the image $\{\text{yes}, \text{no}\}$. In this case, one constructs as many variables as there are languages spoken in Ω_n.

3 Partitions of Ω_n

As in the MEP database, let us construct a database consisting of labels Ω_n. Let X be a variable measured in the database on the set of labels. We explained that a variable is a function $X : \Omega_n \longrightarrow \Im(X)$. The image of X is a set of I distinct values denoted $\{a_1, \ldots, a_I\}$. Consider the *preimage* X^- defined by

$$\begin{aligned} X^- &: \Im(X) \longrightarrow \Omega_n \\ &: a_i \longrightarrow X^-(a_i) = \{\omega \in \Omega_n : X(\omega) = a_i\} \end{aligned}$$

that is, the set of labels for which the variable X takes the value a_i. This set is of course a subset of Ω_n. By considering all the values a_i in $\Im(X)$, we therefore construct different preimages (subsets) of Ω_n that are distinct and whose union is the entire label space. Such a construction is called[3] the **partition of Ω_n induced by the variable X**.

Exemple

Consider the database given in Table 2.2, which shows a set of variables for the space of ten people sharing student housing. As indicated in the caption, the label space is the set $\Omega_{10} = \{1,2,\ldots,10\}$, since the database contains $n = 10$ rows. Let us focus on the variable *Nationality*, whose image is here a set of four elements:

$$\Im(\textit{Nationality}) = \Big\{\text{Belgium}, \text{France}, \text{Japan}, \text{Spain}\Big\} .$$

Since the image contains four elements, the preimages of the variable necessarily partition the labels into four distinct blocks, each sharing the same nationality:

$$\begin{aligned} &\text{Block } 1 = \{1,5,10\} && \text{Block } 3 = \{3,8\} \\ &\text{Block } 2 = \{2,6,9\} && \text{Block } 4 = \{4,7\} . \end{aligned}$$

[3] Reminder: A **partition** of a set Ω_n is a collection of non-empty, disjoint subsets whose union is Ω_n.

ID	First name	Gender	Nationality	Height	Age	Group
1	Amina	Female	Belgium	165	19	BTS
2	Liam	Male	France	178	21	Coldplay
3	Yuna	Female	Japan	160	18	BLACKPINK
4	Carlos	Male	Spain	172	20	Rosalía
5	Fatima	Female	Belgium	168	22	Dua Lipa
6	Ahmed	Male	France	182	19	Imagine Dragons
7	Sofia	Female	Spain	158	20	Arctic Monkeys
8	Kenji	Not specified	Japan	170	18	One Direction
9	Claire	Female	France	174	21	Taylor Swift
10	Malik	Not specified	Belgium	180	22	The Weeknd

TABLE 2.2: Data on the residents of my student housing. *ID* is an index number used to define the label space Ω_{10}. *Height* is in centimeters. *Group* is the person's favorite music group.

The preimage of the value "Belgium" is Block 1 of labels, and so on. These blocks are necessarily disjoint, and their union is necessarily the label space Ω_{10}. We thus assimilate the variable *Nationality* to a partition of Ω_{10}. See also Exercise 2.2 for other observations on partitions of this database.

Each variable therefore induces a partition of the label space Ω_n. For this reason, we may even take this as a definition: a **variable** is a partition of the label space Ω_n. Every partition consists of subsets of Ω_n, also called **parts** of Ω_n. The set of parts of Ω_n constitutes a set that will serve as a reference throughout this course. This set is denoted $\mathcal{P}(\Omega)$.

Exemple

Suppose the label space is $\Omega_3 = \{\text{blue, green, yellow}\}$. The set $\mathcal{P}(\Omega_3)$ of parts of Ω_3 consists of eight elements:

$$\mathcal{P}(\Omega_3) = \Big\{\emptyset, \{\text{blue}\}, \{\text{green}\}, \{\text{yellow}\}, \{\text{blue, green}\}, \{\text{blue, yellow}\}, \{\text{green, yellow}\}, \Omega_3\Big\}$$

where $\emptyset$ is the empty set. The empty set always belongs to the set of parts. Note also that the initial set Ω_3 is likewise included in $\mathcal{P}(\Omega_3)$.

The set of parts $\mathcal{P}(\Omega_n)$ enjoys the following properties, which always hold:

- It always contains the empty set: $\emptyset \in \mathcal{P}(\Omega_n)$
- It always contains Ω_n: $\Omega_n \in \mathcal{P}(\Omega_n)$
- The complement of a part of Ω_n is a part of Ω_n:

$$A \in \Omega_n \Longrightarrow A^c = (\Omega_n \setminus A) \in \Omega_n$$

- The union of parts of Ω_n is a part of Ω_n:

$$\{A_j; j \in J\} \in \Omega_n \Longrightarrow \bigcup_{j \in J} A_j \in \Omega_n$$

Remarque 2.1. Be careful not to confuse the *preimage* of a function X, denoted X^{-}, with its *inverse*. The inverse, denoted X^{-1}, is the function such that $X^{-1}X(\omega) = \omega$; this function does not always exist. The preimage searches in the domain for the values of ω whose image under X is a given object.

4 The Relationship Between Variables

The goal of statistics and econometrics is to determine a relationship that may exist between several variables. In this section we define the two extreme situations we may encounter, namely two variables whose relationship is maximal (we will call such variables *equivalent*) and two variables with no relationship at all (we will call such variables *independent*).

Consider two variables X and Y defined on a common set of labels, Ω_n. Each variable induces a partition of the label set.

Exemple

Let us return to the example of the student housing and Table 2.2. We saw that the variable $X =$ *Nationality* induces a partition into four blocks:

$$\Omega_{10} = \{1{,}5{,}10\} \,\dot{\cup}\, \{2{,}6{,}9\} \,\dot{\cup}\, \{3{,}8\} \,\dot{\cup}\, \{4{,}7\}$$

where the disjoint union is symbolized by a dot above the union symbol. The variable $Y =$ *Age* induces another partition, into six blocks corresponding to the six age values observed in the database:

$$\Omega_{10} = \{3{,}8\} \,\dot{\cup}\, \{1{,}6\} \,\dot{\cup}\, \{4\} \,\dot{\cup}\, \{2{,}9\} \,\dot{\cup}\, \{7\} \,\dot{\cup}\, \{5{,}10\}$$

Every statistical proposition is stated with reference to the set of labels. In particular, the relationship between variables X and Y is expressed through the analysis of their induced partitions. The following definition characterizes the maximal relationship that can exist between these two variables.

Définition 2.1. *Two variables X and Y on Ω_n are* equivalent *if and only if they induce the same partition. In this case, we write $X \sim Y$, or equivalently,*

$$\{X^{-}(\{x\}); x \in \Im(X)\} = \{Y^{-}(\{y\}); y \in \Im(Y)\}.$$

Note that the image of the variables does not enter into this definition, as the following example shows.

Exemple

As an example, on Ω_5, consider the variable $X : \Omega_5 \to \{\blacklozenge, \bigstar\}$ defined by

$$X(\omega_1) = X(\omega_4) = X(\omega_5) = \blacklozenge \quad \text{and} \quad X(\omega_2) = X(\omega_3) = \bigstar$$

which is equivalent to the variable $Y : \Omega_5 \to \{0, 1\}$ defined by

$$Y(\omega_1) = Y(\omega_4) = Y(\omega_5) = 0 \quad \text{and} \quad Y(\omega_2) = Y(\omega_3) = 1.$$

In other words, the equivalence between two variables is verified on their induced partitions (here $\{\{\omega_1, \omega_4, \omega_5\}, \{\omega_2, \omega_3\}\}$).

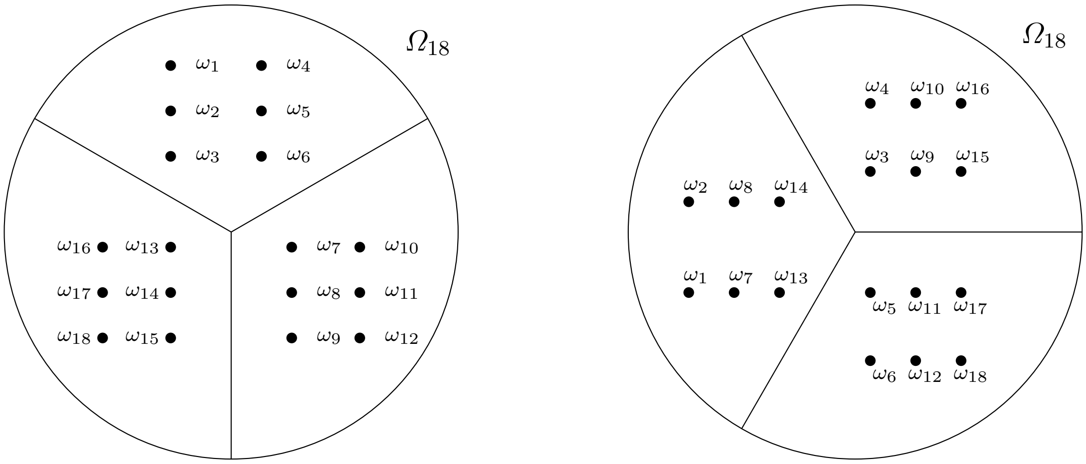


Fig. 2.2: The figure on the left shows a partition of the label set induced by one variable. The figure on the right shows another partition of the same elements induced by a second variable. One observes that the elements of each block of the left partition are uniformly distributed across the blocks of the right partition. This situation characterizes independence between the two variables.

From a statistical and econometric perspective, the equivalence between two variables means that their values are carried by the same labels. If, for example, we analyze a database of firms, and the variable "sector of activity" (industry, services, ...) is equivalent to the variable "legal status" (LLC, corporation, ...), then both refer to the same groups of firms. Using one or the other to analyze a third variable, say "employment," is therefore equivalent in statistical terms. Conversely, if a relationship is established between "employment" and "sector of activity," the same relationship would exist between "employment" and "legal status," since there is no distinction between the labels of two equivalent variables.

Let us now turn to the opposite situation. If we wish to describe *independence* between two variables X and Y, we express a situation in which the partition induced by X is "as contrary as possible" to that induced by Y. This situation is characterized as follows.

Définition 2.2. *Two variables X and Y on Ω_n are* independent *if and only if the labels of any element of the partition induced by X are uniformly distributed among the elements of the partition induced by Y. In this case, we write $X \perp\!\!\!\perp Y$.*

Figure 2.2 illustrates this situation with a particular example on $\Omega_{18} = \{\omega_1, \ldots, \omega_{18}\}$ and two variables, each inducing three partitions.

From a statistical and econometric point of view, independence between two variables means that if a label is in a particular block of the partition induced by X, this information gives no indication about the block of the partition of Y in which this label will be found. Imagine a situation in which X is the secondary school attended by students, while Y is a variable indicating whether these students completed their first year of university in one year. Independence, in this case, means that if a student comes from a school x (he or she belongs to that block of the partition of X), then he or she is equally likely (uniform distribution) to be in any of the blocks of Y, that is, success or failure.

5 Grouping into Classes

Grouping into classes consists of reducing the image of a variable to a smaller number of objects. In the example of Members of the European Parliament, for instance, one might introduce a new variable "Schengen zone," taking three values indicating whether the country belongs to this zone: Schengen, Partial Schengen[4], or Non-Schengen. To do so, one examines each image of the variable *citizenship* and associates it with the value "Schengen" or "Non-Schengen." In this way, the labels have been grouped into a smaller number of objects than the initial nationalities.

The operation of grouping into classes is thus a modification of the function "variable," say X, and this modification defines a new function, say RX. In mathematics, we call *operator* the rule that modifies one function in order to create another. The symbol R designates the "class grouping" operator. This operator does indeed transform the database. It is defined as follows:

$$R : \Omega_n \times \Im(X) \to \Omega_n \times \Im(RX) : (\omega, X(\omega)) \to (\omega, RX(\omega))$$

that is, it modifies the mapping of the function X so that it becomes RX on the same labels ω. Thus, in our example, the values of $(\omega, X(\omega))$ are those shown in Figure 2.1:

$$\begin{gathered} (2268, \text{Sweden}) \\ (34232, \text{Bulgaria}) \\ (33998, \text{Romania}) \\ \vdots \\ (96668, \text{Ireland}) \\ \vdots \end{gathered}$$

and those of $(\omega, RX(\omega))$ are:

$$\begin{gathered} (2268, \text{Schengen}) \\ (34232, \text{Partial Schengen}) \\ (33998, \text{Partial Schengen}) \\ \vdots \\ (96668, \text{Non-Schengen}) \\ \vdots \end{gathered}$$

6 The Probability Measure P_n

The description of partitions of the set of elements of Ω_n can be subject to reduction by calculation, in the etymological sense ("Calculator, from the Latin *calculis*, that is, those small pebbles which the Ancients held in their hand to compose and count numbers."[5]).

[4] Open borders in the air and maritime domains only, but not land borders.

[5] Calculator, a calculis, id est lapillis minutis, quos antiqui in manu tenentes numeros con ponebant (Isidorus Hispalensis, c. 636, Liber X, cap. 43)

This leads us to define the *probability measure*, or simply *probability*, of a preimage as the proportion of labels it contains.

Probability is thus a measure of the relative size of a subset of Ω_n. In the following definition, we denote by $|A|$ the cardinality of the set A, that is, the number of distinct elements it contains.

Définition 2.3. *Let Ω_n be a label space and $\mathcal{P}(\Omega_n)$ the set of parts of Ω_n. The* **probability measure** *P_n is the function*

$$P_n : \mathcal{P}(\Omega_n) \to [0,1] : A \to P_n(A) = \frac{|A|}{n}$$

which assigns to each part A its proportion within Ω_n.

From this definition, the elementary properties of the probability measure follow: $P_n(\emptyset) = 0$, $P_n(\Omega_n) = 1$, and $P_n(A^c) = 1 - P_n(A)$ for every subset $A \in \mathcal{P}(\Omega_n)$. Moreover, if $\{A_j ; j \in J\}$ is a collection of disjoint subsets of $\mathcal{P}(\Omega_n)$ then

$$P_n\left(\bigcup_{j\in J} A_j\right) = \sum_{j\in J} P_n(A_j) \ . \tag{2.1}$$

The relevance of the probability measure in data analysis appears when it is applied to a partition induced by a variable X. Take a_i to be any element of $\Im(X)$ (the image of X). For example, in the parliamentary database, let us take the element a_i = "male," which is a possible value of the variable X = *gender*. The proportion of parliamentarians whose variable *gender* is equal to "male" is given by the following probability measure:

$$P_n(\{\omega \in \Omega_n : X(\omega) = \text{“male”}\}) = \frac{|\{\omega \in \Omega_n : X(\omega) = \text{“male”}\}|}{N} = 0.61 \ .$$

To obtain this proportion for all possible values of the variable *gender*, one uses the following code:

```
proportions = EU['mep_gender'].value_counts(normalize=True)
```

where the option `normalize=True` ensures that the results are proportions between 0 and 1. The result is:

```
masculin     0.613352
féminin      0.385257
inconnu      0.001391
Name: mep_gender, dtype: float64
```

In general, the following definition specifies a probability function defined on any subset of the image of X.

Définition 2.4. *Let $(\Omega_n ; X)$ be a database whose label space is Ω_n and which includes the variable X. The* **probability measure induced by the variable** X *is the function*

$$\begin{aligned} P_X &: \mathcal{P}(\Im(X)) \longrightarrow [0,1] \\ &: A \longrightarrow P_X(A) = P_n(X \in A) := P_n(\{\omega \in \Omega_n : X(\omega) \in A\}) \ . \end{aligned}$$

which assigns to each subset of the image of X the corresponding proportion of labels in the database.

Exemple

Let us return to the student housing database shown in Table 2.2. Take a simple element in the image of the variable *Nationality*: $A = \{\text{Japan}\}$. Its probability measure counts the number of people of this nationality, divided by the total number of people:

$$P_{Nationality}(\{\text{Japan}\}) = P_n(\{\omega \in \Omega_{10} : Nationality(\omega) = \text{Japan}\}) = \frac{2}{10} = 0.2$$

which means that 20% of the residents in this housing are of Japanese nationality.

Note that P_X is defined on any subset of the image. Take a more elaborate set, for example $A = \{\text{Japan or Belgian}\}$. The above calculation applies, by counting in the numerator the number of rows satisfying the condition A, which gives

$$P_{Nationality}(\{\text{Japan or Belgian}\}) = \frac{5}{10} = 0.5$$

which corresponds to a proportion of 50% of individuals. A subset of $\Im(X)$ is always a union of disjoint sets, which is interpreted as an "or" condition and provides a well-defined calculation.

The previous example shows that if we consider several disjoint sets $A_1, \ldots, A_K$ from $\mathcal{P}(\Im(X))$, then we have

$$P_X\left(\bigcup_{i=1}^{K} A_i\right) = \sum_{i=1}^{K} P_X(A_i)\ .$$

What happens when the sets are not disjoint? This is what we consider in the following example.

Exemple

Returning again to the student housing database shown in Table 2.2, in which we have already described $\Im(Nationality)$. Consider two sets among the subsets of the image, for example

$$A = \{\text{Belgium}, \text{France}, \text{Spain}\} \quad (\text{“coming from Europe”})$$

and

$$B = \{\text{Belgium}, \text{France}\} \quad (\text{“coming from the Francophone world”})$$

These two sets are indeed subsets of $\Im(Nationality)$, and they are not disjoint. Their union

$$A \cup B = \{\text{Belgium}, \text{France}, \text{Spain}\}$$

is an element of $\mathcal{P}(\Im(Nationality))$ and we can therefore compute $P_X(A \cup B) = 0.8$. The same holds for their intersection, which here is $A \cap B = B$.

This example shows that the union and intersection of elements of $\mathcal{P}(\Im(X))$ are necessarily elements of $\mathcal{P}(\Im(X))$, and we can therefore compute their probability induced by X according to Definition 2.4.

7 Description of Variables

The first analysis that can be carried out on a database is to describe its variables. This description includes an explanation of what a variable means when its name alone does not suffice to convey its significance. The circumstances of its transcription must be clarified: units of measurement, notational conventions, etc.

7.1 The Distribution Function

The statistical description of a variable begins with an explanation of its image space. Next, it is useful to compute and communicate the probability of each element of the image. This information may be presented in the form of a list, a table, or a graph. All these representations in fact convey the same thing, namely the description of the probability of each element of the image. The **distribution** is the name given to the mathematical function that specifies the probability of each element of the image.

Définition 2.5. *Let $(\Omega_n; X)$ be a database containing a variable denoted X. The* **distribution function of** X *is the function*

$$\begin{aligned} p_X : \Im(X) &\longrightarrow [0,1] \\ : x &\longrightarrow p_X(x) = P_n(\{\omega \in \Omega_n : X(\omega) = x\}) \end{aligned}$$

which assigns to each element of the image of X its induced probability.

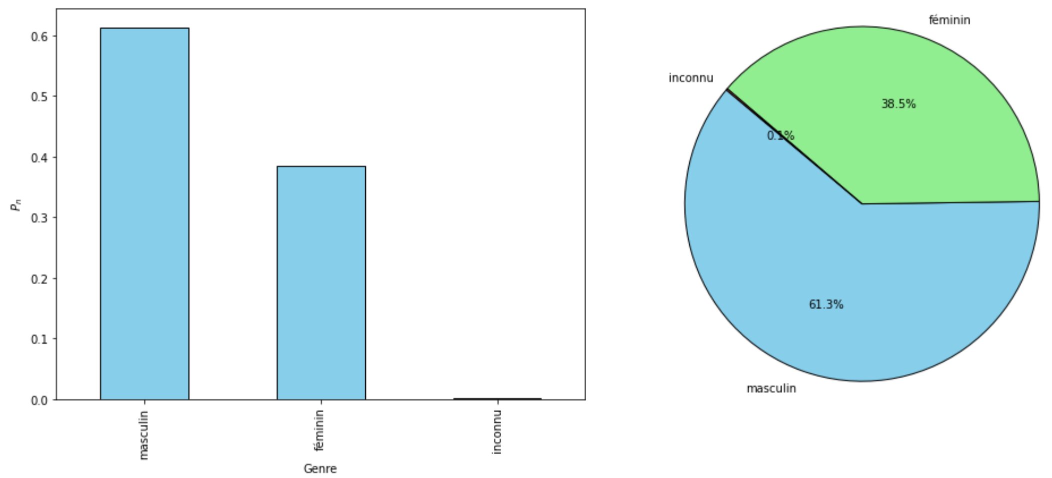


Fig. 2.3: Two graphical representations of the distribution function of the *Gender* of Members of the European Parliament.

As an example, Figure 2.3 shows two possible representations of the function p_X for the variable *gender* in the parliamentary database.

Naturally, computing probabilities reduces the granularity of analysis available from a database. The distribution function provides a summary of the statistical distribution of the variable's image but no longer allows one to link the images with their original labels. One way of observing the reduction introduced by probability calculation is to note that two equivalent variables have the same probability function, whereas two variables with the same probability function are not necessarily equivalent.

7.2 The Joint Distribution Function

When a database includes several variables, the distribution function of each variable can be computed. However, by focusing separately on each variable, this representation does not allow us to examine whether there is a relation or regularity between two or more variables. Is there a different gender distribution depending on the country of origin of the parliamentarians? Is there a link between country and age of parliamentarians? Such questions require a joint analysis of two variables.

Two tools are used to describe the probability distribution on multiple variables. The first tool is a direct extension of the distribution function to the case of multiple variables.

Définition 2.6. *Let* $(\Omega_n; X,Y)$ *be a database containing two variables* X *and* Y. *The* **joint distribution function of** (X,Y) *is the function*

$$\begin{aligned} p_{XY} &: \Im(X) \times \Im(Y) \longrightarrow [0,1] \\ &: (x,y) \longrightarrow p_{XY}(x,y) = P_n(\{\omega \in \Omega_n : X(\omega) = x\} \cap \{\omega \in \Omega_n : Y(\omega) = y\}) \end{aligned}$$

which assigns to each pair (x,y) *from the images of* X *and* Y *its induced probability.*

In this construction, X and Y are variables which induce, by definition, two generally distinct partitions of Ω_n. The pair (X,Y) induces a third partition that is necessarily finer, in the sense that this third partition divides Ω_n into more parts. The following example illustrates this phenomenon.

Exemple

In the student housing database shown in Table 2.2, consider the variables X = *Gender* and Y = *Nationality*. We saw above that *Nationality* induces a partition into four blocks. Since the image of *Gender* contains three values, this variable induces another partition into three blocks: $\{1,3,5,7,9\}$, $\{2,4,6\}$, and $\{8,10\}$. Considering both variables jointly, we must examine all possible cross-classifications of their images, i.e., all pairs in the product

$$\Big\{\text{Female}, \text{Male}, \text{Unspecified}\Big\} \times \Big\{\text{Belgium}, \text{France}, \text{Japan}, \text{Spain}\Big\}$$

which represents $3 \times 4 = 12$ pairs:

$$(\text{Female}, \text{Belgium}), (\text{Female}, \text{France}), \ldots$$

Each pair corresponds to a subset of labels, which in theory yields a partition into twelve blocks, finer than the previous two. It is "in theory" because some blocks may be empty:

Block	**Partition of** Ω_n	$(X(\omega),Y(\omega))$	$p_{XY}(x,y)$
Block 1	$\{1,5\}$	(Female, Belgium)	0.2
Block 2	$\{9\}$	(Female, France)	0.1
Block 3	$\{3\}$	(Female, Japan)	0.1
Block 4	$\{7\}$	(Female, Spain)	0.1
Block 5	$\emptyset$	(Male, Belgium)	0
Block 6	$\{2,6\}$	(Male, France)	0.2
Block 7	$\emptyset$	(Male, Japan)	0
Block 8	$\{4\}$	(Male, Spain)	0.1
Block 9	$\{10\}$	(Unspecified, Belgium)	0.1
Block 10	$\emptyset$	(Unspecified, France)	0
Block 11	$\{8\}$	(Unspecified, Japan)	0.1
Block 12	$\emptyset$	(Unspecified, Spain)	0

The representation of a joint distribution function therefore consists in communicating the probability measure associated with each element of the partition (X,Y). A graphical or tabular representation can be complex or difficult to read. The previous example shows one possible representation of the joint distribution function of $(\mathit{Gender},\mathit{Nationality})$ in the case of the student housing residents. Table 2.3 shows another possible representation of the function in the case of parliamentarians. At first glance, it is not easy to find regularities or salient features linking the variables X and Y from such representations. Some general elements can nevertheless be identified. For example, in Table 2.3, one immediately notices that the person of unspecified gender comes from Germany, and that all Cypriot parliamentarians are male.

	Gender			**Distribution**
Country	**female**	**unknown**	**male**	**marginal**
Germany	0.049	0.001	0.084	0.134
Austria	0.011	0.000	0.017	0.027
Belgium	0.013	0.000	0.018	0.031
Bulgaria	0.006	0.000	0.018	0.024
Cyprus	0.000	0.000	0.008	0.008
Croatia	0.007	0.000	0.010	0.017
Denmark	0.007	0.000	0.014	0.021
Spain	0.042	0.000	0.042	0.084
Estonia	0.003	0.000	0.007	0.010
Finland	0.013	0.000	0.008	0.021
France	0.057	0.000	0.054	0.111
Greece	0.008	0.000	0.021	0.029
Hungary	0.014	0.000	0.015	0.029
Ireland	0.008	0.000	0.011	0.019
Italy	0.035	0.000	0.072	0.107
Latvia	0.003	0.000	0.008	0.011
Lithuania	0.003	0.000	0.013	0.015
Luxembourg	0.003	0.000	0.006	0.009
Malta	0.001	0.000	0.007	0.008
Netherlands	0.021	0.000	0.022	0.043
Poland	0.021	0.000	0.053	0.074
Portugal	0.011	0.000	0.018	0.029
Romania	0.008	0.000	0.038	0.046
Slovakia	0.010	0.000	0.011	0.021
Slovenia	0.004	0.000	0.008	0.012
Sweden	0.018	0.000	0.011	0.029
Czechia	0.011	0.000	0.018	0.029
Marginal distribution	0.432	0.001	0.588	**1.000**

TABLE 2.3: A representation of the joint distribution function (*mep_gender*, *mep_citizenship*) of the 720 Members of the European Parliament.

Table 2.3 presents the joint distribution function in the form of a two-way table, also called a *contingency table*. The margins, at the bottom and to the left of the table, indicate the sum of the joint probabilities found in the corresponding column or row. The following proposition observes that these margins are equal to the probabilities of the marginal distribution functions of X and Y.

PROPOSITION 2.1. *Let p_{XY} be a joint distribution function of (X,Y) arising from a database with label space Ω_n. Then we have the following relations:*

$$p_X(x) = \sum_{y\in\Im(Y)} p_{XY}(x,y) \quad \textit{and} \quad p_Y(y) = \sum_{x\in\Im(X)} p_{XY}(x,y)\ .$$

Proof. We show the first equality, since the other is demonstrated in a similar way. By definition of the function p_X,

$$\begin{aligned} p_X(x) &= P_n(\{\omega \in \Omega_n : X(\omega) = x\}) \\ &= P_n(\{\omega \in \Omega_n : X(\omega) = x\} \cap \Omega_n) \\ &= P_n\left(\bigcup_{y \in \Im(Y)} (\{\omega \in \Omega_n : X(\omega) = x\} \cap \{\omega \in \Omega_n : Y(\omega) = y\})\right) . \end{aligned}$$

This last expression is the probability of the union of disjoint sets. By (2.1) it is therefore equal to the sum of the probabilities of these sets, i.e., to the sum of $p_{XY}(x,y)$, which proves the result. □

In the context where several variables are analyzed, p_{XY} is called the joint distribution function, while p_X and p_Y are called the **marginal distributions**, to recall that their probabilities are found in the margins of the table as illustrated in Table 2.3.

7.3 The Conditional Distribution Function

A second tool allows us to capture the probability distribution across multiple variables. It is based on the notion of conditional probability.

Définition 2.7. *Let Ω_n be a label space and $\mathcal{P}(\Omega_n)$ the set of subsets of Ω_n. Fix a non-empty subset $B \in \Omega_n$. The* **conditional probability given** *B is the function*

$$\begin{aligned} P_n(\cdot|B) : \mathcal{P}(\Omega_n) &\longrightarrow [0,1] \\ : A &\longrightarrow P_n(A|B) = \frac{P_n(A \cap B)}{P_n(B)} = \frac{|A \cap B|}{|B|} \end{aligned}$$

which assigns to each subset A the relative proportion of $A \cap B$ within B.

The conditional probability given a subset B of the database therefore measures the proportion of labels satisfying condition A within this set B. In this reasoning, reading the database is restricted to only those rows contained in B, which thus plays the role of a new, smaller label space. Let us take an example.

Exemple

In the database of 720 Members of the European Parliament, the Belgian delegation includes 22 members. We therefore take B to be the labels of the 22 Belgian MEPs. We then focus only on these 22 rows of the database in order to compute the proportion of females, males, and unknown gender. We must compute

$$P_n(\{\omega \in \Omega_{720} : \textit{Gender}(\omega) = \text{“female”}\}|\{\omega \in \Omega_{720} : \textit{Nationality}(\omega) = \text{“Belgian”}\})$$

which we will write more simply as

$$P_n(\text{“female”}|\text{“Belgian”}) = \frac{\#(\text{“female Belgian”})}{\#(\text{“Belgians”})} = \frac{9}{22} \approx 0.41$$

and

$$P_n(\text{“male”}|\text{“Belgian”}) = \frac{13}{22} \approx 0.59 \qquad P_n(\text{“unknown”}|\text{“Belgian”}) = 0$$

We can perform this calculation for the 27 nationalities of the European Union and obtain 27 different results. Figure 2.4 provides a graphical representation of these proportions for all nationalities.

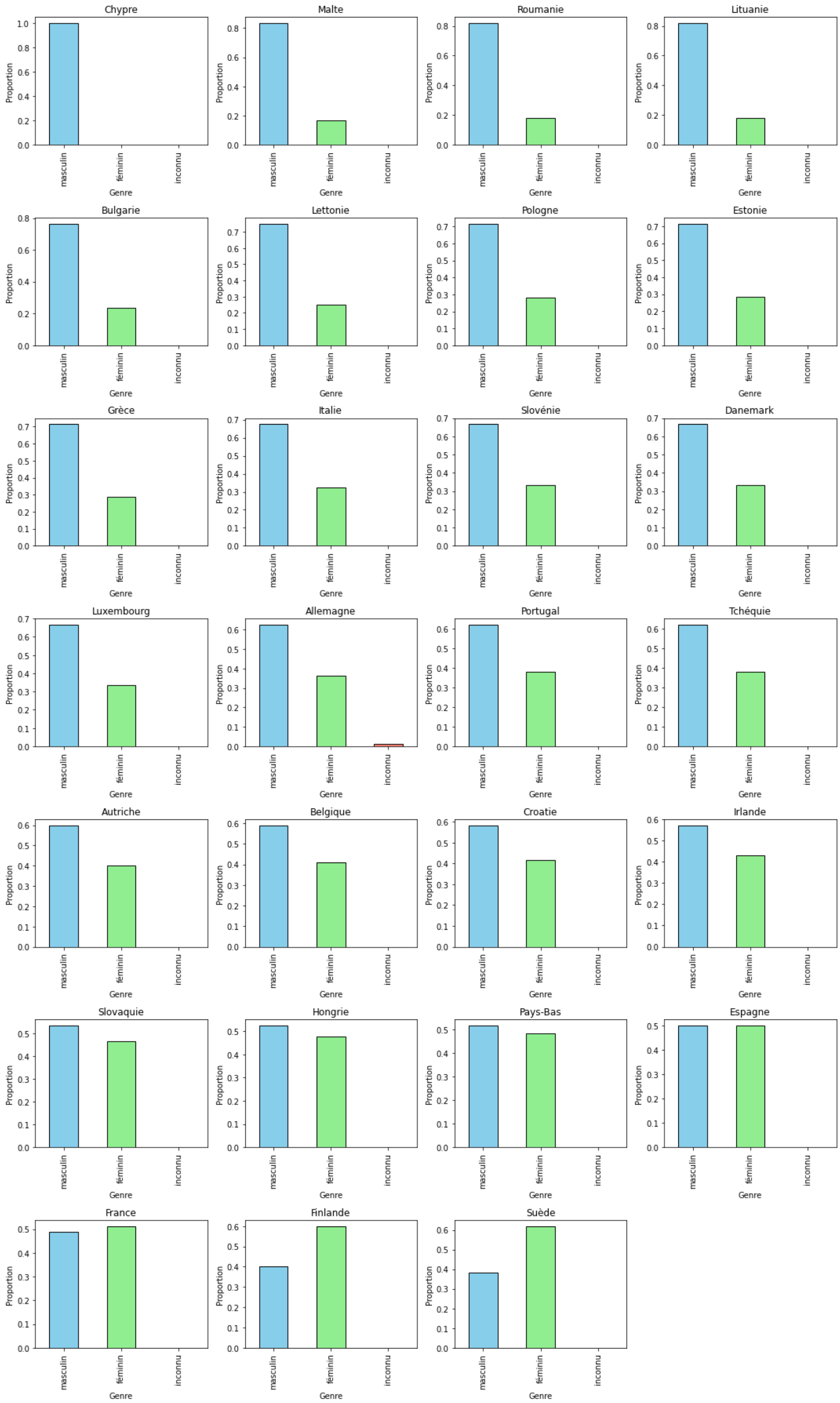


Fig. 2.4: Gender distribution conditional on the nationality of Members of the European Parliament. Countries are ordered by decreasing proportion of males.

As in Figure 2.4, we can define in full generality the conditional distribution function.

Définition 2.8. *Let $(\Omega_n; X,Y)$ be a database containing two variables X and Y, and let $x \in \Im(X)$. The* **distribution function of Y conditional on** $(X = x)$ *is the function*

$$\begin{aligned} p_{Y|X=x} &: \Im(Y) \longrightarrow [0,1] \\ &: y \longrightarrow p_{Y|X=x}(y) = P_n(Y = y|X = x) \end{aligned}$$

assigning to each element of the image of Y its probability conditional on the subset $(X = x)$.

Since this definition is established for all $x \in \Im(X)$, it follows that there exist as many conditional distribution functions as there are elements in the image of X. As an example, Figure 2.4 shows the distribution of gender (Y) conditional on each country of origin of the Members of Parliament (X). As there are 27 elements in the image of the variable *Nationality* on which we condition, it follows that we produce 27 graphical representations.

Conditional distributions play an important role in so-called structural approaches to data analysis. In these approaches, one seeks to examine a link between X and Y that is directional, that is, one seeks to give meaning to the phrase "how does variable X influence variable Y." Exploring the meaning of this phrase will be the subject of the part of the course devoted to structural approaches. The intuition behind this approach is that, by examining how conditional distributions vary when moving from one value of X to another, we obtain a means of quantifying the variation in Y that would be "induced" by a variation in X. To confirm this intuition, however, further concepts will necessarily need to be developed later in the course.

Referring to the gender of parliamentarians and to Figure 2.4, what can you say about the influence of country on the gender distribution within the delegation of Members of the European Parliament? Does this correspond to your intuition?

7.4 Fundamental Relations between Joint and Conditional Distributions

The definition of the conditional distribution $p_{Y|X=x}$ highlights a relation with the joint distribution. Indeed, in Definition 2.8, we have

$$P_n(Y = y|X = x) = \frac{P_n(\{Y = y\} \cap \{X = x\})}{P_n(X = x)} .$$

In this expression, recall that the numerator of the quotient is a simplified form of the probability

$$P_n(\{\omega \in \Omega_n : Y(\omega) = y\} \cap \{\omega \in \Omega_n : X(\omega) = x\})$$

which is therefore the joint probability $p_{XY}(x,y)$ given in Definition 2.6. The denominator of the expression is the marginal distribution of X alone. By moving this denominator to the left-hand side of the equation, we arrive at the following fundamental decomposition.

Proposition 2.2 (Marginal–Conditional Decomposition). *Let $(\Omega_n; X,Y)$ be a database with label space Ω_n and comprising two variables X and Y. Then the joint distribution function of (X,Y) is such that*

$$p_{XY}(x,y) = p_{Y|X=x}(y)p_X(x)$$

for all $x \in \Im(X)$ and $y \in \Im(Y)$.

The relation stated in this proposition is a relation between proportions in a database, as emphasized in the following example.

The next result establishes another relation between proportions in a database. It allows us to establish a link between the marginal distribution function and all conditional distributions given another variable. In our example, this is a relation between the proportions shown in Figure 2.3 and all those found in the conditional distributions represented in Figure 2.4.

PROPOSITION 2.3 (Law of Total Probability). *Let $(\Omega_n; X, Y)$ be a database with label space Ω_n and comprising two variables X and Y. Then the marginal distribution function of Y decomposes into the conditional distributions given $(X = x)$ through the relation*

$$p_Y(y) = \sum_{x \in \Im(X)} p_{Y|X=x}(y) p_X(x)$$

for all $y \in \Im(Y)$.

Proof. By the previous proposition establishing the marginal–conditional decomposition, the right-hand side of the equality is written $\sum_{x \in \Im(X)} p_{XY}(x,y)$ which, according to Proposition 2.1, is equal to the marginal probability $p_Y(y)$. □

Exemple

Figure 2.3 shows the marginal proportion of women among Members of the European Parliament: $p_{Gender}(\text{female}) = 0.39$. The previous proposition allows this proportion to be decomposed across the different countries. Using the law of total probability directly, this proportion is decomposed into 27 terms:

$$p(\text{female}) = p(\text{female}|\text{Germany})p(\text{Germany}) + \cdots + p(\text{female}|\text{Czechia})p(\text{Czechia})$$

In these terms, the conditional probabilities are found in Figure 2.4, while the marginal probabilities are the proportions of Members of Parliament by nationality. These are found in the margin of Table 2.3, for example $p(\text{Germany}) = 0.134$. Thus, we verify

$$0.39 = 0.365 \times 0.134 + \cdots$$

7.5 Independence

The concept of independence between two variables (Definition 2.2) is characterized using the probability function. It suffices to observe that two variables X_1 and X_2 are independent if and only if $P_n(A_1 \cap A_2) = P_n(A_1) \times P_n(A_2)$ for every element A_1 (resp. A_2) of the partition induced by X_1 (resp. X_2). But we already knew that independence between two variables is a proportional concept, perfectly verifiable from the observation of data.

The following definition specifies what is meant by the *independence* of two variables X and Y. This is a notion specific to data analysis and probability. In accordance with the definition of Kolmogorov (1933, Section I.5), it is formulated in terms of two partitions induced on the space Ω_n.

Exemple

> To verify the independence of *Nationality* and *Age* in the previous example, one must check the condition of the Definition for *all* pairs of blocks A_X and A_Y. It follows that, to demonstrate *non*-independence, it suffices to show that this relation fails for just one pair of blocks. Take $A_X = \{2{,}6{,}9\}$ and $A_Y = \{2{,}9\}$, and one easily verifies that the equality in the Definition does not hold. The variables are therefore not independent.

Kolmogorov (1933) generalizes this situation to define independence of more than two variables. If one wishes to define the independence of three variables X,Y, and Z, for example, one requires that the relation

$$P_n(A_X \cap A_Y \cap A_Z) = P_n(A_X)P_n(A_Y)P_n(A_Z)$$

be satisfied for every subset A_X, A_Y, A_Z of the partitions induced by X,Y, and Z. This situation is denoted $X \perp\!\!\!\perp Y \perp\!\!\!\perp Z$. This implies that the three variables are pairwise independent ($X \perp\!\!\!\perp Y$, $Y \perp\!\!\!\perp Z$, and $X \perp\!\!\!\perp Z$). The converse is false, as shown in Exercise 2.6, discovered by the mathematician Sergei Bernstein.

8 Logical Foundations

8.1 What is Statistics?

Statistics is the human activity that consists in formulating necessary propositions about objects, on the basis of the observation of a subset of them.

By *necessary proposition* we mean a proposition that follows logically, without indeterminacy or modality, from the explicit construction of a set of observed objects and the relations they maintain. This form of necessity is assertoric in the Aristotelian sense of the term: it characterizes what must be affirmed within a logical framework in the internal sense. It designates what is deductively entailed within the formal system grounded on the finite construction of observed objects and relations. As Aristotle states, "Demonstrative knowledge proceeds from necessary first principles [. . .] and essential predicates necessarily belong to their subjects."[6] (*APo.* I.74b5). In this framework, "it is being that is the cause of truth, and not the reverse" (Delcomminette, 2018, §50). For a discussion of the independence between truth and necessity in propositions concerning the future, see also Frede (1985). Finally, if the proposition is said to be necessary, it is because it follows from the observed state of the finite constructed system, and not from a principle of universal regularity. The necessity at issue is therefore logical and not hypothetical.

8.2 What is a Set of Labels?

At the origin of statistical activity lies the simple awareness that objects can be grouped together because they possess sensible characteristics allowing them to be compared or classified. The foundation of a statistical science therefore requires an understanding of these objects and of their gathering. Following the constructivist approach to mathematical logic, one defines a *sequence* as a rule associating an integer n with an object Ω_n (Bishop, 1967). The objects Ω_n are constructed by induction. Thus the rule "Human being living in the European Union" observes the objects of our world and successively

[6]... (*APo.* I.74b5; Aristoteles Graece, ed. Bekker; our translation)

integrates into Ω_n those that satisfy the stated rule. The rule is a univocal procedure, explicit and verifiable by a finite number of operations. The elements successively added to Ω_n are identifiable by *distinct* symbols, which we call *labels*. By convention, the set Ω_n contains n labels, so that the sequence is strictly increasing: $\Omega_1 \subset \Omega_2 \subset \dots$

The elements of Ω_n are objects that fall under a concept establishing the rule of inclusion. This means that a pre-existing relation is recognized among the objects gathered in this set: "to bring an object under a concept is simply to acknowledge a relation that already existed beforehand"[7] (Frege). Insofar as they fall under a concept, the objects obey a univocal and *explicit* rule of inclusion, and the statistician's first task is to state this rule. Let us illustrate this point.

If the concept "Region of the European Union" is used to construct Ω_n, one might find objects whose labels are "Normandy," "Lower Normandy," and "Calvados." This example is particular in that, geographically, these three distinct labels refer to nested regions: Calvados belongs to Lower Normandy, which in turn belongs to Normandy. From this point of view, one observes that the existence of a concept does not guarantee the distinction of the subsumed objects. This situation may have consequences in the formulation of statistical propositions, because such propositions always refer to the count (n) of the objects in Ω_n. However, if the concept governing this set is given by the more precise rule "NUTS3 geographical area," the elements are restricted to the 1,165 geographically distinct regions of the nomenclature Eurostat (2025). The description of the concept generating Ω_n is therefore a fundamental requirement in order to construct subsequent statistical statements. We shall return to this point.

8.3 What is a Data Point?

A datum is the measurement of a characteristic obtained (*datum*) on an object in Ω_n. This measurement is formalized by a *function* :

$$X : \Omega_n \longrightarrow \Im(X) : \omega \longmapsto X(\omega)$$

which we call a *variable*. In the sense of Bishop-style constructive analysis, a function generalizes the notion of a sequence: it is a rule consisting of a finite number of explicit operations for computing a unique value $X(\omega)$ from a given label ω (Bishop, 1967). The image of this function, denoted $\Im(X)$, is the set of distinct values taken by X over the set Ω_n. Since Ω_n is a sequence (i.e., $\Omega_1 \subset \Omega_2 \subset \dots$), the set $\Im(X)$ may include more and more elements as n increases. For clarity, we keep the notation X and $\Im(X)$ unindexed, though they are both defined over a growing domain as n ranges over the natural numbers.

A *data point* is therefore a pair $(\omega, X(\omega))$, where ω is a label. The set of data points, or the *database*, is thus the graph of the function X. This definition of a data point is purely descriptive: it describes what appears to us when a datum presents itself, and it presupposes no mechanism of data production. It is defined from a discrete, finite, and growing set of objects, Ω_n, to which values $X(\omega)$ are associated. In line with the project described in the introduction and in reference to the observations of Kolmogorov (1933), the definition of a variable and of a data point from a finite set carries an empirical meaning: that of directly describing the *datum* we receive.

One task of the statistician is to describe the data, that is, to give meaning to the functions X (what is being measured?) and to make explicit the image sets $\Im(X)$. This is where the question of the distinction between the elements of Ω_n must be examined.

[7]das Bringen eines Gegenstandes unter einen Begriff nur Anerkennung einer Beziehung ist, die schon vorher bestand (Frege, 1894, p. 317, our translation)

If the variable X measures, for example, the number of inhabitants, then the object X("Calvados") is itself nested within the object X("Normandy"), and in this sense we would say "the objects to which the labels refer are not distinct with respect to X." The point here is that this statement concerns the aspect of the objects given by the predicate X. If, on the contrary, the variable X measures the number of territorial civil servants, then the objects X("Calvados") and X("Normandy") are perfectly distinct.

8.4 Definition by Abstraction of the Concept of a Variable

Any function defined on Ω_n induces a partition formed by its preimages. For every element x in the image set $\Im(X)$, the *preimage* of x under X is the set $X^-(\{x\}) = \{\omega \in \Omega_n : X(\omega) = x\}$. These preimages are disjoint and their union is Ω_n:

$$X^-[\Im(X)] \equiv \dot{\bigcup_{x\in\Im(X)}} X^-(\{x\}) = \Omega_n. \tag{2.2}$$

The partition induced by X thus classifies the elements of Ω_n according to the values taken by the variable X. A classical combinatorial exercise consists in counting the number of such partitions, leading to the *Bell numbers*. One may also count the partitions into k nonempty subsets, yielding the *Stirling numbers of the second kind* (Conway & Guy, 1996, Chapter 4).

Two variables X and Y on Ω_n are said to be *equivalent* if they induce the same partition of Ω_n. For example, the variable $X : \Omega_5 \to \{\blacklozenge, \bigstar\}$ defined by $X(\omega_1) = X(\omega_4) = X(\omega_5) = \blacklozenge, \quad X(\omega_2) = X(\omega_3) = \bigstar$ is equivalent to $Y : \Omega_5 \to \{0, 1\}$ defined by: $Y(\omega_1) = Y(\omega_4) = Y(\omega_5) = 0, \quad Y(\omega_2) = Y(\omega_3) = 1$. That is, $X \sim Y$ if and only if: $\{X^-(\{x\}) \mid x \in \Im(X)\} = \{Y^-(\{y\}) \mid y \in \Im(Y)\}$. The relation $\sim$ is an equivalence relation on the set of variables defined over Ω_n: it is reflexive, symmetric, and transitive (Halmos, 2018).

An equivalence relation on Ω_n gives rise to a collection of equivalence classes, which correspond precisely to the elements of the partition induced by the variable. Let $[\omega_1]_\sim$ denote the equivalence class containing ω_1, then:

$$\omega \in [\omega_1]_\sim \quad \text{if and only if} \quad \omega_1 \in [\omega]_\sim$$

(e.g. Birkhoff, 1935). A variable is thus defined and fully determined by the equivalence classes $\{[\omega]_\sim, \omega \in \Omega_n\}$, which are disjoint over the set of objects Ω_n. This abstraction-based definition of the concept of variable relies on the notion of equivalence, as proposed by Peano (1894), who used the term "equality" instead of "equivalence".

Two clarifications may be helpful regarding the constructivist status of this abstraction-based definition.

First, our definition does not depend on any particular *representation* of the variable. Two variables are equivalent if and only if they induce the same partition of Ω_n, a condition that can be verified directly and finitely without altering the objects in $\Im(X)$ or $\Im(Y)$. Since X and Y act on the same set Ω_n, governed by a shared conceptual rule, their equivalence entails no appeal to psychological reduction in the Frege meaning (see Li, 2025, for various definitions of psychologism, in particular on p. 151).

Second, the equivalence relation $\sim$ we describe is, for each fixed n, a rule defined over a finite set. It is thus entirely verifiable and operational in the sense of constructive logic. The use of equivalence classes in this context does not violate constructivist principles, as long as they are not postulated axiomatically but explicitly defined through finitely checkable procedures.

9 Exercises

Exercise 2.1 (Python). Complete the last column of Table 2.1.

Exercise 2.2. Using the database shown in Table 2.2, describe the partition of the label space induced by the variable *Age*. Also describe the partition of the label space induced by the variable *Gender*. By comparing these two partitions, examine whether some sets from these two partitions are equal, or whether some sets from these two partitions have an empty intersection. Translate these observations into one or more sentences in French that convey information about the database.

Exercise 2.3. Based on Exercise 2.2, construct the power set of the partition induced by the variable *Gender*. Without constructing it explicitly, evaluate the number of elements contained in the power set induced by the variable *Age*.

Exercise 2.4 (Python). In the parliamentary database, what is the probability induced by the variable of observing a person whose *Gender* is "male" or "unknown"?

Exercise 2.5. Consider arbitrary subsets A,B of Ω_n and the probability measure P_n (Definition 2.3). Prove that

$$P_n(A \cup B) = P_n(A) + P_n(B) - P_n(A \cap B) .$$

Exercise 2.6. We say that two elements A and B of $\mathcal{P}(\Omega_n)$ are independent ($A \perp\!\!\!\perp B$) if the partitions $A \dot\cup A^c$ and $B \dot\cup B^c$ are independent. Consider three elements of $\mathcal{P}(\Omega_4)$ given by

$$A = \{\omega_1, \omega_2\},\ B = \{\omega_1, \omega_3\},\ C = \{\omega_1, \omega_4\} .$$

Show that $A \perp\!\!\!\perp B$, $B \perp\!\!\!\perp C$, $A \perp\!\!\!\perp C$, but that A,B, and C are not jointly independent. This observation, due to Sergei Natanovich Bernstein, is cited by Kolmogorov (1933, footnote 12).

Exercise 2.7. Using Table 2.3 or Figure 2.4, answer the following questions:

- Which country has the largest delegation in the European Parliament, and how many members does it comprise?
- How many Belgian Members of the European Parliament are there?
- Which countries have the most gender-balanced delegations?
- Which countries have a delegation with a strict female majority?

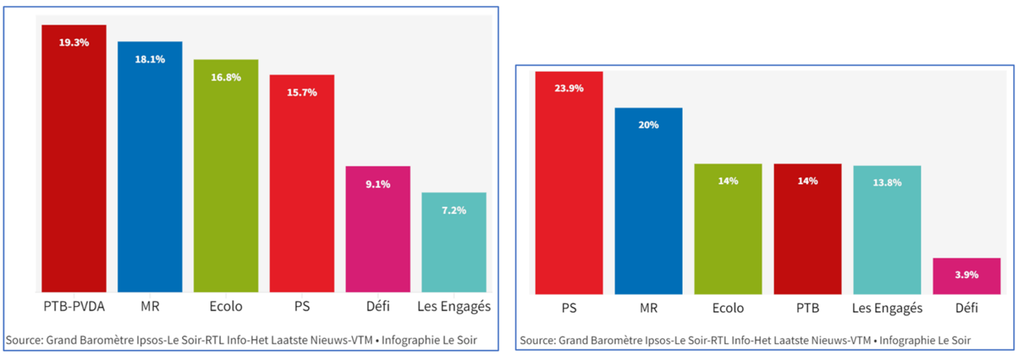


FIG. 2.5: Probability distribution of voting intentions. Left: Brussels-Capital Region; right: Walloon Region. (Le Soir, 15 December 2023)

Exercise 2.8. This exercise refers to Figure 2.5 and Table 2.4.

INS Code	Place of Residence	Men	Women	Total
01000	Belgium	5.761.410	5.936.147	11.697.557
02000	Flemish Region	3.352.319	3.422.488	6.774.807
03000	Walloon Region	1.800.829	1.880.746	3.681.575
04000	Brussels-Capital Region	608.262	632.913	1.241.175

Table 2.4: Population in Belgium and by Region on 1 January 2023 (Source: Statbel).

- Based on the voting intentions by Region shown in the figure, construct a contingency table of voting intentions, with political parties in rows and regions in columns, using the percentages indicated in the figures. For this purpose, you may treat the party "PTB-PVBA" as "PTB." Since each column does not sum to one, add a final row "Other" indicating the complementary percentage.
- Does the existence of the "Other" row, which is not reported by *Le Soir*, introduce any nuance in the ranking of voting intentions by political party in the Brussels-Capital Region or in the Walloon Region? Answer this question with one or more complete sentences in French.
- If voters from the two Regions are pooled, compute the probability distribution of voting intentions by political party in tabular form.

Exercise 2.9. A survey is conducted among 10 residents of the Brussels Region asking whether they are in favor of the LEZ policy (low-emission zone, restricting circulation to less-polluting vehicles). Here are the survey results: The response NA means that the answer was not obtained (no reason

Name	Response
Amira Benali	YES
Thomas Leroy	NA
Sofia Dimitrov	YES
Kwame Mensah	NA
Elena Rossi	NO
Jules Moreau	YES
Fatima Haddad	NO
Lucas Fernández	YES
Mei Chen	NO
Aisha Khan	NA

specified). In this sample, we are interested in the proportion in favor of the LEZ policy and the proportion opposed to it.

1. What is the label space?
2. Ignoring non-responses, what proportions are observed?
3. What is the image of "Response"?
4. Using the class-grouping operator, how can you reduce the image to the values YES and NO only?
5. Based on this (these) regrouping(s), recalculate the proportion in favor or opposed. On this basis, what can you assert with certainty about the distribution of opinion in favor of/against the LEZ policy?

Chapter 3

Operations on a Database

1 Sequences, Functions, and Data

We begin by recalling a few fundamental definitions from real analysis. To do so, we follow the constructivist approach of Bishop (1967).

1.1 Sequences and Functions

A *sequence* is a rule that associates with each natural number k an object a_k. A rule is an unambiguous, explicit, and finitely verifiable procedure. A *set* refers to mathematical objects satisfying a verifiable property. The set of sequences of natural numbers, for example, forms a set.

An *operation* between two sets A and B is a rule f that assigns to each element a of A an element of B, denoted by $f(a)$. The rule f must "allow an explicit, finite, and mechanical reduction of the procedure for constructing $f(a)$ to the procedure for constructing a" (Bishop, 1967). The set A is called the *domain* of the operation. A *function* is an operation f such that $f(a_1) = f(a_2)$ whenever a_1 and a_2 are equal elements of A.

1.2 Real Number

A sequence $\{a_k\}$ of rational numbers is said to be *regular* if

$$|a_j - a_k| \leqslant \frac{1}{j} + \frac{1}{k}$$

for all integers j and k. A *real number* is a regular sequence of rational numbers. Two real numbers $a \equiv \{a_k\}$ and $b \equiv \{b_k\}$ are said to be *equal* if

$$|a_j - b_j| \leqslant \frac{2}{j}$$

for all integers j. The set of real numbers is denoted by $\mathbb{R}$. A real number is therefore a sequence, and the usual operations are defined through the elements of this sequence. For example, addition $a+b$ is defined by the sequence[1] $\{a_{2j}+b_{2j}\}$, and similarly for operations such as multiplication, maximum, etc.

Certain notions concerning real numbers can be expressed as verification criteria on the rational sequence that constitutes them. Thus, we say that $a \equiv \{a_k\}$ is a *positive real number*, denoted $a \in \mathbb{R}^+$, if

$$a_k > \frac{1}{k} \qquad \text{for some natural number } k\ .$$

It is therefore sufficient that this criterion be satisfied for one element k of the sequence for the real number to be positive. Similarly, we say that $a \equiv \{a_k\}$ is a *non-negative real number*, denoted $a \in \mathbb{R}_0^+$, if

$$a_k > -\frac{1}{k} \qquad \text{for every natural number } k\ .$$

[1] explain this

Non-negativity is thus a more restrictive condition to verify.

The rational number a_k is called the *k-th rational approximation* of the real number $a \equiv \{a_k\}$. The operation from $\mathbb{R}$ to $\mathbb{Q}$ that maps a real number to its k-th rational approximation is not a function. The following result, taken from Bishop (1967, Chapter 1, Lemma 3), will be useful later. It provides a bound between a real number and its k-th rational approximation.

LEMME 3.1. *For every real number $a \equiv \{a_k\}$,*

$$|a - a_k| \leqslant \frac{1}{k}$$

for all k.

Proof. The $(2j)$-th approximation of $k^{-1} - |a - a_k|$ satisfies

$$\frac{1}{k} - |a_{2j} - a_k| \geqslant \frac{1}{k} - \frac{1}{2j} - \frac{1}{k}$$

since $\{a_k\}$ is a regular sequence. The right-hand side is greater than or equal to $-(2j)^{-1} > -j^{-1}$ for all j, which shows that $k^{-1} - |a - a_k|$ is a non-negative real number, hence the result. □

1.3 Convergent Sequences

A sequence $\{a_n\}$ of real numbers *converges* to a real number $a_\circ$ if, for every natural number m, there exists a natural number N_m such that

$$|a_n - a_\circ| \leqslant \frac{1}{m} \tag{3.1}$$

for all $n \geqslant N_m$. Note that, in this sequence, each a_n is a real number, that is, itself a regular sequence of rational numbers. The expression "a sequence of real numbers" therefore means a sequence of sequences. If needed, we will use an explicit notation for each $a_n \equiv \{a_{n,k}\}$, in which case expression (3.1) can be written[2]

$$|a_{n,4k} - a_{\circ,4k}| \leqslant \frac{1}{m} + \frac{1}{k}$$

which means that the inequality holds for all k-th rational approximations of the real numbers in the sequence. This will not always be made explicit in order to avoid overloading the notation. Thus, if the sequence $\{a_n\}$ of real numbers converges to a real number $a_\circ$, we will simply write

$$\lim_{n\to\infty} a_n = a_\circ$$

or "$a_n \longrightarrow a_\circ$ as $n \longrightarrow \infty$." A sequence of real numbers is said to be *convergent* if there exists such a real number to which it converges.

The following result provides a useful characterization of convergent sequences of real numbers.

[2] explain this

Proposition 3.1 (Cauchy Criterion). *A sequence $\{a_n\}$ of real numbers converges if and only if, for every natural number j, there exists a natural number M_j such that*

$$|a_m - a_n| \leqslant \frac{1}{j}$$

for all m and $n \geqslant M_j$.

Proof. Necessity. Assume that $\{a_n\}$ converges to $a_\circ$. Then there exists a sequence $\{N_j\}$ such that inequality (3.1) is satisfied. Let $M_j \equiv N_{2j}$. Then, for all m and $n \geqslant M_j$,

$$\begin{aligned} |a_m - a_n| &\leqslant |a_m - a_\circ| + |a_n - a_\circ| \\ &\leqslant \frac{1}{2j} + \frac{1}{2j} \\ &= \frac{1}{j} \end{aligned}$$

which shows that $\{a_n\}$ satisfies the Cauchy criterion.

Sufficiency. Now assume that $\{a_n\}$ satisfies the Cauchy criterion, and we want to verify that this sequence converges. Our proof must exhibit two constructions: First, the limit of $\{a_n\}$ must be explicitly constructed; we shall denote it by α. Second, we must explicitly construct the criterion N_j showing the convergence of a_n to α, as in (3.1).

For the first construction, consider the real subsequence $\{a_{N_i}\}$, where N_i is left arbitrary for now and will be specified as needed later. Let $a_{N_i,k}$ denote the k-th approximation of a_{N_i}. The candidate for α is the sequence $\{a_{N_i,2i}\}$. We now examine under what conditions on N_i this defines a real number. To do so, we must verify that the sequence is regular. We have

$$\begin{aligned} |a_{N_i,2i} - a_{N_k,2k}| &\leqslant |a_{N_i,2i} - a_{N_i}| + |a_{N_i} - a_{N_k}| + |a_{N_k} - a_{N_k,2k}| \\ &\leqslant \frac{1}{2i} + |a_{N_i} - a_{N_k}| + \frac{1}{2k} \end{aligned}$$

where we have used Lemma 3.1. For the middle term, we use the Cauchy criterion. Two situations may occur, depending on whether N_i is larger than N_k or vice versa. If $N_i \geqslant N_k$, we take $N_k \geqslant M_{2k}$, and the Cauchy criterion gives $|a_{N_i} - a_{N_k}| \leqslant (2k)^{-1}$. In the other case, we take $N_i \geqslant M_{2i}$, which gives $|a_{N_i} - a_{N_k}| \leqslant (2i)^{-1}$. Since one of these two cases must hold, we can write

$$|a_{N_i} - a_{N_k}| \leqslant \frac{1}{2k} + \frac{1}{2i}$$

whenever the sequence satisfies $N_x \geqslant M_{2x}$. Continuing the calculation, we obtain

$$|a_{N_i,2i} - a_{N_k,2k}| \leqslant \frac{1}{2i} + \frac{1}{2k} + \frac{1}{2i} + \frac{1}{2k} = \frac{1}{i} + \frac{1}{k}$$

which shows that the sequence $\{a_{N_j,2j}\}$ is regular and, consequently, that $\alpha \in \mathbb{R}$.

For the second construction, in order to examine the convergence of a_n toward α, we consider the decomposition

$$|\alpha - a_n| \leqslant |\alpha - a_{N_n,2n}| + |a_{N_n,2n} - a_{N_n}| + |a_{N_n} - a_n|$$

The first two terms are controlled by Lemma 3.1. The last term is controlled by the Cauchy criterion if, as before, we take $N_n \geqslant n \geqslant M_{2m}$:

$$|\alpha - a_n| \leqslant \frac{1}{n} + \frac{1}{2n} + \frac{1}{2m} .$$

We complete the argument by imposing $N_m \geqslant 3m$, which implies that for all $n \geqslant N(m)$,

$$|\alpha - a_n| \leqslant \frac{1}{3m} + \frac{1}{6m} + \frac{1}{2m} = \frac{1}{m}.$$

This establishes the convergence $a_n \to \alpha$. To summarize, the constraints on N_i are: $N_i \geqslant M_{2i}$ and $N_i \geqslant 3i$. We therefore construct the criterion as $N_i = \max(3i, M_{2i})$. □

1.4 Database

Let us now return to databases, which constitute the fundamental object of statistics and econometrics. At the origin of this activity lies the simple awareness that objects can be gathered together because they possess perceptible characteristics that allow them to be compared or classified. The foundation of a statistical science therefore rests on an understanding of these objects and of their collection. Following the definitions recalled earlier, the label spaces Ω_n form a *sequence*, that is, a rule that associates each integer n with an object Ω_n. The objects Ω_n are constructed inductively. For example, the rule "Human being residing within the European Union" observes the objects of our world and successively includes in Ω_n those that satisfy the stated rule. The rule is an unambiguous, explicit, and finitely verifiable procedure. The elements successively added to Ω_n are identifiable by *distinct* symbols, called *labels*. We stipulate that the set Ω_n contains n labels, so that the sequence is strictly increasing: $\Omega_1 \subset \Omega_2 \subset \dots$

The elements of Ω_n are objects that fall under a concept establishing the rule of inclusion. This means that a pre-existing relation is established between the objects gathered in this set: "bringing an object under a concept is merely the recognition of a relation that already existed"[3] (Frege). Insofar as they fall under a concept, the objects obey an *explicit* and unambiguous rule of inclusion, and the first task of the statistician is to state this rule. Let us illustrate this point.

If the concept of "Region of the European Union" is used to construct Ω_n, one could find within it objects with labels such as "Normandy," "Lower Normandy," and "Calvados." This example is peculiar in that, geographically, these three distinct labels refer to nested regions: Calvados is part of Lower Normandy, which in turn is part of Normandy. From this perspective, we observe that the existence of a concept does not guarantee the distinctness of the objects subsumed under it. This situation can have consequences for the formulation of statistical propositions, since these always refer to the count (n) of objects in Ω_n. However, if the concept governing this set is instead given by the more precise rule "NUTS3 geographical area," the elements are restricted to the 1,165 geographically distinct regions of the official nomenclature Eurostat (2025). The description of the concept that induces Ω_n is therefore a fundamental prerequisite for constructing statistical statements later on. We shall return to this point.

In the previous chapter, we saw that a variable is a function between the label space Ω_n and a set of objects, the image of X, denoted $\Im(X)$. A *datum* is therefore a pair $(\omega, X(\omega))$, where ω is a label. The set of data, or *database*, is thus the graph of the function X. We observe that this definition of data is purely descriptive: it describes what appears to us when data are presented and presupposes no mechanism of data generation. It is defined from the outset on a discrete, finite, and growing set Ω_n, to which values $X(\omega)$ are associated. Consistent with the project outlined in the introduction and in reference to the observations of Kolmogorov (1933), the definition of a variable and of data from

[3] das Bringen eines Gegenstandes unter einen Begriff nur Anerkennung einer Beziehung ist, die schon vorher bestand (Frege, 1894, p. 317, our translation)

a finite set carries an empirical meaning—namely, that of directly describing the *datum* that is received.

One of the statistician's tasks is to describe the data, that is, to give meaning to the functions X (what is being measured?) and to make explicit the image sets $\Im(X)$. It is here that the issue of the distinctness of the elements of Ω_n must be examined. If, for instance, the variable X measures the number of inhabitants, then the object X("Calvados") is itself nested within the object X("Normandy"), and in this sense we may say that "the objects referred to by the labels are not distinct under X." The point is that this statement concerns the aspect of the objects given by the predicate X. Conversely, if the variable X measures the number of local civil servants, then the objects X("Calvados") and X("Normandy") are perfectly distinct.

2 Euclidean Space

Euclidean geometry is the one with which we are most familiar. In this geometry, every point in the plane (or in space) is identified by two coordinates (or three coordinates in the case of a point in space)—for instance, latitude and longitude on a geographic map. These coordinate points presuppose that the plane or space includes perpendicular axes on which a unit length has been marked, as illustrated in Figure 3.1. Once this "setting" is in place, we have a very convenient structure that allows us to define and measure all sorts of objects: we can measure the length separating two points, the shortest path between two objects, the angle between two lines, the distance from a point to a line, or from a point to a plane in space, the set of points equidistant from a given point; we can also define two parallel lines, and so forth. This structure makes it possible to draw architectural plans, and it is precisely this structure that we will use to visualize and navigate data.

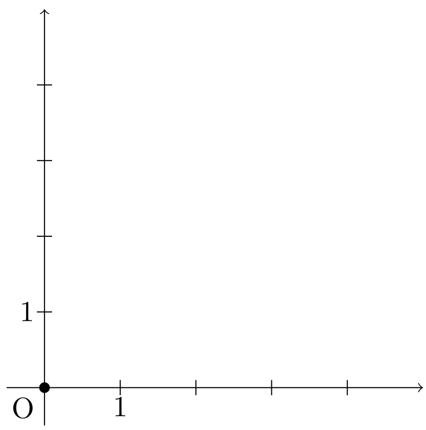


FIG. 3.1: Usual representation of a Cartesian plane with two perpendicular axes and a unit scale.

We first recall the definition of a Euclidean space before making the connection with databases explicit.

To define a Euclidean space, two ingredients are required. The first consists in defining the operations of "addition" and "multiplication by a number." This is the purpose of the next definition, which introduces what is called a "vector space," whose elements are called "vectors." The second ingredient is an operation that allows us to measure the distance between vectors or their length. This operation is called the "inner product," defined just afterwards.

Définition 3.1 (Vector Space). *A vector space is a set $\mathcal{E}$ whose elements are called* vectors, *equipped with two operations*—addition *and* scalar multiplication—*that satisfy the following properties:*

(i) *For all u and v in $\mathcal{E}$, $u+v$ is defined such that $u+v=v+u$ and $u+(v+w)=(u+v)+w$;*

(ii) *$\mathcal{E}$ contains a unique vector, denoted 0 and called the* zero vector, *such that $u+0=u$ for all $u \in \mathcal{E}$;*

(iii) *For each $u \in \mathcal{E}$, there exists a unique vector $-u$ such that $u+(-u)=0$;*

(iv) *For each pair (α,u) where u is a vector in $\mathcal{E}$ and α is a scalar (here meaning $\alpha \in \mathbb{R}$), $\alpha u \in \mathcal{E}$ is defined such that $1u=u$, $\alpha(\beta u)=(\alpha\beta)u$, $\alpha(u+v)=\alpha u+\alpha v$, and $(\alpha+\beta)u=\alpha u+\beta u$.*

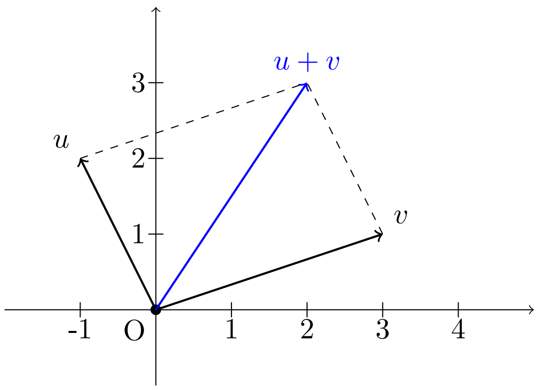


Fig. 3.2: Addition of two vectors in the Cartesian plane.

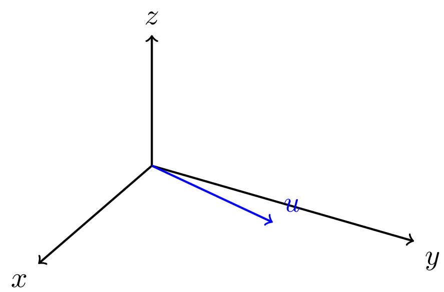


Fig. 3.3: Representation of a vector $u=(2,3,1)$ in three-dimensional Cartesian space.

Exemple

Figure 3.2 shows in the Cartesian plane the two vectors

$$u = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \qquad \text{and} \qquad v = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

Throughout this course, vector coordinates are written in column form. This convention aligns with the usual presentation of databases as tables, where columns correspond to variable values. To save space, we often write $u=(-1,2)^\top$, which reads “the transpose of the row vector $(-1,2)$.” In algebra, the transpose of a vector turns a row into a column, and conversely; hence $(u^\top)^\top = u$.

Figure 3.3 shows a Cartesian space with three perpendicular axes and the vector $u = (2,3,1)^\top$.

In Figure 3.2, the operation of adding the two vectors u and v follows the visual "parallelogram rule," shown with dashed lines. The sum $u + v$ corresponds to the diagonal of the parallelogram. Formally, this vector is obtained by componentwise addition:

$$u + v = \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

which matches the geometric rule in the figure.

To illustrate scalar multiplication, suppose in Figure 3.3 that we multiply the vector u by 2. We obtain

$$2u = 2\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix},$$

i.e., componentwise multiplication. The resulting vector lies in the same direction as u.

Définition 3.2 (Inner Product). *Let $\mathcal{E}$ be a vector space as defined above. An* inner product *is an operation $\langle u,v\rangle$ satisfying the following properties for all $u,v,w \in \mathcal{H}$ and $\alpha \in \mathbb{R}$:*

(i) $\langle u, v\rangle = \langle v, u\rangle$;

(ii) $\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$;

(iii) $\langle \alpha u, v\rangle = \alpha\langle u, v\rangle$;

(iv) $\langle u, u\rangle \geqslant 0$;

(v) $\langle u, u\rangle = 0$ *only if* $u = 0$.

With these two definitions established, we can now define the **Euclidean space**, which is the vector space $\mathbb{R}^n$ endowed with the **Euclidean inner product** defined as follows: If $u = (u_1, \ldots ,u_n)^\top$ and $v = (v_1, \ldots ,v_n)^\top$ are two vectors in $\mathbb{R}^n$, then

$$\begin{aligned}\langle u, v\rangle &= v^\top u \\ &= v_1u_1 + v_2u_2 + \ldots + v_nu_n\,.\end{aligned}$$

The inner product is a very useful operation because it allows us to define the length of a vector and the angle between two vectors. The length, in this context, is called the **norm of a vector** and is defined by

$$\begin{aligned}\|u\| &= \sqrt{\langle u,u\rangle} \\ &= \sqrt{u_1^2 + u_2^2 + \ldots + u_n^2}\end{aligned}$$

where the last equality holds for the Euclidean inner product. Furthermore, if u and v are two vectors in the Euclidean space $\mathbb{R}^n$, there exists a relation between their inner product and the angle between them:

$$\langle u,v\rangle = \|u\| \cdot \|v\| \cdot \cos(u,v) \tag{3.2}$$

which is proved later in this course (Exercise NN).

Exemple

Let us compute the inner product between the vectors u and v shown in Figure 3.2:

$$\langle u, v \rangle = (3{,}1) \begin{pmatrix} -1 \\ 2 \end{pmatrix} = -3 + 2 = -1 \, .$$

In Figure 3.3, the norm of vector u is

$$\|u\| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14} \, .$$

Before returning to databases, we state a few fundamental and widely used results concerning Euclidean spaces.

PROPOSITION 3.2 (Cauchy–Schwarz Inequality). *Let $\mathcal{E}$ be a Euclidean space. Then, for all $u, v \in \mathcal{E}$, we have $|\langle u, v \rangle| \leqslant \|u\| \cdot \|v\|$.*

This inequality states that the inner product, which can be positive or negative, is always less than or equal to the product of the vector lengths. Later in the course, we will sometimes encounter inner products that are difficult to compute. This upper bound is simpler and, in some cases, provides a sufficient approximation. The inequality is consistent with formula (3.2), and equality holds when u and v are collinear (easy exercise). However, since we have not yet proved that formula, we give a proof independent of it.

Proof. If $v = 0$, the result is trivial. Assume $v \neq 0$. For all $\alpha \in \mathbb{R}$,

$$\begin{aligned} 0 &\leqslant \langle u - \alpha v, u - \alpha v \rangle \\ &= \langle u, u \rangle - 2\alpha \langle u,v \rangle + \alpha^2 \langle v, v \rangle \, . \end{aligned}$$

Taking $\alpha = \langle u,v \rangle / \langle v, v \rangle$ yields the result. □

The next result expresses a familiar geometric fact: To go from a point A to a point B, the shortest path in a Euclidean space is the straight line joining them. Taking a detour through another point A' increases the distance traveled. This result is illustrated in Figure 3.4.

PROPOSITION 3.3 (Triangle Inequality). *Let $\mathcal{E}$ be a Euclidean space. Then, for all $u, v \in \mathcal{E}$, $\|u + v\| \leqslant \|u\| + \|v\|$.*

Proof. From the Cauchy–Schwarz inequality,

$$\begin{aligned} \|u + v\|^2 &= \langle u, u \rangle + 2\langle u, v \rangle + \langle v, v \rangle \\ &= \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \\ &\leqslant \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 \\ &= (\|u\| + \|v\|)^2 \end{aligned} \tag{3.3}$$

and the announced result follows. □

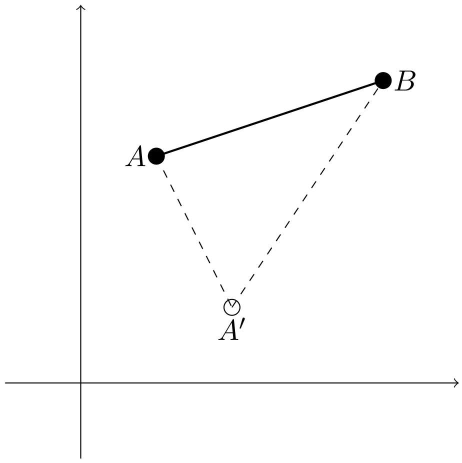


FIG. 3.4: Illustration of the triangle inequality.

In this proof, equality (3.3) shows that the square of the length of the long side of the triangle in Figure 3.4 equals the sum of the squares of the lengths of the two shorter sides plus twice their inner product. When the triangle is right-angled, the side $u + v$ is the hypotenuse, and Pythagoras's theorem states that $\|u + v\|^2 = \|u\|^2 + \|v\|^2$—in other words, when the triangle is right-angled, $\langle u,v\rangle = 0$. This is our first indication that when the angle between u and v is a right angle, their inner product is zero.

3 Euclidean Space of Numerical Variables

3.1 Empirical Inner Product on the Image Space

Databases are composed of a label space Ω_n and a collection of variables, usually denoted by the last capital letters of the alphabet $(U, X_1, X_2, Y, \ldots)$. These variables are functions defined on Ω_n. When a dataset is collected and read, it appears in the form of a table in which each row corresponds to an element of Ω_n. The columns of this table represent the *values of the variables* for each label. If the variable is denoted by X, we write $(x_1, \ldots, x_n)^\top$ for the vector of its values across all labels.

For example, if the variable *Age* refers to members of a parliament, the function could take the values $(37, 62, 57, 43)^\top$ in an assembly of four members. The order of the components is essential: this vector indicates that the member with label ω_1 is 37 years old.

From this example, we can see that collected datasets appear as tables whose columns are vectors in $\mathbb{R}^n$, *provided that the image of the variables is numerical*, that is, $\Im(X) \subset \mathbb{R}$. We can thus think of the variable values as elements of a Euclidean space. However, in what follows, we will not use the usual Euclidean inner product. In the next definition, we introduce a slight modification of it, for a reason that will soon become clear.

DÉFINITION 3.3 (Empirical Inner Product). *Let $(\Omega_n; X, Y)$ be a dataset with label space Ω_n, containing two variables such that $\Im(X)$ and $\Im(Y) \subset \mathbb{R}$. The values taken by these variables are denoted $x = (x_1, \ldots, x_n)^\top$ and $y = (y_1, \ldots, y_n)^\top$, both belonging to $\mathbb{R}^n$. The* *vectors x and y lie in the Euclidean space $\mathbb{R}^n$, endowed with the following inner product:*

$$\begin{aligned}\langle x, y\rangle_n &= \sum_{\omega\in\Omega_n} X(\omega)Y(\omega)P_n(\{\omega\}) \\ &= \frac{1}{n}(y_1x_1 + y_2x_2 + \cdots + y_nx_n) \\ &= \frac{1}{n}y^\top x\ .\end{aligned} \tag{3.4}$$

This is called the empirical inner product.

This definition presents several equivalent expressions of the empirical inner product. They are indeed identical, since the empirical probability measure P_n assigns an equal weight of $\frac{1}{n}$ to each label, that is, $P_n(\{\omega\}) = \frac{1}{n}$ for all $\omega \in \Omega_n$. The adjective "empirical" refers to the data-collection context, as it connects the inner product with the empirical probability measure.

To motivate the introduction of this new inner product, consider a dataset with one numerical variable X and another, special variable that takes the constant value 1 for all labels. We denote by[4] $\iota_n = (1,\ldots,1)^\top$ the values of this constant variable. As before, let $x = (x_1,\ldots,x_n)^\top$ denote the values taken by X. We can then easily verify that the inner product between x and ι_n is

$$\langle x, \iota_n\rangle_n = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

which is precisely the empirical mean of X—a concept that will be defined formally in the next chapter. Since the empirical inner product gives direct access to the mean, it has an appealing statistical interpretation. This observation will be generalized in the next chapter.

3.2 The Space of Numerical Variables

We have just constructed a Euclidean space from the image of variables in a dataset. It is also possible to build a Euclidean space on the variables themselves, taking into account that variables are *functions* rather than merely *numerical values*. The empirical inner product introduced earlier can be used here, but in the previous section it was defined on the values x and y. We can naturally reinterpret its definition (3.4) as an inner product on the variables X and Y themselves, namely:

$$\langle X,Y\rangle = \sum_{\omega\in\Omega_n} X(\omega)Y(\omega)P_n(\{\omega\})\ . \tag{3.5}$$

To distinguish this inner product—defined on functions—from the empirical inner product defined on their values, we omit the subscript. The set of numerical variables equipped with this inner product forms a Euclidean space. Hence, within this space, we can think of the variables X and Y as vectors in a Euclidean space. We can also speak of the *length* (or norm) of X, defined as

$$\|X\| = \sqrt{\sum_{\omega\in\Omega_n} X(\omega)^2P_n(\{\omega\})} = \sqrt{\frac{1}{n}\sum_{\omega\in\Omega_n} X(\omega)^2}\ .$$

[4]Constant vector, pronounced "iota n," where n refers to the length of the vector.

This quantity is finite, since X is a numerical variable and the sum runs over a finite number n of elements. The space of variables plays a central role in data analysis, as it is through the variables that we observe and describe data. This space is given a specific name, formalized below.

DÉFINITION 3.4 (The L_2 Space). *Let Ω_n be a finite set of labels, $\mathcal{P}(\Omega_n)$ the set of all its subsets, and P_n the empirical probability measure (*cf. *Definition 2.3). The set of variables defined on Ω_n equipped with the inner product (3.5) is denoted by $L_2(\Omega_n,\mathcal{P}(\Omega_n),P_n)$, or simply $L_2(\Omega_n)$, and even L_2 when the context makes the underlying elements clear.*

In this notation, the subscript "2" refers to the formula for the length of X, which involves the square of its values. The letter "L" refers to the French mathematician Henri-Léon Lebesgue (1875–1941). The space L_2 is in fact a special case of a much broader class of spaces, defined even when Ω_n is infinite—a situation that brings about significant technical challenges, especially for defining the set of measurable subsets and the probability measure itself.

Of course, when n is finite, the sum appearing in the definition of $\|X\|$ contains only finitely many terms and is therefore finite. For any finite n, the variables therefore belong to L_2. However, if we recall that Ω_n has been viewed as a sequence indexed by n, the statement that X belongs to L_2 becomes an assumption about the convergence of this sum as n grows.

4 Vector Subspaces

In general, a dataset contains several variables. In data analysis and econometrics, one often seeks to establish relationships between these variables. The nature of such relationships will be discussed throughout this course. We often say that we want to see to what extent one variable, say X, "explains" another variable, say Y. More precisely, we might ask whether part of the "variability" of Y can be "explained" by the variability of X. In what follows, we will give these intuitive notions a precise mathematical meaning.

Let us begin with an example of a dataset in which we collect, on the one hand, the gender of a group of individuals who perform the same function and have similar professional experience, and on the other hand, their salaries. An example is provided in Table 3.1. Clearly, salaries are not identical for everyone — they exhibit variability. The same observation applies to gender, which also varies across individuals. Can we explain, that is, find a functional relationship between the variation in salaries and the variation in gender? Many statistical tools can be used to address such a question, and we will discuss them later. The underlying idea behind these tools is to "look at salary data through the lens of the gender variable." Although this expression is vague, it suggests that we must examine salaries by breaking them down across gender categories. That is precisely the idea captured by this section, devoted to subspaces.

Just like *Gender* and *Salary*, variables are functions defined on Ω_n. As seen in the previous chapter, a variable induces a partition of Ω_n—for example, a three-block partition according to gender (male, female, not specified). In general, we denote by $\Im(X)$ the image of the variable X (for *Gender*, this is {male, female, not specified}). The partition induced on Ω_n is denoted by $X^-[\Im(X)]$, meaning the set of all subsets of labels corresponding to the different values taken by X.

The induced partition $X^-[\Im(X)]$ is therefore a collection of subsets of labels; let us call them $A_1,A_2,\ldots,A_p$ if there are p distinct blocks. Focusing on these blocks rather than

Individual	Gender	Gross salary (K€)
1	Male	[100,110[
2	Female	[70,90[
3	Male	[90,100]
4	Female	[50,70[
5	Not specified	[70,90[
6	Male	[110,120[
7	Female	[50,70[
8	Male	[100,110[
9	Not specified	[70,90[
10	Female	[50,70[
11	Male	[110,120[
12	Female	[70,90[
13	Male	[90,100]
14	Female	[50,70[
15	Not specified	[50,70[
16	Male	[110,120[
17	Male	[100,110[
18	Female	[70,90[
19	Male	[90,100]
20	Female	[50,70[

Table 3.1: Dataset of individuals holding the same function within the same sector, with comparable levels of experience.

on individual labels is what we mean by "looking at the dataset through the eyes of X," since X does not distinguish between two labels belonging to the same block. In doing so, we effectively restrict our view by considering these blocks as the only admissible subsets of Ω_n. Instead of working with the entire power set $\mathcal{P}(\Omega_n)$, we construct a new collection of subsets of Ω_n induced by the variable X, denoted by $\sigma(X)$.

Définition 3.5 ($L_2(\Omega_n,\sigma(X),P_n)$). *Let a dataset be defined on the label space Ω_n, and let $X \in L_2(\Omega_n,\mathcal{P}(\Omega_n),P_n)$.*

- *$\sigma(X)$ denotes the set of all possible unions of elements of $X^{-1}[\Im(X)]$. By convention, this collection includes the empty set $\emptyset$, and by definition, it includes Ω_n itself.*
- *The set of variables on Ω_n whose partitions are included in $\sigma(X)$ is denoted $L_2(\Omega_n,\sigma(X),P_n)$.*

In general, another variable Y induces a partition that is not contained in $\sigma(X)$. The paradigm of "explaining" Y by X thus takes on a precise meaning: it consists in analyzing the function Y by examining its approximation by functions belonging to $L_2(\Omega_n,\sigma(X),P_n)$. We will develop this idea further in the next chapter, through the key concept of the conditional expectation.

The following example provides a numerical illustration of this idea—an explicit instance of what it means to "approximate Y by functions belonging to $L_2(\Omega_n,\sigma(X),P_n)$."

Exemple

In the dataset of Table 3.1, the label space Ω_{20} consists of the identifiers of 20 individuals. Let X denote the gender variable and Y the gross salary variable. The distribution of salaries, Y, can be analyzed as before, for instance through a bar chart. To look at Y through the "eyes" of X, we first consider $\sigma(X)$. The gender variable divides Ω_{20} into three blocks:

$$X^{-}[\Im(X)] = \Big\{\{1,3,6,8,11,13,16,17,19\}, \{2,4,7,10,12,14,18,20\}, \{5,9,15\}\Big\} .$$

Let us denote these blocks by A_H, A_F, and A_N (for "hommes," "femmes," and "non spécifié"). The salary variable Y is not defined on this partition, since not all labels in A_H have the same image under Y (for example, individuals 1 and 3 belong to A_H, have the same X value, but different Y values).

How should we understand the idea of "approximating Y by functions defined on $\sigma(X)$"? Several approaches are possible. A simple one consists of taking the mean of Y within each element of $\sigma(X)$ (recall that the elements of $\sigma(X)$ are simply blocks of labels). For instance, consider the salary classes of individuals in the subset A_H. If we take the midpoints of the salary intervals, we have:

- On A_H: 105, 95, 115, 105, 115, 95, 115, 105, 95

and similarly,

- On A_F: 80, 60, 60, 60, 80, 60, 80, 60
- On A_N: 80, 80, 60

By taking the means—i.e., the expectations—within each block, we obtain the average salaries per block: 105 K€ for A_H, 67.5 K€ for A_F, and 73.77 K€ for A_N. These block averages are well-defined on $\sigma(X)$: they form a function of $\sigma(X)$ since they take the same value for all labels in each element of the partition $X^{-}[\Im(X)]$. Hence, we have effectively "approximated" the salary variable by the gender variable. This paradigm will serve as the starting point for the structural approaches introduced later in this course. In the next chapter, we will formalize this notion of approximation and of blockwise means, under the name of *conditional expectation given* $\sigma(X)$.

We conclude with a fundamental result about these subspaces.

THÉORÈME 3.1 (Dynkin's π–λ Theorem). *Let a dataset be defined on the label space Ω_n, with $X \in L_2(\Omega_n,\mathcal{P}(\Omega_n),P_n)$ and $Z \in L_2(\Omega_n,\sigma(X),P_n)$. Then*

$$\sigma(Z) \subseteq \sigma(X) \subseteq \mathcal{P}(\Omega_n) .$$

5 Exercises

Exercise 3.1. Let $\mathcal{E}$ be a Euclidean space. Show that the following properties hold:

$$\begin{aligned}
&\langle 0,v\rangle = 0 \quad \text{for all } v \in \mathcal{E}\\
&\text{For every } v \in \mathcal{E},\ \text{the mapping } u \mapsto \langle u, v\rangle \text{ is linear}\\
&\langle u, \alpha v\rangle = \alpha\langle u, v\rangle \quad \text{for all } u,v \in \mathcal{E} \text{ and } \alpha \in \mathbb{R}\\
&\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle \quad \text{for all } u,v,w \in \mathcal{E}
\end{aligned}$$

Exercise 3.2. Let u and v be two vectors in the Euclidean space $\mathbb{R}^n$. Use the triangle inequality to show that the function $d(u,v) = \|u - v\|$ defines a *distance*.

Recall the mathematical definition of a *distance* (or *metric*). A mapping $d : \mathcal{E} \times \mathcal{E} \to \mathbb{R}^+$ is a *distance* if it satisfies the following three properties:

- Symmetry: $d(u,v) = d(v,u)$,
- Identity of indiscernibles: $d(u,v) = 0 \Leftrightarrow u = v$,
- Triangle inequality: $d(u,v) \leqslant d(u,w) + d(w,v)$.

A space equipped with a distance is called a *metric space*.

Exercise 3.3. For all u,v in the Euclidean space $\mathcal{E}$, prove the *parallelogram identity*:

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2 \ .$$

If $\|u\|$ denotes the length of vector u, this identity expresses a well-known geometric property of parallelograms in Euclidean geometry: the sum of the squares of the lengths of the diagonals equals the sum of the squares of the lengths of the sides.

CHAPTER 4

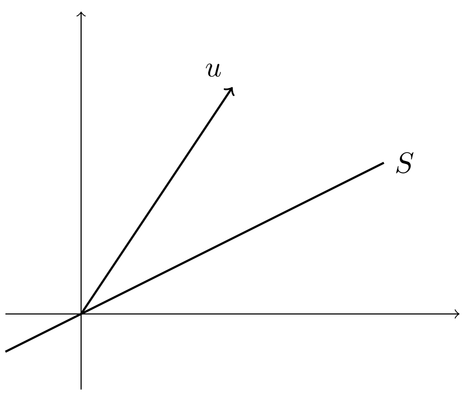
Conditional Expectation

1 Introduction

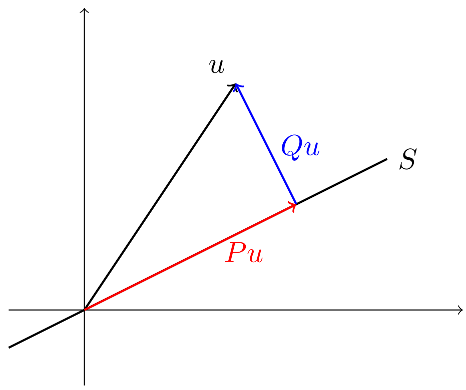
In Chapter 2 we showed that the distribution functions of the variables in a database indicate how the empirical probability is distributed over all possible values of the variables. These distributions form a representation of the variables useful for understanding the phenomena measured by the database. Among these representations, the conditional distribution function made it possible to examine how the probability distribution of one variable evolves according to the value of another variable, as in Figure 2.4. This is a representation that allows us to examine a relationship between two variables. In this chapter we show that representation by conditional distributions is the foundation of the statistical and econometric method for analyzing data and verifying whether a link exists between two variables (or more).

The objective of this chapter is twofold. First, we state and prove the *projection theorem*, often used in statistics and econometrics. Second, we use the projection theorem to obtain, as a corollary, the existence and uniqueness of the conditional expectation. We will immediately apply this result to show how it is used in statistical and econometric methods.

The projection theorem has an intuitive geometric interpretation. It deals with the projection of a vector onto an object, somewhat like the shadow of a person under light is a projection of that person on a wall. A simple situation is that of a vector u on this sheet of paper projected onto a line S of $\mathbb{R}^2$. Here is a drawing of this situation:



The result that we will prove concerns the projection of u onto S. There are several ways to make this projection, but there is only one way to achieve a perpendicular projection, as in this drawing where the vector Qu is *perpendicular* to S:



The projection theorem states that there exists a unique decomposition of the vector $u = Pu + Qu$, where P is the orthogonal projection on S and Q is the projection in the direction perpendicular to S. It provides conditions for the existence and uniqueness of the projection Pu. This vector is a minimal-norm solution, since it is the point closest to u (thus, the length of Qu is minimal compared to any other projection). Below, we will establish a parallel with random variables. Since the variables X and Y belong to the vector space $L_2(\Omega_n)$ endowed with the empirical inner product (3.5), we associate Y with the vector u in the illustration. We will then project Y onto $\sigma(X)$, the set of subsets of Ω_n induced by the variable X. We will thus have defined an object called *conditional expectation*, which is a well-defined concept to represent "looking at the variable Y through the eyes of the variable X." We will of course make this definition more precise in the course of this chapter.

2 The Projection Theorem

2.1 Closed Linear Subset

We address the general situation of a space endowed with an inner product (for example $\mathbb{R}^n$ or $L_2(\Omega_n)$). We will work on this notion of projection of a vector onto a subspace, denoted S. In the result that we will show, S must be a *linear* and *closed* subspace. Let us first define these notions.

A subset S is *linear* if it is itself a vector space. A necessary and sufficient condition is therefore that $u + v \in S$ for all u and $v \in S$, and $\alpha u \in S$ for every scalar $\alpha \in \mathbb{R}$. As an example, the only linear subspaces of $\mathbb{R}^3$ are: (a) $\mathbb{R}^3$, (b) all planes passing through the origin, (c) all straight lines passing through the origin, and (d) $\{0\}$ (?, ?).

In order to define a closed subset, it is necessary to refer to the notion of neighborhood. A *neighborhood space* is a set A and a family of subsets of A called *neighborhoods* such that, if V_1 and V_2 are neighborhoods and if $v \in V_1 \cap V_2$, then there exists a neighborhood V_3 such that

$$v \in V_3 \in V_1 \cap V_2.$$

If v is an element of the neighborhood V, then V is called a "neighborhood of v."

A subset A is *open* if it is equal to the union of a family of neighborhoods. It is therefore open if and only if every a of A lies in a neighborhood V, that is $a \in V \subset A$. A subset S of A is *closed* if every point a of A whose *all* neighborhoods intersect S is necessarily a point of S.

2.2 Statement and Proof of the Projection Theorem

Let A be a subset of $\mathbb{R}$. We have the following definitions:

- Least upper bound (notation: A for *least upper bound*): a real number such that $x \leqslant c$ for all $x \in A$ and if, for every $\epsilon > 0$, there exists an $x \in A$ such that $c - x < \epsilon$.
- Greatest lower bound (notation: A for *greatest lower bound*): a real number such that $x \geqslant c$ for all $x \in A$ and if, for every $\epsilon > 0$, there exists an $x \in A$ such that $x - c < \epsilon$.
- If $f : [a,b] \to \mathbb{R}$ is a continuous function on a bounded and closed interval $[a,b]$, the quantities
$$\{f(x) : x \in [a,b]\} \qquad \text{and} \qquad \{f(x) : x \in [a,b]\}$$
exist[1] and are called, respectively, *supremum* (sup) and *infimum* (inf).

Théorème 4.1. *Let S be a closed linear subspace of a vector space V endowed with an inner product $\langle\cdot,\cdot\rangle$ and a norm $\|\cdot\|$. Then, for every $y \in V$,*

1. *There exists a unique vector s^* in S such that*
$$\|y - s^*\| = \inf\{\|y - s\| : s \in S\} \equiv d \, ,$$
and we write $P_S y = s^*$.
2. *The vector $P_S y$ is the unique vector in S such that $\langle y - P_S y, s\rangle = 0$ for all $s \in S$;*
3. *The function P_S is a linear operator such that $P_S^2 = P_S$, called the projection operator on the subspace S.*

Proof. 1. For the first result, we must construct s^* and show that this object is unique. The idea is to approach this object from within S by a sequence $\{s_n\}$. To do this, note that the function $s \longmapsto \|y - s\|$ is continuous on S, which is itself closed. Consequently,
$$\inf_{s \in S} \|y - s\| =_{s \in S} \|y - s\| \equiv d \in \mathbb{R}$$
is such that $d \leqslant \|y - s\|$ for all s in S and, for every $\epsilon > 0$, there exists s in S such that $\|y - s\| \leqslant d + \epsilon$.

Consider a sequence $\{\epsilon_n\} \searrow 0$ (that is, for every m, there exists M_m such that $\|\epsilon_n\| \leqslant m^{-1}$ for all $n \geqslant M_m$), and consider a sequence $\{s_n\}$ such that $\|y - s_n\| \leqslant d + \epsilon_n$. We then have $\|y - s_n\| \longrightarrow d$, since, for every m and $n \geqslant M_n$, we have $\|y - s_n\| \leqslant d + \epsilon_n \leqslant d + m^{-1}$ and therefore $|\|y - s_n\| - d| \leqslant m^{-1}$.

We then construct $s^* = \lim_{n\to\infty} s_n$ and must show that this limit exists. It is sufficient to show that $\{s_n\}$ satisfies the Cauchy criterion[2]: Let $k \in \mathbb{N}$, then for all m and $n > M_k$ (to be constructed), we use the parallelogram identity (Exercise 3.3) in the following reasoning:
$$\begin{aligned}
\|s_n - s_m\|^2 &= \|(s_n - y) - (s_m - y)\|^2 \\
&= 2\|s_n - y\|^2 + 2\|s_m - y\|^2 - \|s_n + s_m - 2y\|^2 \\
&= 2\|s_n - y\|^2 + 2\|s_m - y\|^2 - 4\|\frac{s_n + s_m}{2} - y\|^2 \\
&\leqslant 2\left(\|s_n - y\|^2 - d^2\right) + 2\left(\|s_m - y\|^2 - d^2\right),
\end{aligned}$$

[1] Bishop (1967, Corollary, p. 89)

[2] ...

the two terms in parentheses converging to zero. The sequence is therefore Cauchy and s^* exists.
To prove its uniqueness, suppose $\tilde{s}$ such that $\|y - s^*\| = \|y - \tilde{s}\|$. Then, by another use of the parallelogram identity,

$$\begin{aligned}\|s^* - \tilde{s}\|^2 &= \|(s^* - y) - (\tilde{s} - y)\|^2 \\ &= 2\|s^* - y\|^2 + 2\|\tilde{s} - y\|^2 - 4\|\frac{s^* - \tilde{s}}{2} - y\|^2 \\ &= 4d^2 - 4\|\ldots\|^2\end{aligned}$$

the squared norm in this expression is greater than d^2 because $(s^* - \tilde{s})/2$ is in the linear subspace S. Consequently $\|s^* - \tilde{s}\|^2 \geqslant 0$, which shows that $s^* = \tilde{s}$, hence uniqueness.

2. For every s in S and for every real a we have

$$\langle y - P_S y + as, y - P_S y + as\rangle \geqslant d^2 = \langle y - P_S y, y - P_S y\rangle$$

and therefore

$$2a\langle s, y - P_S y\rangle + a^2\|s\|^2 \geqslant 0 \ .$$

If we take $a = -(\langle s, y - P_S y\rangle)/\|s\|^2$ this leads to $(-\langle s, y - P_S y\rangle^2) \geqslant 0$, thus $\langle s, y - P_S y\rangle = 0$.

3. Exercise.

□

Note that $P_S y$ is read "P_S of y" or "P_S applied to y." Indeed, P_S is a *mapping* that takes a vector y and transforms it into another vector denoted $P_S y$. We could therefore have written $P_S(y)$ but the parentheses are superfluous.

Subsequently, we will denote by Q the projection on the space orthogonal to S, that is, $Qu = u - P_S u$.

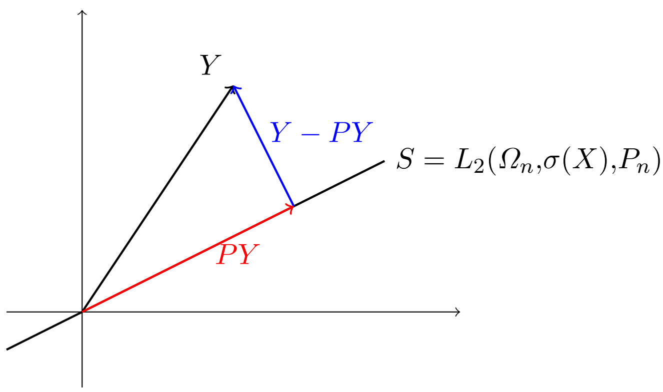
3 Conditional Expectation

In the previous chapter we defined the notion of vector space (Definition 3.1) and that of inner product (Definition 3.2) in complete generality. In practice, we have seen that the column vectors of a numerical database are vectors of the Euclidean space $\mathbb{R}^n$ endowed with the empirical Euclidean product (3.4). This observation allowed us to induce an inner product (3.5) on the variables themselves. The variables are functions and are also vectors of a vector space denoted $L_2(\Omega_n)$ (Definition 3.4). Indeed, if the two variables X and Y are in $L_2(\Omega_n)$, their inner product is defined by $\langle X, Y\rangle = \mathbb{E}(XY)$.

In the vector space of variables, *conditioning on* X means applying the orthogonal projection of variables onto the closed convex space $L_2(\Omega_n,\sigma(X),P_n)$ (see Exercise 4.5). This projection exists and is unique thanks to the result of the projection theorem. Indeed, by Theorem 4.1, there exists a unique mapping P such that PY ("P of Y", or "P applied to Y") is the orthogonal projection of Y onto $L_2(\Omega_n,\sigma(X),P_n)$. In other words, the mapping P is such that

$$(Y - PY) \perp \xi$$

for every variable ξ lying in the projection subspace $L_2(\Omega_n,\sigma(X),P_n)$, as in this drawing:



Perpendicularity is a condition on the inner product that can also be written as

$$\langle Y - PY, \xi \rangle = 0$$

for every variable $\xi \in L_2(\varOmega_n,\sigma(X),P_n)$, or, by the definition of the inner product that we have recalled,

$$\mathbb{E}(Y\xi) = \mathbb{E}((PY)\xi) \ .$$

The variable PY thus defined is called the *expectation of Y conditional on X*. In Chapter 1, we already encountered probabilities and distributions conditional on the values of a variable (see, for example, Figure 2.4). The novelty here is that we have defined an expectation conditional on a variable X, hence on a function whose value has not been fixed. The conditional expectation is therefore a function of the values of X that we have called the image of X, $\Im(X)$. We summarize the new definition in the following paragraph.

Définition 4.1. *Let X be a variable of $L_2(\varOmega_n,\mathcal{P}(\varOmega_n),P_n)$. The expectation conditional on X is the operator*

$$\begin{aligned} \mathbb{E}(\bullet|\sigma(X)) &: L_2(\varOmega_n,\mathcal{P}(\varOmega_n),P_n) \longrightarrow L_2(\varOmega_n,\sigma(X),P_n) \\ &: Y \longrightarrow \mathbb{E}(Y|\sigma(X)) := P_X Y \end{aligned}$$

where P_X denotes the orthogonal projection onto $L_2(\varOmega_n,\sigma(X),P_n)$.

When the context is clear, the notation is sometimes simplified by writing $\mathbb{E}(Y|X)$ instead of $\mathbb{E}(Y|\sigma(X))$. However, it must remain clear that the operation consists of projecting onto the functions that are constant on the partition induced by X. In this definition we have used the term *operator* to denote a function between two spaces of functions. The result of the conditional expectation is indeed a function of $L_2(\varOmega_n,\sigma(X),P_n)$. The conditional expectation $\mathbb{E}(Y|X)$ is therefore itself a variable, that is, a function $\varOmega_n \longrightarrow \mathbb{R}$ whose partitions are included in those of X. In other words, the functions $\mathbb{E}(Y|X)$ are constant on all the preimages (blocks $X^{-}[\Im(X)]$) of X.

Exemple

To illustrate this point, let us take again the database of members of the European Parliament. We are interested in the two variables *Age* (Y) and *Nationality* (X). The distribution function of *Age* is presented in Figure 4.1 and the distribution conditional on four nationalities is presented in Figure 4.2. The conditional expectation

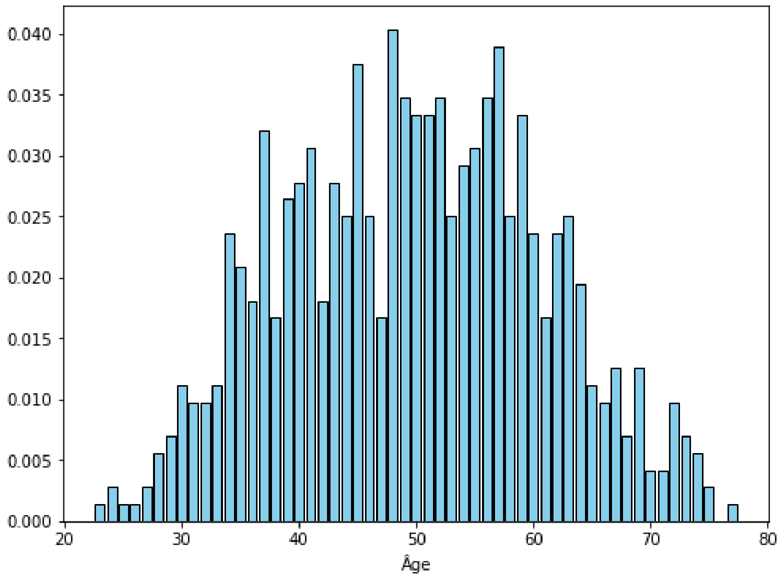


FIG. 4.1: Distribution of the age of members of the European Parliament

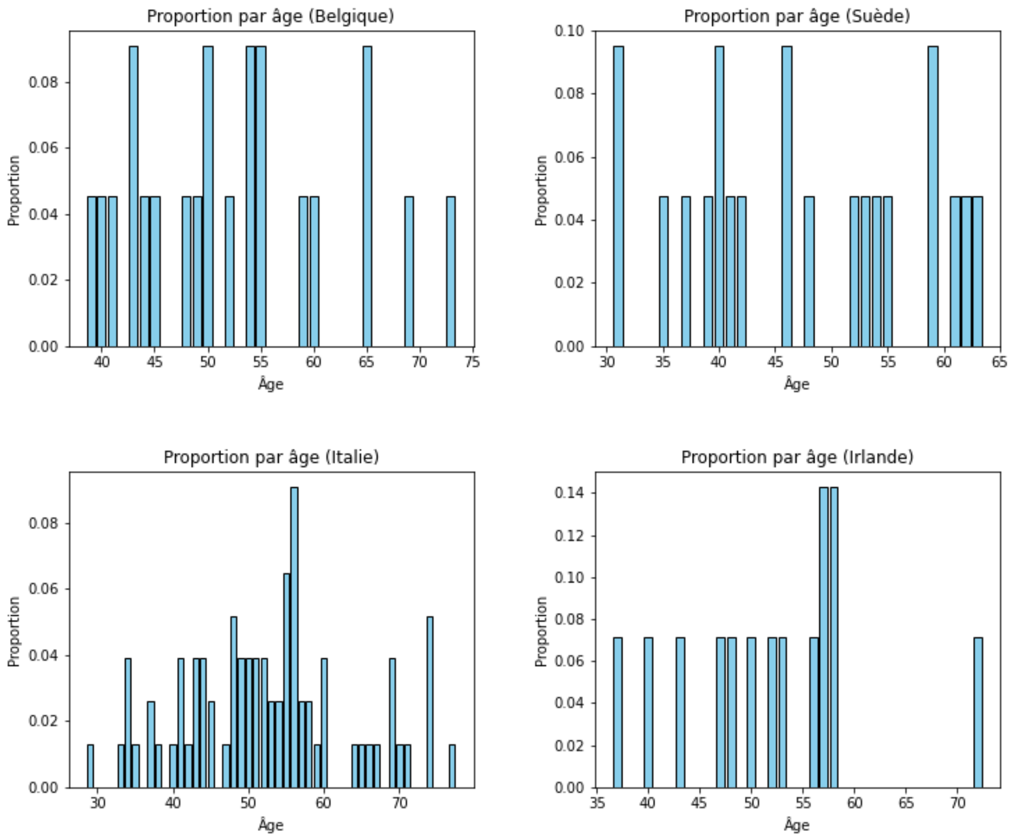


FIG. 4.2: Distribution of age conditional on the nationality of members of the European Parliament, for four nationalities among the 27.

$\mathbb{E}(Age|Nationality)$ is a variable, hence a function on the space of labels Ω_{720} of the members of the European Parliament. This means that for each member ω we have a value

$$\mathbb{E}(Age|Nationality)(\omega) \ .$$

The definition of conditional expectation indicates that this value is identical for all members of the same nationality (because the function is constant on the preimages of *Nationality*, which we have denoted $\sigma(X)$). We therefore read Figure 4.2 as follows: each ω has a nationality, and conditional on this fixed nationality we consider the expectation of age conditional on that nationality. The expectation is calculated as a simple inner product with ι_n,. This result gives the following function:

$$\begin{aligned}\mathbb{E}(Age|Nationality) : \Omega_{720} &\longrightarrow \mathbb{R}\\ : \omega &\longrightarrow \begin{cases} 0 & \text{if } Nationality(\omega) = \text{“Belgium”}\\ 0 & \text{if } Nationality(\omega) = \text{“Sweden”}\\ 0 & \text{if } Nationality(\omega) = \text{“Italy”}\\ \vdots & \vdots \end{cases}\end{aligned}$$

The complete list of conditional means is presented in Figure 4.3. We clearly see that this function is constant by nationality, that is, constant on the labels sharing the same image through X.

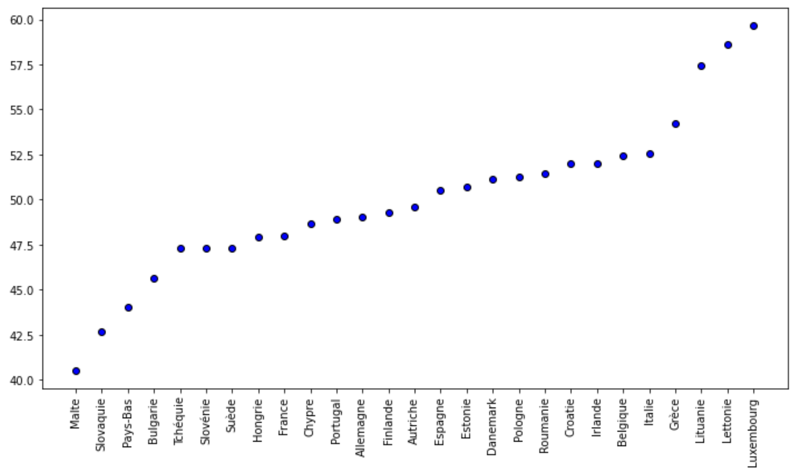


FIG. 4.3: Expectation of age conditional on the nationality of members of the European Parliament

4 Fundamental Mean–Variance Decomposition

Let us first define the constant variable.

DÉFINITION 4.2 (Constant Variable). *Let Ω_n be a set of labels. We define the constant variable, denoted $\mathbb{1}$, as the function*

$$\mathbb{1} : \Omega_n \to \{1\} : \omega \to \mathbb{1}(\omega) = 1$$

that is, the function which associates the value 1 to every label of Ω_n. We necessarily have $\Im(\mathbb{1}) = \{1\}$.

The constant variable $\mathbb{1}$ has the particularity that all its values are equal to 1. The vector of values, denoted ι_n, is therefore the bisector of the Cartesian axes in $\mathbb{R}^n$, which means that ι_n is equidistant from all the axes.

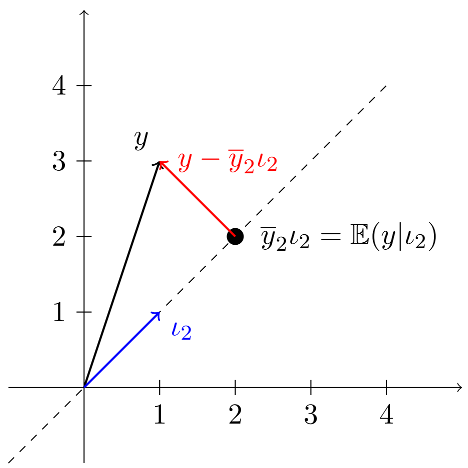


FIG. 4.4: In dimension 2, orthogonal projection of the vector $y = (1{,}3)^\top$ onto the diagonal of direction $\iota_2 = (1{,}1)^\top$. This projection is the vector with coordinates $\overline{y}\iota_2 = (\overline{y}_2, \overline{y}_2)^\top = (2{,}2)^\top$, since the arithmetic mean of (1,3) equals (2,2). The length of the orthogonal vector $y - \overline{y}_2\iota_2$, in red on the figure, represents the distance from the vector y to the diagonal. This length is the standard deviation of y.

By the projection theorem, any variable Y can be decomposed into the sum of two perpendicular vectors (functions)

$$Y = \mathbb{E}(Y|\mathbb{1}) + \{Y - \mathbb{E}(Y|\mathbb{1})\} \; . \tag{4.1}$$

Figure 4.4 illustrates this decomposition and the reasoning below on the image of Y (denoted y) in dimension $n = 2$. The first term of this decomposition is the orthogonal projection of Y onto the subspace $L_2(\Omega_n,\sigma(\mathbb{1}),P_n)$ consisting of the functions that are constant on Ω_n (Exercise A.5). The functions of $L_2(\Omega_n,\sigma(\mathbb{1}),P_n)$ are therefore written $c\mathbb{1}$ for some real number c. The orthogonal projection is thus such that

$$\{Y - \mathbb{E}(Y|\mathbb{1})\} \perp (c\mathbb{1})$$

for every real number c. This condition can be written using the inner product

$$\langle Y - \mathbb{E}(Y|\mathbb{1}), c\mathbb{1}\rangle = 0$$

or, by dividing by c assumed nonzero and rearranging the terms according to the rules of calculation of the inner product,

$$\langle Y, \mathbb{1}\rangle = \langle \mathbb{E}(Y|\mathbb{1}), \mathbb{1}\rangle \; .$$

The projection $\mathbb{E}(Y|\mathbb{1})$ is a function of $L_2(\Omega_n,\sigma(\mathbb{1}),P_n)$, which is therefore written $k\mathbb{1}$ for some real number k, allowing us to continue the calculation using the definition of the inner product (3.5):

$$\frac{1}{n}\sum_{\omega\in\Omega_n} Y(\omega) = k \; .$$

The first term of the decomposition (4.1) is therefore the variable *mean of* Y

$$\mathbb{E}(Y|\mathbb{1}) = \overline{Y}_n \, \mathbb{1}$$

where we have denoted $\overline{Y}_n = \sum_\omega Y(\omega)/n$.

Since the decomposition (4.1) is orthogonal, the Projection Theorem also implies an equality concerning the length of the vectors:

$$\|Y\|^2 = \|\mathbb{E}(Y|\mathbb{1})\|^2 + \|Y - \mathbb{E}(Y|\mathbb{1})\|^2 . \tag{4.2}$$

The first term is simply equal to $\overline{Y}_n^2$, which means that the length of the first term is $\|\overline{Y}_n\|$. The second term is

$$\begin{aligned} \|Y - \mathbb{E}(Y|\mathbb{1})\|^2 &= \mathbb{E}(Y - \mathbb{E}(Y|\mathbb{1}))^2 \\ &= \mathbb{E}(Y - \overline{Y}_n \mathbb{1})^2 \\ &= \frac{1}{n} \sum_{\omega} (Y(\omega) - \overline{Y}_n)^2 \end{aligned}$$

which is called the *variance of* Y and denoted $V(Y)$. Its square root is the length of the residual vector $(Y - \mathbb{E}(Y|\mathbb{1}))$ of the decomposition and bears the name *standard deviation*. By the identity (4.2) we immediately see that the variance of Y is also equal to

$$\frac{1}{n} \sum_{\omega} Y(\omega)^2 - \overline{Y}_n^2 .$$

As illustrated in Figure 4.4, the mean variable (or the vector of means) is a projection that can be interpreted as an approximation of Y. In statistics and econometrics, an approximation is a simplification of Y allowing one to have an idea of Y that is more or less rough without reading all its details. The mean variable is a simple approximation, because it is of the form $k\mathbb{1}$. Is this approximation close to or far from Y? In other words, is it a good approximation, capturing the essential part, or on the contrary an excessive simplification of Y? We can use Figure 4.4 to answer this question by noting that it depends on the distance between Y and its projection space. Since the standard deviation measures this distance, and by virtue of the decomposition (4.2), we use the ratio

$$r^2 = \frac{\|\mathbb{E}(Y|\mathbb{1})\|^2}{\|Y\|} = 1 - \frac{\|Y - \mathbb{E}(Y|\mathbb{1})\|^2}{\|Y\|}$$

to quantify the quality of the approximation by the mean. As the projection is orthogonal, it is the cosine of the angle between Y and its projection space (see Figure 4.4). This ratio is therefore always between 0 and 1. If it is zero, it means that Y is perpendicular to the projection space. In that case the mean is a very poor approximation. If it equals 1, then the angle is zero. This means that Y belongs to the projection space. The mean is therefore a perfect approximation of the variable, since it is itself a constant function of $L_2(\Omega_n,\sigma(\mathbb{1}),P_n)$.

5 Generalization: Decomposition by Explanatory Variable

We now examine how to generalize the decomposition of the previous section when the variable Y is projected onto a subspace induced by another variable X which is not necessarily constant. In this case, the decomposition

$$Y = \mathbb{E}(Y|X) + \{Y - \mathbb{E}(Y|X)\} \tag{4.3}$$

is also a decomposition into two perpendicular terms by virtue of the Projection Theorem. It can be shown (Exercise 4.4) that the first term of this decomposition is a function that is constant on the preimages of X. It is an approximation of the variable Y by the variable X in the sense we discussed in the previous section. This approximation is referred to in common language as "the part of Y that is explained by X," and X is called an *explanatory variable* of Y. This decomposition will be studied in more detail in the last part of the course devoted to econometrics.

Already we can ask whether $\mathbb{E}(Y|X)$ is a satisfactory approximation of Y. To measure this, we again use the Projection Theorem to write

$$\|Y\|^2 = \|\mathbb{E}(Y|X)\|^2 + \|\{Y - \mathbb{E}(Y|X)\}\|^2 \,. \tag{4.4}$$

As before, we can start from this equality to construct a measure of the quality of approximation of the orthogonal projection. The usual practice follows another path and introduces the variance in the decomposition (4.3). Let us first consider the variance of each term of (4.3):

- The variance of Y is the square of the norm of

$$Y - \langle Y, \mathbb{1}\rangle \mathbb{1} \,.$$

- The variance of $\mathbb{E}(Y|X)$ is the square of the norm of

$$\mathbb{E}(Y|X) - \langle \mathbb{E}(Y|X), \mathbb{1}\rangle \mathbb{1} = \mathbb{E}(Y|X) - \langle Y, \mathbb{1}\rangle \mathbb{1}$$

  which is shown in Exercise 4.6.

- The last term of (4.3) has zero mean since, using the result of Exercise 4.6,

$$\begin{aligned}\langle Y - \mathbb{E}(Y|X), \mathbb{1}\rangle &= \langle Y, \mathbb{1}\rangle - \langle \mathbb{E}(Y|X), \mathbb{1}\rangle \\ &= \langle Y, \mathbb{1}\rangle - \langle Y, \mathbb{1}\rangle \\ &= 0\end{aligned}$$

  Therefore the variance of this term is the square of its norm.

Starting from (4.3) and from these three observations, we can therefore write

$$\begin{aligned}Y - \langle Y, \mathbb{1}\rangle \mathbb{1} &= \mathbb{E}(Y|X) - \langle Y, \mathbb{1}\rangle \mathbb{1} + \{Y - \mathbb{E}(Y|X)\} \\ &= \mathbb{E}(Y|X) - \langle \mathbb{E}(Y|X), \mathbb{1}\rangle \mathbb{1} + \{Y - \mathbb{E}(Y|X)\}\end{aligned}$$

and by applying the Projection Theorem, the equality between the squares of the norms implies

$$V(Y) = V[\mathbb{E}(Y|X)] + V[Y - \mathbb{E}(Y|X)] \,.$$

This equality decomposes the variance of Y into two parts. The first part is the variance of the approximation, sometimes called "variance of Y explained by X," whereas the second part is the residual variance or the variance not explained by X. The perspective is then to measure the quality of approximation by the amount of variance explained by X. Thus the usual measure of the quality of approximation of Y by the conditional expectation is defined by

$$R^2 := \frac{V[\mathbb{E}(Y|X)]}{V(Y)} \,.$$

This quantity is sometimes called the *coefficient of determination* in the literature.

6 Correlation and Partial Correlation

We conclude this chapter with the definition of some computational tools. Let X, Y, and Z be three variables of $L_2(\Omega_n)$. We call *covariance between X and Y* the real number

$$C(X,Y) := \langle X - \mathbb{E}(X|\mathbb{1}), Y - \mathbb{E}(X|\mathbb{1})\rangle$$

which is therefore the inner product of X and Y centered on their mean. It is obvious that $V(X) = C(X,X)$. The following normalization by the product of the standard deviations is called the correlation between X and Y:

$$\rho_{XY} := \frac{C(X,Y)}{\|X - \mathbb{E}(X|\mathbb{1})\|\|Y - \mathbb{E}(X|\mathbb{1})\|} .$$

It is a number between -1 and 1 (Exercise 4.9). Finally, we call *partial correlation of X and Y given Z* the number $\rho_{XY \cdot Z}$ given by the correlation between $X - \mathbb{E}(X|Z)$ and $Y - \mathbb{E}(Y|Z)$. It is therefore the correlation between two residuals of an orthogonal projection onto $L_2(\Omega_n, \sigma(Z), P_n)$. Its interpretation is as follows: once X and Y have been approximated by the variable Z, is there still a correlation, a link, between these variables? If the answer is no, then we can conclude that Z suffices to approximate (or explain) X and Y. The notion of partial correlation was introduced by the famous psychologist Charles Spearman (1843–1945). His objective was to determine whether Z is a common cause of the variables X and Y. Being a common cause is a condition of independence between X and Y once these variables have been stripped of their part approximated (explained) by Z. The condition is then written

$$X \perp\!\!\!\perp Y|Z$$

and under this condition the partial correlation $\rho_{XY \cdot Z}$ is equal to zero (Exercise 4.10).

7 Exercises

Exercise 4.1. In a vector space $\mathcal{E}$ endowed with an inner product, when the inner product between two vectors u and v is zero, that is $\langle u, v\rangle = 0$, we say that u and v are *orthogonal*, which we write $u \perp v$. We also denote by $u^\perp$ the set of all $v \in \mathcal{E}$ that are orthogonal to u. Show that (i) $u^\perp$ is a vector subspace of $\mathcal{E}$ and that (ii) this set is closed.

Exercise 4.2. If S is a subspace of $\mathcal{E}$, we denote by $S^\perp$ the set of all $v \in \mathcal{E}$ that are orthogonal to every $u \in S$. Show that $S \cap S^\perp = \{0\}$.

Exercise 4.3 (Law of Iterated Expectations). Let X, Y be two variables of $L_2(\Omega_n)$. Show that $\mathbb{E}(\mathbb{E}(Y|X)|\mathbb{1}) = \mathbb{E}(Y|\mathbb{1})$.

Exercise 4.4. Let X, Y be two variables of $L_2(\Omega_n)$. Suppose that $\Im(X)$ has cardinality $J \leqslant n$ and denote by $B_1 \dot\cup B_2 \dot\cup \cdots \dot\cup B_J$ the partition induced by X. Show that

$$\mathbb{E}(Y|X) = \sum_{j=1}^{J} \frac{1}{|B_j|} \sum_{\omega \in B_j} Y(\omega) \mathbb{1}_{B_j}$$

Exercise 4.5. In a vector space, recall that a set is *closed* if and only if it contains the limit of every convergent sequence of vectors. Prove that the vector subspace $L_2(\Omega_n, \sigma(X), P_n)$ is a closed set of $L_2(\Omega_n, \mathcal{P}(\Omega_n), P_n)$.

Exercise 4.6. Let X, Y, and Z be three variables of $L_2(\Omega_n)$. Show that $\langle \mathbb{E}(Y|X), Z\rangle = \langle Y, \mathbb{E}(Z|X)\rangle$. Deduce that $\langle \mathbb{E}(Y|X), \mathbb{1}\rangle = \overline{Y}_n$.

Exercise 4.7. Show that the vector subspace $L_2(\Omega_n,\sigma(\mathbb{1}),P_n)$ consists of the functions that are constant on Ω_n.

Exercise 4.8. If $X = \mathbb{1}$ in (4.1), show that $R^2 = 0$.

Exercise 4.9. Show that $-1 \leqslant \rho_{XY} \leqslant 1$ (Hint: use the Cauchy–Schwarz inequality).

Exercise 4.10 (Correlation and Causality). In this exercise we attempt to clarify what is meant by the common expression *"correlation is not proof of causation."* Consider three variables of $L_2(\Omega_n)$ denoted X, Y, Z, the last of which has zero mean, that is $\langle Z,\mathbb{1}\rangle = 0$.

(i) Show that

$$C(X - \mathbb{E}(X|Z),Y - \mathbb{E}(Y|Z)) = C(X,Y) - C(\mathbb{E}(X|Z),\mathbb{E}(Y|Z)) \; .$$

(ii) Add the hypothesis that Z is the unique common cause of X and Y, which is formulated by the conditional independence $X \perp\!\!\!\perp Y|Z$. In this case, and using the previous identity, show that

$$C(X - \mathbb{E}(X|Z),Y - \mathbb{E}(Y|Z)) = 0.$$

(iii) Using the previous result and maintaining all the hypotheses stated so far, construct an example showing that $C(X,Y) > 0$.

With this sequence of results, you have shown that if Z is the unique common cause of X and Y, the partial correlation is zero, but the covariance between X and Y is not zero, despite the fact that these variables have no causal link. This is the argument advanced by Spearman to use partial correlation in the verification of causality.

References

Birkhoff, G. (1935). On the structure of abstract algebras. *Proc. Camb. Philos. Soc.*, *31*, 433–454.

Bishop, E. (1967). *Foundations of constructive analysis.* New York: McGraw-Hill.

Borel, E. (1909). *Éléments de la Théorie des Probabilités.* Librairie Scientifique A. Hermann & Fils, Paris.

Borel, É. (1949). Probabilité et certitude. *Dialectica*, *3*(1/2), 24–27.

Chow, Y. S., & Teicher, H. (1997). *Probability theory: Independence, interchangeability, martingales* (3rd ed.). New York: Springer.

Conway, J. H., & Guy, R. K. (1996). *The Book of Numbers.* New York: Copernicus, An Imprint of Springer-Verlag.

Cramér, H. (1946). *Mathematical methods of statistics.* Princeton: Princeton University Press.

Davidson, J. (1994). *Stochastic limit theory.* Oxford: Oxford University Press.

Delcomminette, S. (2018). *Aristote et la nécessité. Tradition de la pensée classique.* Paris: Vrin.

Eurostat. (2025). *Nomenclature of Territorial Units for Statistics (NUTS).* https://ec.europa.eu/eurostat/web/nuts.

Fisher, R. (1922). On the mathematical foundations of theoretical statistics. *Philos. Trans. R. Soc. Lond. A*, *222*, 309–368.

Fisher, R. (1970). *Statistical methods for research workers* (14th ed.). Edinburgh: Oliver and Boyd.

Frede, D. (1985). The sea-battle reconsidered: A defense of the traditional interpretation. *Oxford Stud. Anc. Philos.*, *3*, 31–87.

Frege, G. (1894). Rezension von E. Husserl, Philosophie der Arithmetik. *Z. Philos. Philos. Krit.*, *103*, 313–332.

Frisch, R. (1933). Editor's note. *Econometrica*, *1*(1), 1–4.

Frisch, R. (1934). Statistical confluence analysis by means of complete regression systems. *Skr. utg. Det Norske Vidensk.-Akad. i Oslo, II. Hist.-Filos. Kl.*, 1–117.

Gauss, C. F. (1821). Theoria combinationis observationum erroribus minimis obnoxiae. *Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores*, *5*, 33–76.

Haavelmo, T. (1944). The probability approach in econometrics. *Econometrica*, *12*(0), 1–115.

Halmos, P. (2018). *Naive Set Theory.* New York: Dover Publciations, Inc.

Hilbert, D. (1900). Mathematische Probleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, *1900*, 253-297. Retrieved from http://eudml.org/doc/58479

Hilbert, D. (1902). Mathematical problems. lecture delivery before the international congress of mathematics at paris in 1900. *Bulletin of the American Mathematical Society*, *8*, 437-479.

Isidorus Hispalensis. (c. 636). *Etymologiae.* Monasterii Sancti Galli.

Karr, A. (1993). *Probability.* Springer.

Khinchin, A. (1949). *Mathematical foundations of statistical mechanics.* New York: Dover Publications.

Kolmogorov, A. (1933). *Grundbegriffe der wahrscheinlichkeitsrechnung.* Berlin: Springer.

Kolmogorov, A. (1956). *Foundations of the theory of probability.* Chelsea Publishing Company, New Yor.

Kolmogorov, A. (2006). Contemporary Debates on the Nature of Mathematics. *Problems of Information Transmission*, *42*(4), 379–389.

Koopmans, T. (1937). *Linear regression analysis of economic time series.* Haarlem: De Erven F. Bohn.

Lazarsfeld, P. (1959). Latent structure analysis. In S. Koch (Ed.), *Psychology: A study of a science. Volume 5: The process areas, the person and societal processes* (pp. 476–543). New York: McGraw-Hill.

Li, D. (2025). A search for the definitions of logical (anti-)psychologism. *Synthese*, *205*(4), 1–19.

Lindley, D. V. (1965). *Introduction to probability and statistics from a bayesian viewpoint* (Vol. 1). Cambridge: Cambridge University Press.

Lord, F., & Novick, M. (1968). *Statistical theories of mental test scores.* Reading, MA: Addison-Wesley.

Lord, F. M., & Novick, M. R. (1968). *Statistical theories of mental test scores.* Reading, MA: Addison-Wesley.

Neyman, J. (1937). Outline of a theory of statistical estimation based on the classical theory of probability. *Philos. Trans. R. Soc. Lond. A*, *236*(767), 333–380.

Novick, M. R. (1969). The axioms and principal results of classical test theory. *Journal of Mathematical Psychology*, *6*, 1–18.

Peano, G. (1894). Notations de logique mathématique: Introduction au formulaire de mathématique. *Riv. Mat. Univ. Parma*, *4*(1), 1–15.

Quetelet, A. (1848). Sur la statistique morale et les principes qui doivent en former la base. *Mem. Acad. r. Sci. Lett. Beaux-Arts Belg.*, *21*, 1–68.

Rényi, A. (1962). *Foundations of probability.* San Francisco: Holden-Day.

Rényi, A. (1970). *Probability theory.* Amsterdam: North-Holland.

Rowe, D. E. (2022). Brouwer and Hausdorff: On reassessing the foundations crisis. *Sci. Context.*, *35*(4), 395–413.

Appendix A

Additional Exercises

Additional Exercises to chapter 2

Exercise A.1. A survey observes consumer purchasing behavior and their exposure to an advertising channel (TV or Social Media). The data are summarized in table A.1. Check for independence.

	Buy (A)	Do not Buy (A^c)	Total
Social Medias (B^c)	20	40	60
Total	50	50	100

Table A.1: joint distribution of purchasing behavior and exposure to an advertising channel

Exercise A.2. A company measures customer satisfaction (Satisfied or Unsatisfied) based on delivery time (Fast or Slow). The results are given in table A.2. Are the variables independent?

	Satisfied	Unsatisfied	Total
Fast delivery	40	10	50
Slow delivery	20	30	50
Total	60	40	100

Table A.2: Joint distribution of customer satisfaction and delivery time

Exercise A.3. A store wants to determine if the type of product purchased (T) is independent of the customers' age group (A). The product types are: Technology (T1), Fashion (T2), and Home (T3). The age groups are: 18-30 years (A1) and 31-50 years (A2). The sales data are given in table A.3.

	A1	A2	Total
T1	30	20	50
T2	40	30	70
T3	30	50	80
Total	100	100	200

Table A.3: Joint distribution of product types purchased and customers' age group

Additional Exercises to chapter 3

Exercise A.4. Calculate:

(i) $\langle \mathbb{1}, \mathbb{1} \rangle$

Exercise A.5. Show that the subspace $L_2(\Omega_n, \sigma(\mathbb{1}), P_n)$ consists of the constant functions on Ω_n.

Exercise A.6. Let $x := \{x_n\}$ and $y := \{y_n\}$ be two real numbers. Show that the operation $\cdot * \cdot : \mathbf{R}^2 \to \mathbf{R}$ defined by

$$x * y = \{x_n\} * \{y_n\} \equiv \{x_n y_n\}$$

cannot define the multiplication between real numbers since it is not well defined for all couple of real numbers. Then, show that the operation $(\cdot;\cdot) : \mathbf{R}^2 \to \mathbf{R}$ defined by

$$(x; y) = (\{x_n\}; \{y_n\}) \equiv \{x_{2kn}y_{2kn}\},$$

where $k \equiv \max\{K_x, K_y\}$ can define the multiplication between real numbers since $\{x_{2kn}y_{2kn}\}$ is a real number if both $\{x_n\}$ and $\{y_n\}$ are and since it is coherent with the multiplication between rational numbers.

Exercise A.7. The table A.4 indicates the income and the experience of 10 individuals. Answer the following questions:

- Calculate $\langle X_1, X_2\rangle$ and $\langle X_1, \mathbb{1}\rangle$
- Give an example of a variable in $L_2(\Omega_n, \sigma(X_2), P_n)$

Individual	**X_1: Income (in thousand USD)**	**X_2: Years of Experience**
ω_1	55	5
ω_2	43	3
ω_3	70	10
ω_4	35	5
ω_5	80	3
ω_6	60	10
ω_7	95	3
ω_8	50	5
ω_9	45	10
ω_{10}	65	3

Table A.4: Database of the income and the experience in years for 10 individuals

Additional Exercises to chapter 4

Exercise A.8 (Markov's Inequality). Let X be a random variable on a label set Ω_n. Suppose that X takes positive values, i.e., $\Im(X) \subseteq \mathbb{R}^+$. Show that, for all $a > 0$, the inequality

$$P_X(X \geq a) \leqslant \frac{\bar{X}_n}{a}$$

always holds.

Exercise A.9. Let X be a numerical random variable on the label set Ω_n. Show that

$$E(X|\mathbb{1}) = \sum_{x\in\Im(X)} xP_X(X = x)\ .$$

Exercise A.10. Let X be a random variable on the label set Ω_n whose image is given by the natural numbers, i.e., $\Im(X) \subseteq \{1,2,3,\ldots\}$. Show that

$$E(X|\mathbb{1}) = \sum_{x=1}^{\infty} P_X(X \geqslant x)\ .$$

Exercise A.11. We have shown that the projection of $\mathbb{1}$ on the subspace $L_2(\Omega_n,\sigma(X),P_n)$ did not modified the constant function. It is then in $L_2(\Omega_n,\sigma(X),P_n)$ and hence orthogonal to the space orthogonal to $L_2(\Omega_n,\sigma(X),P_n)$. By the same argument, we can argue that $\mathbb{1} \in L_2(\Omega_n,\sigma(Y - P_XY),P_n)$. Is is contradictory with the first argument? Justify.

Exercise A.12. Consider a vector space H equipped with a scalar product $\langle .,. \rangle$ and assume that every cauchy sequences converge in H. Let S be a closed subspace of H and P the orthogonal projection onto S. Show that $\langle Px,y \rangle = \langle x,Py \rangle$.

Exercise A.13. Consider the database A.5 and answer the following questions:

- What is the label set (as a mathematical set and in words)?
- Compute $\mathbb{E}(\text{Annual Salary}|\mathbb{1})$ and r^2. What is the geometrical interpretation of r^2?
- Compute $\mathbb{E}(\text{Annual Salary}|\sigma(\text{Year, Gender}))$ and the coefficient of determination R^2. Comment.

ID	Year	Annual salary (k€)	Gender	Country
A01	2022	45	Female	Germany
A01	2023	47	Female	Germany
A01	2024	50	Female	Germany
A02	2022	38	Female	Spain
A02	2023	41	Female	Spain
A02	2024	44	Female	Spain
A03	2022	62	Female	France
A03	2023	65	Female	France
A03	2024	67	Female	France
A04	2022	55	Male	Italy
A04	2023	57	Male	Italy
A04	2024	59	Male	Italy
A05	2022	40	Female	Spain
A05	2023	42	Female	Spain
A05	2024	45	Female	Spain
A06	2022	70	Male	Germany
A06	2023	73	Male	Germany
A06	2024	75	Male	Germany
A07	2022	52	Female	France
A07	2023	54	Female	France
A07	2024	57	Female	France
A08	2022	60	Male	Italy
A08	2023	61	Male	Italy
A08	2024	63	Male	Italy
A09	2022	47	Female	Spain
A09	2023	49	Female	Spain
A09	2024	51	Female	Spain
A10	2022	66	Male	Germany
A10	2023	68	Male	Germany
A10	2024	70	Male	Germany
A11	2022	58	Male	France
A11	2023	60	Male	France
A11	2024	62	Male	France
A12	2022	44	Female	Italy
A12	2023	46	Female	Italy
A12	2024	47	Female	Italy

Table A.5: Dataset: wage, year, sex and country of residence for 12 individuals

Additional exercises to chapter ??

Exercise A.14. Calculate:

(i) 2!

(ii) 3!

(iii) 4!

(iv) 5!

(v) 6!

(vi) $\binom{4}{2}$

(vii) $\binom{6}{3}$

(viii) $\binom{5}{2}$

Exercise A.15. Solve:

(i) A marketing agency sends a promotional email to 8 subscribers. It is expected that, for each subscriber, there is a 70% chance of opening the email. For a performance study, how many scenarios are possible if exactly 5 subscribers open the email?

(ii) A consulting firm wants to form a consumer panel to test a product. It has a group of 15 people and needs to choose 4 for an in-depth study. How many different combinations can it form to create this panel of 4 people?

Exercise A.16. Solve:

(i) We flip a coin 4 times, and get 1 heads and 3 tails. What is the likelihood that this coin is fair?

(ii) We flip a coin 6 times and get 1 heads and 5 tails. What is the likelihood that this coin is fair?

Exercise A.17. We flip a coin 6 times and get 1 heads and 5 tails. We don't know if the coin is fair. Several people have different hypotheses about the situation.

a. John thinks the coin is fair. What is the likelihood?

b. Marie thinks the coin favors tails 25 out of 100 times. What is the likelihood?

c. Ingrid thinks the coin favors heads 25 out of 100 times. What is the likelihood?

d. Calculate the likelihood for 2 other priors on the coin.

e. Draw the graph where the prior value is on the x-axis and the likelihood is on the y-axis

Exercise A.18. A coffee machine produces a correct espresso if it contains between 2.99cl and 3.01cl of coffee. 10 espressos are made with a machine, and the data obtained are given in table A.6

a. We assume that every espresso machine must meet this standard for all the coffees it produces. What is the likelihood that the machine is working correctly?

b. We assume that a machine is considered to be working if it meets the standard in 99 out of 100 cases. What is the likelihood that the machine is working correctly?

c. We assume that a machine is considered to be working if it meets the standard in 8 out of 10 cases. What is the likelihood that the machine is working correctly?

Exercise A.19. Solve:

(i) We flip a coin 100 times and get 40 heads and 60 tails. What is the likelihood that this coin is fair? Use the normal approximation.

(ii) We flip a coin 120 times and get 40 heads and 80 tails. What is the likelihood that this coin

Trials	cl	compliance
1	3.005	1
2	3.012	0
3	2.994	1
4	3.008	1
5	3.002	1
6	3.000	1
7	2.994	1
8	3.018	0
9	3.004	1
10	2.999	1

Table A.6: data table of the size and the compliance of the 10 expressos

is fair? Use the normal approximation.

(iii) We flip a coin 250 times and get 50 heads and 200 tails. What is the likelihood that this coin is fair? Use the normal approximation.

(iv) A company is launching a new product and sends samples to 500 people. It expects that 40% of the people will prefer this product over others. After the survey, 215 people indicate that they prefer the new product. Use the normal approximation to determine if the observed preference percentage matches the company's initial forecast.

Exercise A.20. We roll a 6-sided die and get: 3, 4, 2, 3, 6, 1. What is the partition of the set of labels induced by this variable? What would be the partition corresponding to the bet that the die is fair? How many point movements are needed (at minimum) to reach the bet?

Exercise A.21. An online store observes its sales distributed by hourly intervals over a 24-hour day. The observed sales for 300 transactions are shown in table A.7. Calculate the contingent distance and the quadratic distance to the bet.

hourly intervals	Observations	bet
Morning (6h-12h)	100	0.30
Afternoon (12h-18h)	120	0.40
Evening (18h-24h)	80	0.30

Table A.7: Observed sales and bet

Exercise A.22. A company tests three versions of an advertisement (red, blue, green) with 200 consumers. The empirical choices of the consumers are given in table A.8. Calculate the contingent distance and the quadratic distance to the bet.

Color	Observations	Bet
Red	90	0.50
Blue	70	0.30
Green	40	0.20

Table A.8: Observed choices and bet

Exercise A.23. A store observed 100 customers and recorded their preferred purchasing channel. The actual (empirical) observations are given in table A.9. Calculate the contingent distance and the quadratic distance to the bet.

Channel	Observations	Bet
Online Advertising	50	0.4
Marketing E-mail	30	0.30
Social Medias	20	0.30

TABLE A.9: Observed preferred channel and bet

Exercise A.24. An online store observed 200 transactions. It wants to know if the preferred payment method of customers is independent of their age group. The collected data are given in table A.10. Measure if the hypothesis of independence between the variables is plausible.

	credit card	PayPal	Bank transfer	Total
18-30	30	40	10	80
31-50	40	30	20	90
51+	10	15	5	30
Total	80	85	35	200

TABLE A.10: Joint Distribution of the preferred payment method and the age group of customers

Exercise A.25. An e-commerce company launched two types of promotions (price discount, free gift) and observed their impact on three purchasing channels (website, mobile app, social media). Table A.11 gives the data obtained from 300 sales. Measure if the hypothesis of independence between the variables is plausible.

	Price discount	Free gift	Total
Website	70	30	100
Mobile app	50	50	100
Social Media	40	60	100
Total	160	140	300

TABLE A.11: Joint distribution of the types of promotions and the purchasing channels

Exercise A.26. A company wants to compare preferences for three types of promotions (price discount, free gift, free shipping) between two online sales platforms (mobile app and website). The collected data are given in table A.12:

	mobile app	website	total
Price discount	60	40	100
Free gift	50	70	120
Free shipping	40	40	80
Total	150	150	300

TABLE A.12: promotion types preferences in different sales platforms

Additional Exercises to chapter ??

Exercise A.27. Consider the following graph of two variables defined on $\Omega_5 = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\}$, with two missing values $Y(\omega_4)$ and $Y(\omega_5)$. Assuming the principle of uniformity, the integrity of the past and the preservation of structure, predicts these two values.

	X	Y
ω_1	0	-1
ω_2	1	1
ω_3	2	2
ω_4	3	
ω_5	4	

TABLE A.13: Graph of X and Y

Exercise A.28. Let (Ω_n, Y_1, Y_2, X) be a dataset on the label set Ω_n. Consider the structural form of the simultaneous equation

$$E((Y_1,Y_2)A|l_n(X)) = XB$$

Where $A = \begin{pmatrix} 1 & \alpha \\ \beta & 1 \end{pmatrix}$ and $B = \begin{pmatrix} \gamma & \delta \end{pmatrix}$.

1. What are the dimensions of XB ?
2. Under which conditions does the reduced form exists?
3. Assuming these conditions are fullfiled, give the reduced form.
4. Assuming these conditions are fullfiled, are all the parameters identified ? If no propose a restriction to identified them.

Exercise A.29. Consider the following graph of three variables T,X and Y.

ω	T	X	Y
ω_1	0	0	0
ω_2	0	0	2
ω_3	1	1	0
ω_4	1	1	2

- Given Y and X, is T a statistics ?
- We want to study the of Y on X. Is X exogenous for p_T?

Exercise A.30. Let (Ω_8, X,Y,Z) and consider their graph.

	X	Y	Z
ω_1	1	10	A
ω_2	2	20	A
ω_3	2	10	B
ω_4	5	25	B
ω_5	6	30	B
ω_6	8	32	C
ω_7	10	40	C
ω_8	14	42	D

(a) Compute the linear approximation of $E(Y|\sigma(X))$

(b) You know that the data were generated as follow: given X and Z, Y is given by :$Y = \beta(Z)X$. What object is $\beta(Z)$? Compute it.

(c) Does the statistics you computed in question (a) generates a cut in the joint distribution of X, Y and Z?

Exercise A.31. Consider the dataset (Ω_4,Y,X,Z) defined on $\Omega_4 = \{\omega_1,\omega_2,\omega_3,\omega_4\}$:

ω	X	Z	Y
ω_1	0	0	0
ω_2	0	1	2
ω_3	1	0	1
ω_4	1	1	3

1. Compute $E(Y \mid l_n(X,Z))$ and $E(Y \mid l_n(X))$. Are they equal?
2. Does the statistic $T = E(Y \mid l_n(X))$ generate a cut of the joint distribution of (Y,X,Z)? Justify your answer.

Appendix B

Exercise Solutions

Exercises of chapter 2

Solution to Exercise 2.1. The Python code is given in figure B.1

Solution to Exercise 2.2. Age is a variable $Age : \Omega_n \rightarrow \Im(Age)$, where

$$\Im(Age) = \{18{,}19{,}20{,}21{,}22\}$$

Its preimages are:

$$\begin{aligned}
Age^{-}(18) &= \{3{,}8\}\\
Age^{-}(19) &= \{1{,}6\}\\
Age^{-}(20) &= \{4{,}7\}\\
Age^{-}(21) &= \{2{,}9\}\\
Age^{-}(22) &= \{5{,}10\}
\end{aligned}$$

Similarly, $Gender$ is a variable $Gender : \Omega_n \rightarrow \Im(Gender)$, where

$$\Im(Gender) = \{\text{Female}, \text{Male}, \text{Not Specified}\}$$

And its preimages are:

$$\begin{aligned}
Gender^{-}(\text{Female}) &= \{1{,}3{,}5{,}7{,}9\}\\
Gender^{-}(\text{Male}) &= \{2{,}4{,}6\}\\
Gender^{-}(\text{Not Specified}) &= \{8{,}10\}
\end{aligned}$$

No part of these two partitions are equal. The following parts have an empty intersection:

$$\begin{aligned}
Age^{-}(18) \ \cap \ Gender^{-}(\text{Male}) &= \{3{,}8\} \ \cap \ \{2{,}4{,}6\} = \emptyset\\
Age^{-}(22) \ \cap \ Gender^{-}(\text{Male}) &= \{5{,}10\} \ \cap \ \{2{,}4{,}6\} = \emptyset\\
Age^{-}(19) \ \cap \ Gender^{-}(\text{Not Specified}) &= \{1{,}6\} \ \cap \ \{8{,}10\} = \emptyset\\
Age^{-}(20) \ \cap \ Gender^{-}(\text{Not Specified}) &= \{4{,}7\} \ \cap \ \{8{,}10\} = \emptyset\\
Age^{-}(21) \ \cap \ Gender^{-}(\text{Not Specified}) &= \{2{,}9\} \ \cap \ \{8{,}10\} = \emptyset
\end{aligned}$$

Exemples of sentence:

- There is no Male aged 18 or 22.
- There is more Female than Male.
- There is the same number of individuals aged 18, 19, 20, 21 and 22.
- ...

```python

import os
import pandas as pd
os.chdir("path to the field in which you work")

df = pd.read_csv('_meps_10_2_en.csv')


df['mep_identifier'].unique()
df['mep_honorific_prefix'].unique()
df['mep_given_name'].unique()
df['mep_family_name'].unique()
df['mep_official_given_name'].unique()
df['mep_official_family_name'].unique()
df['mep_gender'].unique()
df['mep_citizenship'].unique()
df['mep_place_of_birth'].unique()
df['mep_birthday'].unique()
df['mep_death_date'].unique()
df['mep_image'].unique()
df['mep_email'].unique()
df['mep_homepage'].unique()
df['mep_twitter'].unique()
df['mep_facebook_page'].unique()
df['mep_facebook_profile'].unique()
df['mep_instagram'].unique()
df['mep_URI'].unique()

```

FIG. B.1: Python code for exercise 2.1

```python

df['mep_gender'].isin(['male','not known']).sum()/len(df)
```

FIG. B.2: Caption

Solution to Exercise 2.3.

$$\sigma(Gender) = \Big\{\emptyset,\ \{1,3,5,7,9\},\ \{2,4,6\},\ \{8,10\},\ \{1,2,3,4,5,6,7,9\},\\ \{1,3,5,7,8,9,10\},\ \{2,4,6,8,10\}, \Omega_{10}\Big\}$$

$$\mathrm{Card}(\mathcal{P}(\Im^{-1}(Age))) = 2^{\mathrm{Card}(\Im(Age))} = 2^5 = 32$$

Solution to Exercise 2.4.

$$P_{gender}(\{\text{male}, \text{not known}\}) = P_n(\{Gender(\omega) \in \{\text{male}, \text{not known}\}\}) = 0.615$$

Solution to Exercise 2.5. The set $A \cup B$ can be rewrite as follow:

$$\begin{aligned} A \cup B &= (A \cup B) \cap \Omega_n \\ &= (A \cup B) \cap (A \dot{\cup} A^c) \\ &= ((A \cup B) \cap A) \dot{\cup} ((A \cup B) \cap A^c) \\ &= A \dot{\cup} (B \cap A^c) \end{aligned}$$

Where A and $B \cap A^c$ are disjoint. Then by equation 2.1,

$$P_n(A \cup B) = P_n(A) + P_n(B \cap A^c) \tag{B.1}$$

Moreover, B can be decomposed in the following disjoint union:

$$\begin{aligned} B &= B \cap \Omega_n \\ &= B \cap (A \dot{\cup} A^c) \\ &= (B \cap A) \dot{\cup} (B \cap A^c) \end{aligned}$$

Then, using again 2.1,

$$\begin{aligned} P_n(B) &= P_n(B \cap A) + P_n(B \cap A^c) \\ \Leftrightarrow P_n(B \cap A^c) &= P_n(B) - P_n(B \cap A) \end{aligned} \tag{B.2}$$

Using B.1 and B.2, one gets

$$P_n(A \cup B) = P_n(A) + P_n(B) - P_n(B \cap A)$$

Solution to Exercise 2.6. By the definition of a probility measure (definition 2.3), one can compute the following probabilities:

$$P_4(A) = P_4(B) = P_4(C) = \frac{1}{2}$$

$$P_4(A \cap B) = P_4(A)P_4(B) = \frac{1}{4} \tag{B.3}$$

$$P_4(A \cap C) = P_4(A)P_4(C) = \frac{1}{4} \tag{B.4}$$

$$P_4(B \cap C) = P_4(B)P_4(C) = \frac{1}{4} \tag{B.5}$$

$$P_4(A \cap B \cap C) = \frac{1}{4} \neq P_4(A)P_4(B)P_4(C) = \frac{1}{8} \tag{B.6}$$

From the definition of independance (definition 2.2), (B.3) implies that $A \perp\!\!\!\perp B$, (B.4), implies that $A \perp\!\!\!\perp C$ and (B.4) implies that $B \perp\!\!\!\perp C$. However, the inequality in (B.6) implies that A, B and C are not jointly independent.

Solution to Exercise 2.7. Which country has the largest delegation in the European Parliament, and how many male and female members does it have ?
Germany has the largest delegation in the European Parliament, with a proportion of 0.134, which corresponds to 96 parliamentarians. 60 of them are males and 35 females.

How many Belgian parliamentarians are there ?
There are 22 Belgian parliamentarians.

Which countries have the most gender-equal delegation ?
Spain has the most gender-equal delegation, with as many females than males

Which countries have a strictly female-majority delegation ?
Three countries have a strictly female-majority delegation: France, Finland, Sweden.

Solution to Exercise 2.8. To construct the contingency table B.1 based on Figure2.5 and Table 2.4, we have first to compute the marginal probability of being in the Walloon Region or in Brussels-Capital (we only consider these two regions).

$$p_{regions}(\text{Walloon Region}) = \frac{3{,}681{,}575}{3{,}681{,}575 + 1{,}241{,}175} = 0.748$$
$$p_{regions}(\text{Brussels-Capital Reion}) = \frac{1{,}241{,}175}{3{,}681{,}575 + 1{,}241{,}175} = 0.252$$

We can also compute the conditional probability of not expressing a voting intention for one of these parties ('other') by substracting to 1 the conditional probability to express a voting intention for one of these parties. After, since the figure 2.5 gives the conditionnal distribution, we can compute the joint distribution using proposition 2.1. The results are given in table B.1. The marginal distribution of voting for one of the parties is the sum of the joint distribution of voting for this party in each region (see proposition 2.1).

	Walloon Region	Brussels-Capital Region	marg. dist.
PTB-PVDA	0.105	0.048	0.153
MR	0.150	0.046	0.196
Ecolo	0.105	0.042	0.147
PS	0.179	0.040	0.219
Defi	0.029	0.023	0.052
Les Engagés	0.103	0.018	0.121
Other	0.078	0.034	0.112
marg. dist.	0.748	0.252	1

TABLE B.1: contingency table of voting intentions

Since the percentage of "Other" is higher than some of the differences of percentages between parties, both conditionally on regions and marginally, it may then change the ranking.

Solution to Exercise 2.9.

1. The label space is Ω_{10} = {Amira Benali, Thomas Leroy, Sofia Dimitrov, Kwame Mensah, Elena Rossi, Jules Moreau, Fatima Haddad, Lucas Fernández, Mei Chen, Aisha Khan}
2. Ignoring the non-responses NA, the proportion of residents in the sample in favor of the LEZ policy is $\frac{4}{7}$ and the proportion against is $\frac{3}{7}$.
3. $\Im(\text{Response}) = \{\text{YES, NO, NA}\}$
4. Such an operator is an operator $R : \Omega_{10} \times \{\text{NO, YES, NA}\} \to \Omega_{10} \times \{\text{YES, NO}\} : (\omega, \text{Response}(\omega)) \to (\omega, R\text{Response}(\omega))$ such that

$$R\text{Response}(\omega) = \begin{matrix} \text{Response}(\omega) & \text{if Response}(\omega) \in \{\text{Yes, No}\} \\ \text{Yes or No} & \text{if Response}(\omega) \notin \{\text{Yes, No}\} \end{matrix}$$

Since the $\text{card}\{\omega \in \Omega_{10} : \text{Response}(\omega) \notin \{\text{Yes, No}\}\} = 3$ and $\text{card}\{\text{Yes, No}\} = 2$, the number of different operators of that kind is $2^3 = 8$ (see Table B.2).

5. We do not have prior knowledge about which of these operators is the correct $R := R_i$ for some i. However, for all i we have $P_n(\text{Response} = x) \le P_n(R_i\text{Response} = x)$ for all i and for $x \in \{\text{YES, NO}\}$.Hence necessarily,

Name	**Res.**	R_1Res.	R_2Res.	R_3Res.	R_4Res.	R_5Res.	R_6Res.	R_7Res.	R_8Res.
Amira Benali	YES	YES	YES	YES	YES	YES	YES	YES	YES
Thomas Leroy	NA	YES	YES	YES	YES	NO	NO	NO	NO
Sofia Dimitrov	YES	YES	YES	YES	YES	YES	YES	YES	YES
Kwame Mensah	NA	YES	YES	NO	NO	YES	YES	NO	NO
Elena Rossi	NO	NO	NO	NO	NO	NO	NO	NO	NO
Jules Moreau	YES	YES	YES	YES	YES	YES	YES	YES	YES
Fatima Haddad	NO	NO	NO	NO	NO	NO	NO	NO	NO
Lucas Fernández	YES	YES	YES	YES	YES	YES	YES	YES	YES
Mei Chen	NO	NO	NO	NO	NO	NO	NO	NO	NO
Aisha Khan	NA	YES	NO	YES	NO	YES	NO	YES	NO

Table B.2: The different possible grouping operators

$$P_n(\text{Response} = x) \leq P_n(R\text{Response} = x) \leq \max_i P_n(R_i\text{Response} = x)$$

Which gives,

$$\frac{4}{10} \leq P_n(R\text{Response} = \text{YES}) \leq \frac{7}{10}$$
$$\frac{3}{10} \leq P_n(R\text{Response} = \text{NO}) \leq \frac{6}{10}$$

Solution to Exercise A.1. From the definition of independence (definition 2.2), the two variables are independent iff $P_n(A)P_n(B) = P_n(A \cap B)$. It is not necessary to check for the other combination of the parts of the partitions induced by the variables since they are binary. Indeed For all subsets of Ω_n:

$$\begin{aligned}
&P_n(A)P_n(B) = P_n(A \cap B)\\
\Leftrightarrow\ &P_n(A^c)P_n(B) = P_n(A^c \cap B)\\
\Leftrightarrow\ &P_n(A)P_n(B^c) = P_n(A \cap B^c)\\
\Leftrightarrow\ &P_n(A^c)P_n(B^c) = P_n(A^c \cap B^c)
\end{aligned}$$

These equivalences are easy to prove. We will show here only the first, since it is possible to prove the others with the same way. Using equation 2.1 of the definition of a probability measure.

$$\begin{aligned}
P_n(A)P_n(B) = P_n(A \cap B) &\Leftrightarrow (1 - P_n(A^c))P_n(B) = P_n(B) - P_n(A^c \cap B)\\
&\Leftrightarrow P_n(B) - P_n(A^c)P_n(B) = P_n(B) - P_n(A^c \cap B)\\
&\Leftrightarrow P_n(A^c)P_n(B) = P_n(A^c \cap B)
\end{aligned}$$

$P_n(A) = \frac{50}{100} = 0.5$ and $P_n(B) = \frac{100-60}{100} = 0.4$. $P_n(A \cap B) = \frac{50-20}{100} = 0.3$. Then, $P_n(A)P_n(B) \neq P_n(A \cap B)$

Solution to Exercise A.2. As the precedent exercise, the two variables are independent iff $P_n(A)P_n(B) = P_n(A \cap B)$, where A is the set of all the satisfied customers and B the set of all costumers that had a fast delivery time.

$P_n(A) = \frac{60}{100} = 0.6$ and $P_n(B) = \frac{50}{100} = 0.5$. $P_n(A \cap B) = \frac{40}{100} = 0.4$. Then, $P_n(A \cap B) \neq P_n(A)P_n(B)$.

Solution to Exercise A.3. The two variables are independent if and only if $P_n(A_i \cap T_j) = P_n(A_i)P_n(T_j) \quad \forall i \in \{1{,}2\}, \forall j \in \{1{,}2{,}3\}$. Let us compare the table of the joint probabilities (table **??**) and the table of the multiplication of the probabilities.

	$P_n(A_1 \cap \dot{})$	$P_n(A_2 \cap \dot{})$	$P_n(\dot{})$
T_1	0.15	0.10	0.25
T_2	0.20	0.15	0.35
T_3	0.15	0.25	0.40
Ω_n	0.50	0.50	1.00

Table B.3: Contingence table for exercise A.3

	$P_n(A_1)\times$	$P_n(A_2)\times$	$1\times$
$\dot{\times} P_n(T_1)$	0.125	0.125	0.25
$\dot{\times} P_n(T_2)$	0.175	0.175	0.35
$\dot{\times} P_n(T_3)$	0.20	0.20	0.40
$\dot{\times} 1$	0.5	0.5	1

Table B.4: Table of the multiplication of marginal probabilities for exercise A.3

Since the two tables (B.3 and B.4) are not the same, the two variables are not independent.

Exercices of chapter 3

Solution to Exercise 3.1. Reminder: An Euclidean Space $\mathcal{E}$ is a Vector Space in $\mathbb{R}^n$ equipped with the Euclidean scalar product $\langle u{,}v\rangle := v^T u$.

$\langle 0, v\rangle = 0 \quad \forall v \in \mathcal{E}$

$$\langle 0, v\rangle = v^T 0 = v_1 0 + v_2 0 + ... + v_n 0 = 0$$

$u \mapsto \langle u, v\rangle$ is linear $\forall v \in \mathcal{E}$. Reminder: If U and V are two real vector spaces, function $f : U \to V$ is linear iff:

$$\forall a_1, a_2 \in U, \quad f(a_1 + a_2) = f(a_1) + f(a_2)$$
$$\forall a \in U \text{ and } \alpha \in \mathbb{R}, \quad f(\alpha a) = \alpha f(a)$$

We have to show that $f_v : \mathbb{R}^n \to \mathbb{R}^n : u \to f_v(u) := \langle u, v\rangle := v^T u$ has these two properties

$$\begin{aligned}
\langle u+w, v\rangle &= v^T(u+w) \\
&= v_1(u_1+w_1) + ... + v_n(u_n+w_n) \\
&= v_1u_1 + v_1w_1 + ... + v_nu_n + v_nw_n \\
&= v_1u_1 + ... + v_nu_n + v_1w_1 + ... + v_nw_n \\
&= \langle u, v\rangle + \langle w, v\rangle \\
\langle \alpha u, v\rangle &= v^T(\alpha u) \\
&= \alpha v_1u_1 + ... + \alpha v_nu_n \\
&= \alpha(v_1u_1 + ... + v_nu_n) \\
&= \alpha\langle u,v\rangle
\end{aligned}$$

$\langle u, \alpha v\rangle = \alpha\langle u,v\rangle$

$$\begin{aligned}
\langle u, \alpha v\rangle &= (\alpha v)^T u \\
&= \alpha v_1u_1 + \alpha v_2u_2 + ... + \alpha v_nu_n \\
&= \alpha(v_1u_1 + ... + v_nu_n) \\
&= \alpha\langle u,v\rangle
\end{aligned}$$

$\langle u, v+w\rangle = \langle u,v\rangle + \langle u, w\rangle \quad \forall u,v,w \in \mathcal{E}$

$$\begin{aligned}
\langle u, v+w\rangle &= (v+w)^T u \\
&= (v_1+w_1)u_1 + ... + (v_n+w_n)u_n \\
&= v_1u_1 + ... + v_nu_n + w_1u_1 + ... + w_nu_n \\
&= \langle u,v\rangle + \langle u, w\rangle
\end{aligned}$$

Solution to Exercise 3.2. We have to show that $d(u,v) := \|u-v\|$ satisfies the three properties of a distance.

First property: $\|u-v\| = \|v-u\|$

$$\begin{aligned}
\|u-v\| &= \sqrt{\langle u-v, u-v\rangle} \\
&= \sqrt{(u_1-v_1)^2 + ... + (u_1-v_1)^2} \\
&= \sqrt{(v_1-u_1)^2 + ... + (v_1-u_1)^2} \\
&= \|v-u\|
\end{aligned}$$

Second property: $\|u-v\| = 0 \Leftrightarrow u = v$

$$\begin{aligned}
&\|u-v\| = 0 \\
\Leftrightarrow &(u_1-v_1)^2 + ... + (u_1-v_1)^2 = 0
\end{aligned}$$

Since all the terms of the sum are squared, they are either positive or null. Then, for the sum to equal to 0, all the terms as to be null. This implies $u_j = v_j \quad \forall j = 1,...,n$, and then $u = v$.

Third property: $\|u-v\| \leq \|u-w\|\|w-v\|$

$$\begin{aligned}\|u-v\| &= \|(u-w)+(w-v)\| \\ &\leq \|u-w\|+\|w-v\| \qquad \text{by the triangle inequality}\end{aligned}$$

Solution to Exercise 3.3.

$$\begin{aligned}\|u+v\|^2+\|u-v\|^2 &= \langle u+v,u+v\rangle + \langle u-v,u-v\rangle \\ &= \langle u,u\rangle + 2\langle u,v\rangle + \langle v,v\rangle + \langle u,u\rangle - 2\langle u,v\rangle + \langle v,v\rangle \\ &= 2\langle u,u\rangle + 2\langle v,v\rangle \\ &= 2\|u\|^2+2\|v\|^2\end{aligned}$$

Solution to Exercise A.4. (i) Let us use the definition of the scalar product on the variables (3.5).

$$\begin{aligned}\langle \mathbb{1},\mathbb{1}\rangle &= \sum_{\omega\in\Omega_n} \mathbb{1}(\omega)\mathbb{1}(\omega)P_n(\{\omega\}) \\ &= \sum_{\omega\in\Omega_n} \frac{1}{n} \\ &= n\frac{1}{n} = 1\end{aligned}$$

Solution to Exercise A.5. From the definition of $L_2(\sigma(X))$ (definition 3.5), we know that the variables in $L_2(\sigma(\mathbb{1}))$ are the variables that induce a partition contained in $\sigma(\mathbb{1})$, i.e $\{\emptyset, \Omega_n\}$. The only possible partition of $Omega_n$ in $\sigma(\mathbb{1})$ is Ω_n, which is the partition induced by a constant variable. Then $L_2(\sigma(\mathbb{1}))$ are the space of all variables that can be written as:

$$L_2(\sigma(\mathbb{1})) = \{a\mathbb{1} : a \in \mathbb{R}\}$$

Solution to Exercise A.6. Let $x = \{n^{-1}\}$ and $y = \{2,2,2,2,2,2,2,...\}$. Both are real numbers, i.e. regular sequences of rational numbers. The sequence $\{2n^{-1}\}$ is not regular. Take $n = 1$ and $m = 4$,

$$|2-\frac{1}{2}| = \frac{3}{2} > 1+\frac{1}{4} = \frac{5}{4}$$

Let $x = \{x_n\}$ and $y = \{y_n\}$ be two real numbers with K_x and K_y their respective canonical bounds. Write $z_n \equiv x_{2kn}y_{2kn}$. Then $xy \equiv \{z_n\}$. $\forall m,n \in \mathbf{Z}^+$,

$$\begin{aligned}|z_m-z_n| &= |x_{2km}(y_{2km}-y_{2kn})+y_{2kn}(x_{2km}-x_{2kn})| \\ &\leq k|y_{2km}-y_{2kn}|+k|x_{2km}-x_{2kn}| \\ &\leq k((2km)^{-1}+(2kn)^{-1}+(2km)^{-1}+(2kn)^{-1}) = m^{-1}+n^{-1}\end{aligned}$$

Then, xy is a real number.

Solution to Exercise A.7.

$$\begin{aligned}\langle X_1, X_2\rangle =& \sum_{\omega\in\Omega_{10}} X_1(\omega)Y_2(\omega)P_n(\{\omega\})\\ =&\frac{55\times 5+43\times 3+70\times 10+35\times 5+80\times 3+60\times 10+95\times 3+50\times 5+45\times 10}{10}\\ =&330.9\end{aligned}$$

$$\begin{aligned}\langle X_1, \mathbb{1}\rangle =& \sum_{\omega\in\Omega_{10}} X_1(\omega)\mathbb{1}(\omega)P_n(\{\omega\})\\ =&\frac{55+43+70+35+80+60+95+50+45}{10}\\ =&53.3\end{aligned}$$

The set $\sigma(X_2)$ is composed of all the possible unions of elements of $X_2^{-1}[\Im(X_2)]$ and of the empty set $\emptyset$.

$$\begin{aligned}\sigma(X_2) = \{&\emptyset,\{\omega_2,\omega_5,\omega_7,\omega_{10}\},\{\omega_1,\omega_4,\omega_8\},\{\omega_3,\omega_6,\omega_9\},\{\omega_1,\omega_2,\omega_4,\omega_5,\omega_7,\omega_8,\omega_{10}\},\\ &\{\omega_2,\omega_3,\omega_5,\omega_6,\omega_7,\omega_9,\omega_{10}\},\{\omega_1,\omega_3,\omega_4,\omega_6,\omega_8,\omega_9\},\Omega_{10}\}\end{aligned}$$

$L_2(\sigma(X_2))$ is the variable space containing all the variables that are constant over the blocks $\{\omega_2,\omega_5,\omega_7,\omega_{10}\}$, $\{\omega_1,\omega_4,\omega_8\}$ and $\{\omega_3,\omega_6,\omega_9\}$ of the partition induced by X_2, i.e. the variables whose partitions are included in $\sigma(X_2)$.

Examples of such variables are:

Individual	Z_1	Z_2	Z_3
ω_1	14	8	-174
ω_2	2,5	8	-174
ω_3	-4	-2	-174
ω_4	14	8	-174
ω_5	2,5	8	-174
ω_6	-4	-2	-174
ω_7	2,5	8	-174
ω_8	14	8	-174
ω_9	-4	-2	-174
ω_{10}	2,5	8	-174

TABLE B.5: Examples of variables in $L_2(\sigma(X_2))$

Exercices of chapter 4

Solution to Exercise 4.1. Let $u\in\mathcal{E}$, $u^\perp := \{v\in\mathcal{E} : \langle v,u\rangle = 0\}$

(i) To show that $u^\perp$ is a vector subspace of $\mathcal{E}$, it is only necessary to show that (definition 3.1) :

$$\forall v,w\in u^\perp, \forall \alpha,\beta\in\mathbb{R} : \alpha v+\beta w\in u^\perp$$

Let v and w be two vectors in $u^\perp$ and α and β two real numbers. Then

$$\langle \alpha v+\beta w, u\rangle = \alpha\langle v,u\rangle + \beta\langle w,u\rangle = 0$$

Hence, $z \equiv \alpha v + \beta w$ is in $u^\perp$.

(ii) $u^\perp$ is a closed set. Closeness in a topologic notion, so let's recall what is a neighborhood structure. Let X a set. A neighborhood structure $\mathcal{V}$ on X is a family of subsets of X, called neighborhoods, such that if V_1 and V_2 are neighborhoods and $x \in V_1 \cap V_2$, then there exists a neighborhood V_3 such that

$$x \in V_3 \subset V_1 \cap V_2$$

A subset A of X is closed if and only if evert $a \in X$ whose all neigborhoods intesect A is necessarily in A.

When a $(X,\|\cdot\|)$ is a metric space, the metric induces a neighborhood structure. The open sphere $S(x,r)$ of radius $r > 0$ about a point x in X is the subset of X:

$$S(x,r) \equiv \{y \in X : \|y - x\| < r\} \tag{B.7}$$

The neighborhood structure induced by $\|\cdot\|$ is the set of all the open spheres in X. To see that this family of subset of X is indeed a neighborhood structure, let x, x_0 and x_1 be three elements of X and r_0 and r_1 be two strictly positive real numbers such that $x \in S(x_0, r_0) \cap S(x_1,r_1)$. Let also define $t \equiv \min\{r_0 - \|x - x_0\|, r_1 - \|x - x_1\|\}.^{I} tisclearthat$t>0.$Ifz$∈ $S(x,t)$, then,

$$\|z - x_0\| \leq \|z - x\| + \|x - x_0\| < t + \|x - x_0\| \leq r_0$$

Hence, $z \in S(x_0,r_0)$ and $S(x,t) \subset S(x_0,r_0)$. Similarly, $S(x,t) \subset S(x_1,r_1)$. It follows that $x \in S(x,t) \subset S(x_0,r_0) \cap S(x_1,r_1)$.

It follows that a subset A of a metric space $(X,\|\cdot\|)$ equipped with the neighborhood structure induced by its metric (the set of the open spheres) is closed if for all $x \in X$,

$$\forall r > 0, \exists y \in A : y \in S\check{r}(x,r) \Rightarrow x \in A \tag{B.8}$$

Or, equivalently, if for all $x \in X$,

$$\exists \{x_n\} \subset A : \lim_{n \to \infty} x_n = x \Rightarrow x \in A,$$

i.e. A is closed if it contains the limit of every convergent sequence of vectors in A.

Hence, to show that $u^\perp$ is closed, it is sufficient to show that if $w \in \mathcal{E}$ and $\{w_n\} \subset u^\perp$ is a sequence converging to w, then $w \in u^\perp$, i.e. $\langle w,u \rangle = 0$.

$$\begin{aligned}
|\langle w,u \rangle| &= |\langle w - w_n, u\rangle + \langle w_n, u\rangle| \text{ for some n} \\
&= |\langle w - w_n, u\rangle + 0| \\
&= |\langle w - w_n, u\rangle| \\
&\leq \|w - w_n\| \|u\| \text{ by Cauchy-Schwarz inequality}
\end{aligned}$$

Since $\|u\|$ is finite and since, by the definition of a convergent sequence, there always exists a n for which $\|w - wn\|$ is as close of 0 as we want, it follows that $\langle w, u\rangle = 0$.

Solution to Exercise 4.2. By definition of a vector subspace, $0 \in S$. Moreover, by definition of the scalar product, $\langle u, 0\rangle = 0$ for all vector u in S. It follows that $0 \in S^\perp$. Hence, $0 \in S \cap S^\perp$. It remains to show that if $u \in S \cap S^\perp$, then $u = 0$. It follows the definition of the scalar product, since $\langle u,u \rangle = 0 \Leftrightarrow u = 0$.

Solution to Exercise 4.3. Using definition 4.1, the equality we have to prove is equivalent to:

$$\begin{aligned}
&\mathbb{E}(\mathbb{E}(Y|X)|\mathbb{1}) = \mathbb{E}(Y|\mathbb{1})\\
\Leftrightarrow&\langle P_X Y, \mathbb{1}\rangle \mathbb{1} = \langle Y, \mathbb{1}\rangle \mathbb{1}\\
\Leftrightarrow&\langle Y, \mathbb{1}\rangle \mathbb{1} - \langle PY, \mathbb{1}\rangle \mathbb{1} = 0\\
\Leftrightarrow&\langle Y - P_X Y, \mathbb{1}\rangle \mathbb{1} = 0\\
\Leftrightarrow&\langle Y - P_X Y, \mathbb{1}\rangle = 0
\end{aligned}$$

To prove the equality, it is sufficient to show that $Y - P_X Y \perp \mathbb{1}$.

From the projection theorem, the application P_X is such that $Y - P_X Y \perp W$ for all $W \in L_2(\Omega_n, \sigma(X), P_n)$. Since $\mathbb{1}$ is a constant variable (definition **??**), the partition of Ω_n it induces is a subset of $\sigma(X)$. Hence, $\mathbb{1} \in L_2(\Omega_n, \sigma(X), P_n)$. (Since $\mathbb{1}$ is constant on Ω_n, it is also constant on each parts of the partition of Ω_n induced by the variable X).

Solution to Exercise 4.4. By definition 4.1,

$$\mathbb{E}(Y|X) = P_X Y \in L_2(\sigma(X)) := \{ \sum_{j=1}^{J} a_j \mathbb{1}_{B_j} : a_1,\ldots,a_J \in \mathbb{R}\}$$

So $\mathbb{E}(Y|X) = \sum_{j=1}^{J} a_j \mathbb{1}_{B_j}$ for some $\{a_j\}$. How to find them?

By the projection theorem (4.1),

$$\begin{aligned}
&\langle Y - \mathbb{E}(Y|X), W\rangle = 0 \quad \forall W \in L_2(\sigma(X))\\
\Leftrightarrow&\langle Y - \sum_{j=1}^{J} a_j \mathbb{1}_{B_j}, W\rangle = 0\\
\Leftrightarrow&\langle Y,W\rangle = \langle \sum_{j=1}^{J} a_j \mathbb{1}_{B_j}, W\rangle
\end{aligned}$$

Since for all $k = 1,\ldots,J$, the partitions induced by $\mathbb{1}_{B_k}$ are included in $\sigma(X)$, it follows

$$\begin{aligned}
&\langle Y, \mathbb{1}_{B_k}\rangle = \langle \sum_{j=1}^{J} a_j \mathbb{1}_{B_j}, \mathbb{1}_{B_k}\rangle \forall k = 1,\ldots,J\\
\Leftrightarrow&\langle Y, \mathbb{1}_{B_k}\rangle = \sum_{j=1}^{J} a_j \langle \mathbb{1}_{B_j}, \mathbb{1}_{B_k}\rangle
\end{aligned}$$

Where $\langle \mathbb{1}_{B_j}, \mathbb{1}_{B_k}\rangle$ equals 0 when $j \neq k$ and $\frac{|B_k|}{n}$ when $j = k$. Moreover, based on the scalar product in the variables space (3.5),

$$\begin{aligned}
\langle Y, \mathbb{1}_{B_k}\rangle &= \sum_{\omega\in\Omega_n} Y(\omega)\mathbb{1}_{B_k}(\omega)P_n(\{\omega\})\\
&= \frac{1}{n}\sum_{\omega\in B_k} Y(\omega)
\end{aligned}$$

Then,

$$\begin{aligned}
&\frac{1}{n}\sum_{\omega\in B_k} Y(\omega) = a_k \frac{|B_k|}{n}\\
\Leftrightarrow&a_k = \frac{1}{|B_k|}\sum_{\omega\in B_k} Y(\omega)
\end{aligned}$$

And then,

$$\mathbb{E}(Y|X) = \sum_{j=1}^{J} \frac{1}{|B_j|} \sum_{\omega \in B_j} Y(\omega) \mathbb{1}_{B_j}$$

Solution to Exercise 4.5. Let W be a variable in $L_2(\Omega_n, \mathcal{P}(\Omega_n), P_n)$ and $\{W_j\}$ be a sequence of variables in $L_2(\Omega_n, \sigma(X), P_n)$ converging to W.

By the projection theorem, we have $W = P_X W + (1 - P_X)W$ where the first term is in $L_2(\sigma(X))$ and the second term is orthogonal to it. To show that $W \in L_2(\sigma(X))$, it is sufficient to show that $W = P_X W$, i.e. $\|(1 - P_X)W\| = 0$.

$$\begin{aligned} \|(1 - P_X)W\| &= \|(1 - P_X)(W - W_n + W_n)\| \\ &\leq \|(1 - P_X)(W - W_n)\| + \|(1 - P_X)W_n\| \end{aligned}$$

The first term in the right-hand side of this inequality is bounded by $\|W - W_n\|$. Indeed, by the projection theorem, we have that for all vector $\zeta \in L_2(\Omega_n)$, $\|\zeta\|^2 = \|P_X \zeta\|^2 + \|(1 - P_X)\zeta\|^2$, and hence $\|\zeta\| \geq \|(1 - P_X)\zeta\|$. The second term is null since W_n belongs to $L_2(\sigma(X))$ by hypothesis. It follows that

$$\|(1 - P_X)W\| \leq \|W - W_n\|$$

Since there always exists a n for which this bound is as close to 0 as we want, it follows that $\|(1 - P_X)W\| = 0$, and hence $W \in L_2(\sigma(X))$.

Solution to Exercise 4.6.

$$\begin{aligned} \langle \mathbb{E}(Y|X), Z \rangle &= \langle \mathbb{E}(Y|X), Z - \mathbb{E}(Z|X) + \mathbb{E}(Z|X) \rangle & \text{(B.9)} \\ &= \langle \mathbb{E}(Y|X), Z - \mathbb{E}(Z|X) \rangle + \langle \mathbb{E}(Y|X), \mathbb{E}(Z|X) \rangle & \text{(B.10)} \\ &= \langle \mathbb{E}(Y|X), \mathbb{E}(Z|X) \rangle & \text{(B.11)} \\ &= \langle \mathbb{E}(Y|X) - Y + Y, \mathbb{E}(Z|X) \rangle & \text{(B.12)} \\ &= \langle Y, \mathbb{E}(Z|X) \rangle - \langle Y - \mathbb{E}(Y|X), \mathbb{E}(Z|X) \rangle & \text{(B.13)} \\ &= \langle Y, \mathbb{E}(Z|X) \rangle & \text{(B.14)} \end{aligned}$$

Where the third equality holds from the theorem of Projection (4.1) and the definition 4.1 because we know that $Z - \mathbb{E}(Z|X)$ is orthogonal to every variables in $L_2(\sigma(X))$ and $\mathbb{E}(Y|X) \in L_2(\sigma(X)$. The same reasoning applies to the sixth equality.

Let us apply this result on $\langle \mathbb{E}(Y|X), \mathbb{1} \rangle$:

$$\begin{aligned} \langle \mathbb{E}(Y|X), \mathbb{1} \rangle &= \langle Y, \mathbb{E}(\mathbb{1}|X) \rangle \\ &= \langle Y, \mathbb{1} \rangle \\ &= \bar{Y}_n \end{aligned}$$

Where the second equality holds because $\mathbb{1} \in L_2(\sigma(X))$ since it is a constant function (and then it is constant on all parties of the partition induces by X).

Solution to Exercise 4.8. Let recall first the definition of R^2

$$R^2 = \frac{V(\mathbb{E}(Y|X)}{V(Y)}$$

When $X = \mathbb{1}$, $\mathbb{E}(Y|\mathbb{1}) = \bar{Y}_n\mathbb{1}$. Using the definition of variance and the fact that $\mathbb{E}(\mathbb{1}|\mathbb{1}) = \mathbb{1}$, one gets:

$$V(\mathbb{E}(Y|X)) = \|\bar{Y}_n\mathbb{1} - \mathbb{E}(\bar{Y}_n\mathbb{1}|\mathbb{1})\|^2 = \|\bar{Y}_n\mathbb{1} - \bar{Y}_n\mathbb{E}(\mathbb{1}|\mathbb{1})\|^2 = 0$$

Solution to Exercise 4.9. Let us first recall the definition of ρ_{XY}:

$$\rho_{XY} = \frac{C(X,Y)}{\|X - \mathbb{E}(X|\mathbb{1})\|\|Y - \mathbb{E}(Y|\mathbb{1})\|}$$

Where $C(X,Y) = \langle X - \mathbb{E}(X|\mathbb{1}), Y - \mathbb{E}(Y|\mathbb{1})\rangle$.

Using the Cauchy-Schwarz inequality,

$$|C(X,Y)| \leq \|X - \mathbb{E}(X|\mathbb{1})\|\|Y - \mathbb{E}(Y|\mathbb{1})\| \Leftrightarrow \qquad \rho_{XY} \leq 1$$

Solution to Exercise 4.10. (i) We will first prove a more general formula: $C(X,Y) = \langle X,Y\rangle - \langle X,\mathbb{1}\rangle\langle Y,\mathbb{1}\rangle$.

$$\begin{aligned} C(X,Y) &= \langle X - \langle X, \mathbb{1}\rangle\mathbb{1}, Y - \langle Y, \mathbb{1}\rangle\mathbb{1}\rangle \\ &= \langle X, Y\rangle - \langle X, \mathbb{1}\rangle\langle \mathbb{1}, Y\rangle - \langle X, \mathbb{1}\rangle\langle Y, \mathbb{1}\rangle + \langle X, \mathbb{1}\rangle\langle Y, \mathbb{1}\rangle \\ &= \langle X, Y\rangle - \langle X, \mathbb{1}\rangle\langle Y, \mathbb{1}\rangle \end{aligned}$$

Now, we can apply this formula to $C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z))$.

$$C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) = \langle X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)\rangle - \langle X - \mathbb{E}(X|Z), \mathbb{1}\rangle\langle Y - \mathbb{E}(Y|Z), \mathbb{1}\rangle$$

Since, as shown in exercise 4.3, $\langle X - \mathbb{E}(X|Z), \mathbb{1}\rangle = 0$ and $\langle Y - \mathbb{E}(Y|Z), \mathbb{1}\rangle = 0$, this equality simplifies in:

$$\begin{aligned} C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) &= \langle X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)\rangle \\ &= \langle X, Y\rangle - \langle X, \mathbb{E}(Y|Z)\rangle - \langle \mathbb{E}(X|Z), Y\rangle + \langle \mathbb{E}(X|Z), \mathbb{E}(Y|Z)\rangle \end{aligned}$$

In exercise 4.6, we have shown that $\langle \mathbb{E}(Y|Z), X\rangle = \langle Y, \mathbb{E}(X|Z)\rangle = \langle \mathbb{E}(Y|Z), \mathbb{E}(X|Z)\rangle$. Then,

$$\begin{aligned} C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) &= \langle X, Y\rangle - \langle \mathbb{E}(X|Z), \mathbb{E}(Y|Z)\rangle \\ &= \langle X, Y\rangle - \langle X,\mathbb{1}\rangle\langle Y,\mathbb{1}\rangle - \langle \mathbb{E}(X|Z), \mathbb{E}(Y|Z)\rangle + \langle X, \mathbb{1}\rangle\langle Y, \mathbb{1}\rangle \\ &= C(X,Y) - \langle \mathbb{E}(X|Z), \mathbb{E}(Y|Z)\rangle + \langle \mathbb{E}(X|Z), \mathbb{1}\rangle\langle \mathbb{E}(Y|Z), \mathbb{1}\rangle \\ &= C(X,Y) - C(\mathbb{E}(X|Z), \mathbb{E}(Y|Z)) \end{aligned}$$

Where the third equality has been shown in exercise 4.3.

(ii) The definition of conditional independence between 2 variables (X and Y) given a third one (Z), noted $X \perp\!\!\!\perp Y|Z$, is given p.44. It means that $\forall B \in Z^-(\Im(Z))$, $\forall x \in \Im(X)$, $\forall y \in \Im(Y)$:

$$\begin{aligned} &P_n(\{\omega \in \Omega_n : X(\omega) = x\} \cap \{\omega \in \Omega_n : Y(\omega) = y\}|B) \\ &= P_n(\{\omega \in \Omega_n : X(\omega) = x\}|B)P_n(\{\omega \in \Omega_n : Y(\omega) = y\}|B) \end{aligned}$$

We have to show that the independance of two variables X and Y conditionnally on Z implies $C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) = 0$. We have shown in the previous point that

$(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) = \langle X,Y\rangle - \langle \mathbb{E}(X|Z) - \mathbb{E}(Y|Z)\rangle$. Hence, it remains to show that $\langle X,Y\rangle = \langle \mathbb{E}(X|Z) - \mathbb{E}(Y|Z)\rangle$.

$$\begin{aligned}
\langle X,Y\rangle &= \sum_{\omega\in\Omega_n} X(\omega)Y(\omega)P_n(\{\omega\})\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)}\sum_{y\in\Im(Y)}\sum_{\omega\in\{X(\omega)=x\}\cap\{Y(\omega)=y\}\cap B} xyP_n(\{\omega\})\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)} x \sum_{y\in\Im(Y)} y \sum_{\omega\in\{X(\omega)=x\}\cap\{Y(\omega)=y\}\cap B} P_n(\{\omega\})\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)} x \sum_{y\in\Im(Y)} yP_n(\{X(\omega)=x\}\cap\{Y(\omega)=y\}\cap B)\\
&= \sum_{B\in Z^-(\Im(Z))} P_n(B) \sum_{x\in\Im(X)} x \sum_{y\in\Im(Y)} yP_n(\{X(\omega)=x\}\cap\{Y(\omega)=y\}|B)\\
&= \sum_{B\in Z^-(\Im(Z))} P_n(B) \sum_{x\in\Im(X)} xP_n(\{X(\omega)=x\}|B) \sum_{y\in\Im(Y)} yP_n(\{Y(\omega)=y\}|B)\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)} xP_n(\{X(\omega)=x\}|B) \sum_{y\in\Im(Y)} yP_n(\{Y(\omega)=y\}\cap B)\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)} xP_n(\{X(\omega)=x\}|B) \sum_{y\in\Im(Y)} y \sum_{\omega\in\{Y(\omega)=y\}\cap B} P_n(\{\omega\})\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)} xP_n(\{X(\omega)=x\}|B) \sum_{\omega\in B} Y(\omega)P_n(\{\omega\})\\
&= \sum_{B\in Z^-(\Im(Z))}\sum_{x\in\Im(X)} xP_n(\{X(\omega)=x\}|B)P_n(B)\frac{1}{|B|}\sum_{\omega\in B} Y(\omega)\\
&= \sum_{B\in Z^-(\Im(Z))} P_n(B)\frac{1}{|B|}\sum_{\in B} X(\omega)\frac{1}{|B|}\sum_{\omega\in B} Y(\omega)\\
&= \sum_{\omega\in\Omega_n} \mathbb{E}(X|Z)(\omega)\mathbb{E}(Y|Z)(\omega)P_n(\{\omega\})
\end{aligned}$$

$= \langle \mathbb{E}(X|Z), \mathbb{E}(Y|Z)\rangle$

(iii) from (i), we have:

$$\begin{aligned}
&C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) = C(X,Y) - C(\mathbb{E}(X|Z), \mathbb{E}(Y|Z))\\
\Leftrightarrow &C(X,Y) = C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) + C(\mathbb{E}(X|Z), \mathbb{E}(Y|Z))
\end{aligned}$$

Since $X \perp\!\!\!\perp Y|Z$, $C(X - \mathbb{E}(X|Z), Y - \mathbb{E}(Y|Z)) = 0$. Then,

$$C(X,Y) = C(\mathbb{E}(X|Z), \mathbb{E}(Y|Z)).$$

We construct an example where $Z = \mathbb{E}(X|Z) = \mathbb{E}(Y|Z)$. Then,

$$C(X,Y) = \|Z\|^2 > 0$$

Solution to Exercise A.8. Let us first recall the definition of the mean of a variable X, which is the projection of X on the space of constant variables:

$$\begin{aligned}
\bar{X}_n &= \sum_{\omega\in\Omega_n} X(\omega)P_n(\{\omega\})\\
&= \sum_{\{\omega\in\Omega_n: X(\omega)\geq a\}\dot\cup\{\omega\in\Omega_n: X(w)<a\}} X(\omega)P_n(\{\omega\})
\end{aligned}$$

The last equality is correct since $\{\omega \in \Omega_n : X(\omega) \geq a\}$ and $\{\omega \in \Omega_n : X(\omega) < a\}$ form a partition of Ω_n. Since this union is therefore a disjoint union, we can divide the sum in two parts.

$$\begin{aligned}\bar{X}_n &= \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)\geq a\}} X(\omega)P_n(\{\omega\}) + \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)<a\}} X(\omega)P_n(\{\omega\}) \\ &\geq \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)\geq a\}} aP_n(\{\omega\}) + \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)<a\}} 0P_n(\{\omega\})\end{aligned}$$

The last inequality comes from two facts: (i) The first sum is a sum over the ω when $X(\omega)$ is greater or equal to a. Then, a lower bound of this sum is when $X(\omega)$ is replaced by a. (ii) Since the image set of X is a subset of the positive real numbers, the lowest value that X can have in the second sum is 0. Then a lower bound of this sum is when $X(\omega)$ is replaced by 0. Then, the inequality becomes

$$\begin{aligned}\bar{X}_n &\geq a \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)\geq a\}} P_n(\{\omega\}) \\ &= aP_n(\{\omega \in \Omega_n : X(\omega) \geq a\}) \\ &= aP_X(X \geq a)\end{aligned}$$

The first equality comes from equation 2.1 of the definition of the probability measure. The sum of the probabilities of disjoint sets is equal to the probability of the union of these sets. The equality comes from the definition of the probability measure induced by a variable (definition 2.4). Then, it follows

$$P_X(X \geq a) \leq \frac{\bar{X}_n}{a}$$

Solution to Exercise A.9. Let's begin by recalling the definition of the mean of a variable, i.e. the projection of the variable on the space of the constant variables, or, in other words, the mean of a variable X is the constant variable the closest to X with respect to the norm $\|.\|$.

$$\begin{aligned}\mathbb{E}(X|\mathbb{1}) &= \sum_{\omega\in\Omega_n} X(\omega)P_n(\{\omega\}) \\ &= \sum_{x\in\Im(X)} \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)=x)} xP_n(\{\omega\}) \\ &= \sum_{x\in\Im(X)} x \sum_{\omega\in\{\omega\in\Omega_n:X(\omega)=x)} P_n(\{\omega\}) \\ &= \sum_{x\in\Im(X)} xP_n(\{\omega \in \Omega : X(\omega) = x\} \\ &= \sum_{x\in\Im(X)} xP_X(X = x)\end{aligned}$$

The second inequality comes from the fact that the set $\{\{\omega \in \Omega_n : X(\omega) = x\}\}_{x\in\Im(X)}$ form a partition of Ω_n. The fourth equality comes from equation 2.1 and the fifth from the definition of a probability measure induced by a variable (definition 2.4).

Solution to Exercise A.10. Let's begin with the result we obtained at the exercise A.9:

$$\begin{aligned}\mathbb{E}(X|\mathbb{1}) &= \sum_{x\in\Im(X)} xP_X(X=x)\\ &= \sum_{x\in\Im(X)} xP_X(X=x) + \sum_{x\in\mathbb{N}_0\setminus\Im(X)} xP_X(X=x)\\ &= \sum_{x=1}^{\infty} xP_X(X=x)\end{aligned}$$

Where the second equality holds since $P_X(X=x) = P_n(\{\omega \in \Omega_n : X(\omega) = x\})$ equals 0 for all naturals number not in $\Im(X)$. Then,

$$\begin{aligned}\mathbb{E}(X|\mathbb{1}) &= \sum_{x=1}^{\infty} xP_X(X=x)\\ &= \sum_{x=1}^{\infty}\sum_{k=1}^{x} P_X(X=x)\\ &= \sum_{k=1}^{\infty}\sum_{x=k}^{\infty} P_X(X=x)\\ &= \sum_{k=1}^{\infty} P_X(X\geq k)\\ &= \sum_{x=1}^{\infty} P_X(X\geq x)\end{aligned}$$

The second equality comes from the fact that $xP_X(X=x) = P_X(X=x) + P_X(X=x) + ... + P_X(X=x)$ x times. The third equality comes from the rule of inversion of double sums. The fourth comes from the definition of a probability induced by a variable (definition 2.4) and from equation 2.1 of the definition of a probability measure. The fifth equality is just a change of subscript.

Solution to Exercise A.11. $L_2(\sigma(Y - P_X Y))$ is not a subset of the space orthogonal to $L_2(\sigma(X))$. The first of the spaces is the composed of all the variables that are constant on the blocks of the partition induced by the variable $Y - P_X Y$, whereas the second is the space composed of all the variables that are orthogonal to every variables in $L_2(\sigma(X))$, i.e. the variables that on each block of the partition induced by X are in average equal to 0. The variable $\mathbb{1}$ is in the first space since it is constant over Ω_n, and hence constant over each blocks of the partition induced by $Y - P_X Y$, but it is not in the second since it equals one on average on each block of the partition induced by X.

Solution to Exercise A.12. From the projection theorem, we have that for all $x \in H$

$$\langle x - Px, \zeta\rangle = 0 \quad \forall \zeta \in S$$

In particular, $\langle x - Px, Py \rangle$ for any $y \in H$. It follows that,

$$
\begin{aligned}
\langle Px,y \rangle &= \langle Px, y - Py + Py \rangle \\
&= \langle Px, y - Py \rangle + \langle Px, Py \rangle \\
&= \langle Px, Py \rangle \quad \text{(in fact, by symmetry, this is already sufficient to prove the result)} \\
&= \langle Px - x + x,Py \rangle \\
&= \langle Px - x, Py \rangle + \langle x, Py \rangle \\
&= \langle x,Py \rangle
\end{aligned}
$$

Solution to Exercise A.13. We can construct the label set as : $\Omega_{36} = \{A01, A02, A03, A04, A05, A06, A07, A08, A09, A10, A11, A12\} \times \{2022, 2023, 2024\}$. The label set is composed of 12 individuals at 3 periods.

The projection of Annual Salary on the subspace of constant variables is the function $\mathbb{E}(\text{Annual Salary}|\mathbb{1}) : \Omega_n \to \Im(\mathbb{E}(\text{Annual Salary}|\mathbb{1}))$ given by :

$$
\mathbb{E}(\text{Annual Salary}|\mathbb{1})(\omega) = \frac{\sum_{\omega \in \Omega_{36}} \text{Annual Salary}(\omega)}{36} = 55.27 \text{ for all } \omega \in \Omega_{36}
$$

The r^2 is given by :

$$
\begin{aligned}
r^2 &= \frac{\|\mathbb{E}(\text{Annual Salary}|\mathbb{1})\|^2}{\|\text{Annual Salary}\|^2} \\
&= \frac{\langle \mathbb{E}(\text{Annual Salary}|\mathbb{1}); \mathbb{E}(\text{Annual Salary}|\mathbb{1}) \rangle}{\langle \text{Annual Salary; Annual Salary} \rangle} \\
&= \frac{57.31^2}{\frac{1}{36} \sum_{\omega \in \Omega_{36}} \text{Annual Salary}(\omega)^2} = \frac{3054.77}{3305.53} = 0.92
\end{aligned}
$$

The r^2 gives the cosine of the angle between the variable Annual Salary and the space of the constant variables.

The partition induced by the variables Year and Annual Salary contains 6 blocks: $\{(A01,2022), (A02,2022), (A03,2022), (A05,2022), (A07,2022),(A09,2022),(A12,2022)\}$, $\{(A01,2023), (A02,2023), (A03,2023), (A05,2023), (A07,2023),(A09,2023),(A12,2023)\}$, $\{(A01,2024), (A02,2024$ $(A03,2024), (A05,2024), (A07,2024),(A09,2024),(A12,2024)\}$, $\{(A04,2022), (A06,2022), (A08,2022), (A10,2022), (A11,2022)\}$, $\{(A04,2023), (A06,2023), (A08,2023), (A10,2023), (A11,2023)\}$, $\{(A04,2024), (A06,2024), (A08,2024), (A10,2024), (A11,2024)\}$. $\mathbb{E}(\text{Annual Salary}|\sigma(\text{Year, Gender})) : \Omega_{36} \to \Im(\mathbb{E}(\text{Annual Salary}|\sigma(\text{Year, Gender})))$ is then the variable resulting from the orthogonal projection of Annual Salary on the space of the variables constant on these blocks. We can compute it using the formula given in exercise 4.4. It is given by :

$$
\mathbb{E}(\text{Annual Salary}|\sigma(\text{Year, Gender}))(\omega) = \begin{cases}
46.86 & \text{when Year}(\omega) = 2022 \text{ and Gender}(\omega) = \text{Female} \\
49.14 & \text{when Year}(\omega) = 2023 \text{ and Gender}(\omega) = \text{Female} \\
51.57 & \text{when Year}(\omega) = 2024 \text{ and Gender}(\omega) = \text{Female} \\
61.8 & \text{when Year}(\omega) = 2022 \text{ and Gender}(\omega) = \text{Male} \\
63.8 & \text{when Year}(\omega) = 2023 \text{ and Gender}(\omega) = \text{Male} \\
65.8 & \text{when Year}(\omega) = 2024 \text{ and Gender}(\omega) = \text{Male}
\end{cases}
$$

The coefficient of determination is given by

$$\begin{aligned} R^2 &= \frac{V(\mathbb{E}(\text{Annual Salary}|\sigma(\text{Year, Gender})))}{V(\text{Annual Salary})} \\ &= \frac{55{,}15}{250{,}76} = 0.22 \end{aligned}$$

The coefficient of determination is small but not close to 0, meaning that the variables Year and Gender explain a unnegligible part of the variance of Annual Salary, but not all of it.

1 Exercises of chapter ??

Solution to Exercise ??. Let us first recall what is $p_{Y|X=x}$.

Solution to Exercise ??. $E(\cdot|\sigma(\mathbb{1}))$ is the projection operator from $L_2^2(\Omega_n)$ to $L_2^2(\sigma(\mathbb{1}))$, where $L_2^2(\Omega_n)$ denotes the space of real bivariate functions on Ω_n and $L_2^2(\sigma(1))$ the space of bivariate constant functions, i.e:

$$L_2^2(\sigma(1)) := \{V \in L_2^2(\Omega_n) : V = \mathbb{1}\begin{pmatrix} a & b \end{pmatrix}, \quad a,b \in \mathbb{R}\}$$

From the projection theorem, we know that $(Y_1 - Y_2) - E((Y_1,Y_2)|\sigma(\mathbb{1}))$ is orthogonal to any function in $L_2^2(\sigma(1))$. Therefore, we have to find the values a_1 and a_2 such that

$$\begin{aligned} &\langle (Y_1,Y_2) - \mathbb{1}\begin{pmatrix} a_1 & a_2 \end{pmatrix}; \mathbb{1}C\rangle = 0 \quad \text{for any } C = \begin{pmatrix} c_1 & c_2 \end{pmatrix} \in \mathbb{R} \\ &\langle Y_1 - \mathbb{1}a_1; c_1\mathbb{1}\rangle + \langle Y_2 - \mathbb{1}a_2; c_2\mathbb{1}\rangle = 0 \\ &\langle Y_1 - \mathbb{1}a_1; c_1\mathbb{1}\rangle + \langle Y_2 - \mathbb{1}a_2; c_2\mathbb{1}\rangle = 0 \\ &c_1\langle Y_1; \mathbb{1}\rangle - a_1c_1\langle \mathbb{1}; \mathbb{1}\rangle + c_2\langle Y_2; \mathbb{1}\rangle - a_2c_2\langle \mathbb{1}; \mathbb{1}\rangle = 0 \\ &c_1(\bar{Y}_{1n} - a_1) + c_2(\bar{Y}_{2n} - a_2) = 0 \end{aligned}$$

Since this has to be true for all c_1 and c_2, we obtain $a_1 = \bar{Y}_{1n}$ and $a_2 = \bar{Y}_{2n}$. Hence,

$$E((Y_1,Y_2)|\sigma(\mathbb{1})) = \mathbb{1}\begin{pmatrix} \bar{Y}_{1n} & \bar{Y}_{2n} \end{pmatrix}$$

Solution to Exercise A.27. Let us use proposition **??** on prdiction. We have first to compute the 5×2 matrix $\tilde{H}$ and the vector $\tilde{G}$. For that we first need to compute $P_x y$. For that, the ordinary least squares gives us :

$$\begin{aligned} P_x y &= (\bar{y} - \frac{\langle y,x\rangle - \bar{y}\bar{x}}{\|x\|^2 - (\bar{x})^2}\bar{x}) + \frac{\langle y,x\rangle - \bar{y}\bar{x}}{(\bar{x})^2 - \|x\|^2}x \\ &= -\frac{5}{6} + \frac{3}{2}\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \\ &= \frac{1}{6}\begin{pmatrix} -5 \\ 4 \\ 13 \end{pmatrix} \end{aligned}$$

Thus, $\tilde{H}$ and $\tilde{G}$ are:

$$\tilde{H} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} \left(\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} \right)^{-1} \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 0 & -2 \\ 1 & 0 \\ 2 & 2 \\ 3 & 4 \\ 4 & 6 \end{pmatrix}$$

$$\tilde{G} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} \left(\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} \right)^{-1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} \frac{1}{6} \begin{pmatrix} -5 \\ 4 \\ 13 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix}$$

Hence, the predicted values of Y for ω_4 and $Y(\omega_5)$ are

$$\hat{y}_o = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{10} \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix} - \frac{1}{10} \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix} + \left(\frac{1}{10}\right)^2 \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ -2 & 0 & 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 0 & -2 \\ 1 & 0 \\ 2 & 2 \\ 3 & 4 \\ 4 & 6 \end{pmatrix} \right)^{-1}$$

$$\left(\frac{1}{10} \begin{pmatrix} 0 & 1 & 2 \\ 2 & 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{3}{5} \end{pmatrix} - \left(\frac{1}{10}\right)^2 \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ -2 & 0 & 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix} \right) = \frac{1}{6} \begin{pmatrix} 22 \\ 31 \end{pmatrix}$$

Solution to Exercise A.28.

1. XB is a function from Ω_n to a subset $\Im(XB)$ of $\mathbb{R}^2$. Since $|\Omega_n| = n$, it can be represented as an $n \times 2$ matrix.
2. Matrix A has to be invertible. Hence, the determinant has to be different from 0.

$$\begin{aligned} \det A = 1 - \alpha\beta &\neq 0 \\ \Leftrightarrow \alpha\beta &\neq 1 \end{aligned}$$

3. Using the linearity of conditional, we obtain

$$E((Y_1,Y_2) \mid l_n(X))A = XB.$$

Since A is invertible, the reduced form is

$$E((Y_1,Y_2) \mid \mathcal{L}_n(X)) = XBA^{-1}.$$

Where matrix $A^{-1} = \frac{1}{1-\alpha\beta} \begin{pmatrix} 1 & -\alpha \\ -\beta & 1 \end{pmatrix}$. Therefore,

$$\begin{aligned} E((Y_1,Y_2)|l_n(X)) &= X \frac{1}{1-\alpha\beta} \begin{pmatrix} \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & -\alpha \\ -\beta & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{\gamma-\beta\delta}{1-\alpha\beta} X & \frac{\delta-\alpha\gamma}{1-\alpha\beta} X \end{pmatrix} \end{aligned}$$

4. By projecting (Y_1, Y_2) on $l_n(X)$, i.e. the space of functions (giving in $\mathbb{R}^2$) that are linear in X, we get a vector of coefficients $a = \begin{pmatrix} a_1 & a_2 \end{pmatrix} \in \mathbb{R}^2$ such that $E((Y_1,Y_2)|l_n(X)) = Xa$. Hence,

$$\begin{cases} a_1 = \frac{\gamma-\beta\delta}{1-\alpha\beta} \\ a_2 = \frac{\delta-\alpha\gamma}{1-\alpha\beta} \end{cases}$$

Therefore, from a, one cannot obtains unique values for α, β, δ and γ.
Hence, the reduced form identifies only the composite coefficients a_1 and a_2, not the structural parameters $(\alpha,\beta,\gamma,\delta)$.
To identify them, I propose two restrictions (since there are for unknown parameters but only two reduced-form coefficients).

a) $\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \beta \end{pmatrix} = 0$, i.e. $\beta = 0$

b) $\begin{pmatrix} -1 & 1 \end{pmatrix} \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = 0$, i.e. $\gamma = \delta$

Solution to Exercise A.29. T is $\sigma(X,Y)$-measurable, and hence is a statistics. However, X is not exogenous for p_T for studying the impact of X on Y. Indeed, T does not generate a cut into $p_{X,Y}$ since $T \neq E(Y|\sigma(X))$.

Solution to Exercise A.30. Using the OLS, $E(Y|l_n(X))(\omega) = a + bX(\omega)$, with a and b given by

$$b = \frac{\langle Y,X\rangle - \bar{Y}\bar{X}}{\|X\|^2 - (\bar{X})^2} = \frac{\frac{10+40+30+125+180+256+400+588}{8} - \frac{10+20+15+25+30+32+40+42}{8}\frac{1+2+2+5+6+8+14}{8}}{\frac{1+4+4+25+36+64+100+196}{8} - (\frac{1+2+2+5+6+8+10+14}{8})^2}$$
$$= 1.503$$
$$a = \bar{Y} - b\bar{X}$$
$$= 26.75 - 1.503 * 4.75 = 19.609$$

$\beta(Z)$ is a $\sigma(Z)$-measurable variable. For a given ω: $\beta(Z)(\omega) = \frac{Y(\omega)}{X(\omega)}$ No, $T := E(Y|l_n(X))$

	X	Y	Z	$\beta(Z)$
ω_1	1	10	A	10
ω_2	2	20	A	10
ω_3	2	10	B	5
ω_4	5	25	B	5
ω_5	6	30	B	5
ω_6	8	32	C	4
ω_7	10	40	C	4
ω_8	14	42	D	3

can be comuted but it does not correspond to the genuine conditionnal expectation $E(Y|\sigma(X,Z))$.

Solution to Exercise A.31.

1. We first compute $E(Y \mid l_n(X,Z))$. Since (X,Z) fully determines each observation, we have:

ω	X	Z	Y	$E(Y \mid l_n(X,Z))$
ω_1	0	0	0	0
ω_2	0	1	2	2
ω_3	1	0	1	1
ω_4	1	1	3	3

   Hence,

$$E(Y \mid l_n(X,Z)) = Y.$$

   We now compute $E(Y \mid l_n(X))$.
   For $X = 0$, we consider ω_1 and ω_2, so:

$$E(Y \mid X = 0) = \frac{0+2}{2} = 1.$$

   For $X = 1$, we consider ω_3 and ω_4, so:

$$E(Y \mid X = 1) = \frac{1+3}{2} = 2.$$

   Therefore,

ω	X	Z	Y	$E(Y \mid l_n(X))$
ω_1	0	0	0	1
ω_2	0	1	2	1
ω_3	1	0	1	2
ω_4	1	1	3	2

   Thus,

$$E(Y \mid l_n(X)) = \begin{cases} 1 & \text{if } X = 0, \\ 2 & \text{if } X = 1. \end{cases}$$

   The two conditional expectations are not equal:

$$E(Y \mid l_n(X,Z)) \neq E(Y \mid l_n(X)).$$

2. No, the statistic $T = l_n(X)$ does not generate a cut of the joint distribution of (Y,X,Z).
   Indeed, conditioning on X alone does not contain as much information as conditioning on (X,Z). More precisely,

$$E(Y \mid l_n(X,Z)) \neq E(Y \mid l_n(X)).$$

   This means that X alone is not sufficient to recover the relevant conditional expectation: the variable Z still matters once X is given.
   Therefore, $T = l_n(X)$ does not generate a cut.